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# Best Engineered For Energy Absorption Technology

Shock Absorber  
Hydraulic Buffer





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# Best Engineered For Energy Absorption Technology







## Overview

In many of industrial fields, production speed increased in high face for better productivity. In addition, the needs for high energy capacity with compact shock absorber has increased whereas the application product size is getting bigger.

Through intensive technical development, IZMAC upgrade all shock absorber models more than 200% capacity of previous ones and improve collision speed allowance range as 0.1m/sec~5 m/sec. Thanks to production management, IZMAC can provide low cost, long life products with short delivery time.

IZMAC would do its best to supply products that customer satisfy by way of on-going research and development with advanced production & quality management.

## Best Engineered For Energy Absorption Technology



### **DECELERATION CONTROL SOLUTION :**

Provide customer deceleration control solution  
customer satisfactory policy

### **CREATE MAX PRODUCTIVITY :**

Create maximum productivity by supplying best  
product to customers

### **INNOVATION :**

On-going innovation to maintain its high level quality







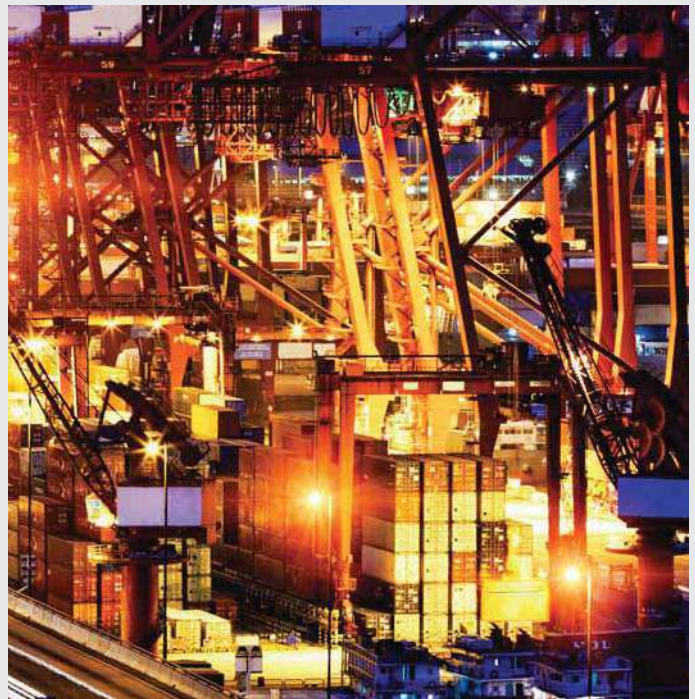


## Hydraulic Buffers

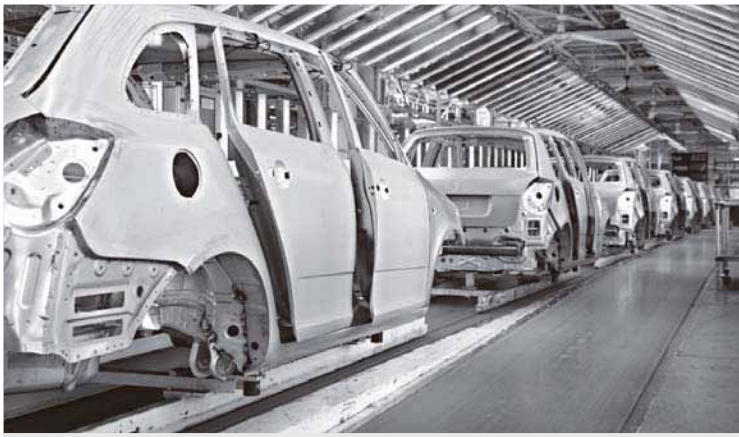
Military, Bridge, Crane & Trolley, Rail End Stop, Railway Coupler & Side Buffers, Elevator

## Logistic Automation

Warehouse Automation, Mining, Conveyor Systems











# Applications

## Factory Automation

### Automotive

- Car manufacturing(Assembly/Transfer/ Inspection/Pick & Place robot line)
- Car-Welding, Assembly line, Tire line
- Electric actuator, Rodless cylinder, High speed cylinder, Guide cylinder, Stopper cylinder, Handling system, Linear module protection
- Pneumatic components  
Grippers, Pneumatic cylinder, Pneumatic valve, Fluid control components
- Electronics  
Semi-conductor equipment, Circuit breaker
- Factory  
Steel factory, Paper factory
- Safety : Brake cylinder, Tie rod cylinder, Compact cylinder, Brake unit, Linear guide, Emergency stop

### \* Warehouse Automation



### Features

- Upgrade precision degree
- Secure stability, Optimizing control
- Reliability of production lines
- Max productivity
- Decrease failure ratio
- Increase output quality level



# Production Process

01

## Design

What is technology? It will be the one which makes differences. IZMAC always open ears to hear customer's voices for renovation. IZMAC makes endless R&D and it supports to develop quality of products for customer's productivity. On time customizing is also one of helpful features of IZMAC.

02

## Machining process

Clean machining clean products. IZMAC has its own fleets to doing machining process. To minimizes defect ratio into "0", skillful engineers manage all over the process. Advanced machineries are helpful to carry out complicate & difficult output process.

03

## Part inspection

IZMAC only supply products after total inspection process. If there are any small defects found it goes directly to the beginning process. Perfect product delivery is what IZMAC quality assurance system is aiming at.

04

## Assembly

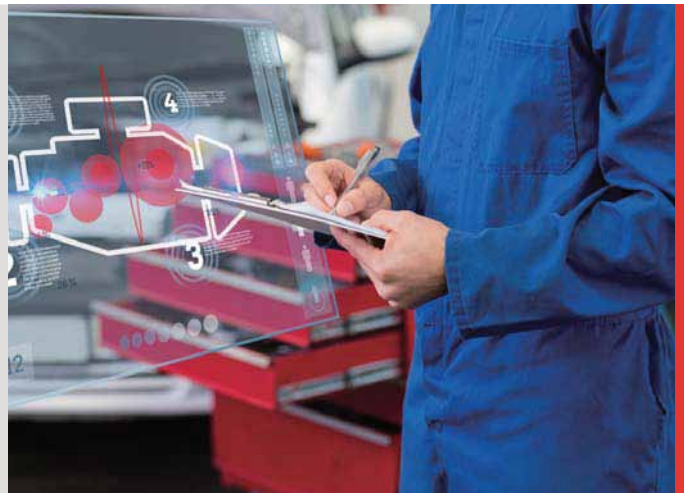
Assembly is one of very important process of production. Every part should be inserted into the right position to do its right function.



05

## Product inspection / Test

Through product inspection In the final qualifying process, IZMAC products are ready to deliver customer's places. Defective products are automatically transferred to the beginning process of production or trashed to prevent quality trouble.



06

## Shipping

IZMAC only supply products in perfect condition of packing. If there are any small defects found or manager feels that package is not settled well, the packaging system is reset for safe delivery. Perfect product delivery is what IZMAC quality assurance system is looking for.



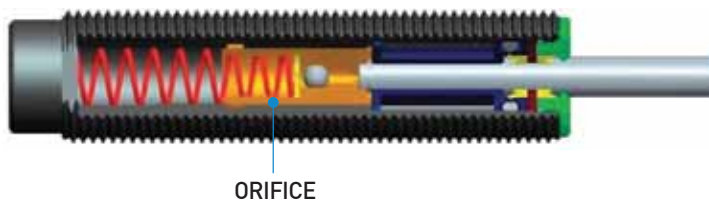


# Shock Absorber

## Orifice features

### SINGLE ORIFICE

- 1 Single orifice is called as "Dashpot" and it has only one orifice on the tube as a hole or circled shape gap. It has the features that orifice dimension is maintained for total stroke when it has flow activity.
- 2 As per the diagram, resistance force is going high suddenly after collision and it dramatically down when the stroke is in progress. And the force is decreasing step by step.

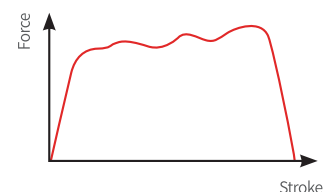
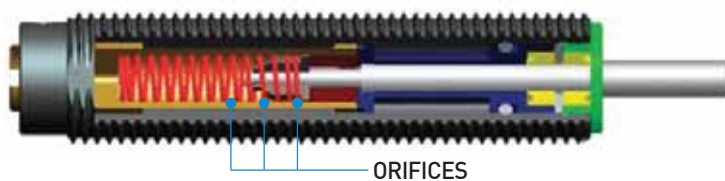


### MULTIPLE ORIFICES

On the other hands, by using multiple orifices, total oil route dimension is decreasing when the stroke is on-going. It enables to maintain total shock absorption safely and softly.

#### ■ Adjustable type

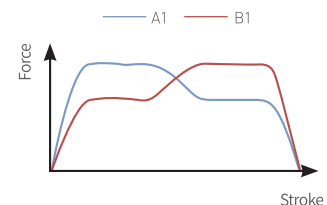
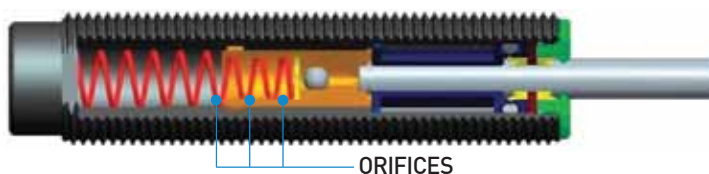
By using adjustment dial, it can provide best deceleration features against various collision cases.



#### ■ Self compensating

Self compensating type shock absorber is optimized for repeated work conditions and it makes ideal deceleration features by using highly optimized orifices on customized shock conditions.

Diagram A1 shows the condition of high speed/low propelling force, B1 shows low speed/high propelling forces and they show resist forces are not increased excessive manner on those contrast conditions.



## Principal of energy absorption

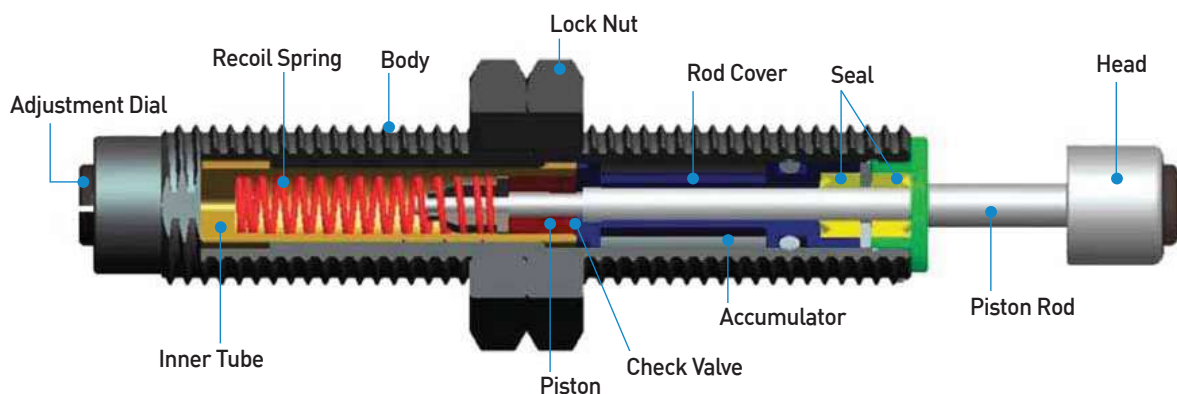
After impact piston is inserted into inner tube, oil flows through orifice holes made in the tube. The impact kinetic energy is changed into thermal energy by way of rebounding activity. And the thermal energy is dissipated outside through threaded body.



## Internal Structure

- 1 Shock absorber consists of 5 major parts : Head, Piston rod, Body, Adjustment dial and Lock nut.
- 2 Bumper head is the part which contact colliding object directly, and it has the function of stopper at the end of stroke with contacting body.
- 3 Piston rod transfer impact energy to the piston through full stroke.
- 4 Body outside is fully threaded for easy installation at any position and for more heat dissipation.
- 5 Adjustment dial is for optimum operation by setting Knob to the various impact conditions.
- 6 Lock nut is used for installing and fixing shock absorber tightly coupled with body thread on the machine.

Inside of shock absorber consist of piston, check valve, inner tube, multiple orifices, accumulator, return spring and oil. On impact the piston rod is inserted into inner tube and check valve is closed. And it forces oil in the chamber to flow through orifice. At the same time, damping force is generated into piston. Compressed oil which flowed into accumulator position is compensating same volume of piston inside. After one cycle ended, return process is made by coil spring and check valve is opened when it starts return process. Because the oil is moved from oil tube into inner tube, piston rod is returned to its original position for proper operation.



## Ordering information

| Ordering Info |                 | Previous    | Current | Explanation                                           |
|---------------|-----------------|-------------|---------|-------------------------------------------------------|
| Series        | Absorber        | IAAS<br>SDA | IAA     | Izmac mini Absorber Adjustable type                   |
|               |                 | IASS        | IMS     | Izmac Mini & Middle class absorber Self-compensating  |
|               |                 | IAAH        | IHA     | Izmac Heavy type Absorber                             |
|               |                 | IAAD        | ISA     | Izmac Stright type Absorber (middle class)            |
|               | Buffer          | IBSG        | IHG     | Izmac Hydraulic Gas & oil type buffer                 |
|               |                 | IBAW        | ICA     | Izmac hydraulic & oil stacker Crane Adjustable buffer |
|               |                 | IBSW        | ISC     | Izmac hydraulic & oil Stcker Crane buffer             |
|               | Special product | ISAA        | IHC     | Izmac Hydro Check                                     |
|               |                 | IDSM        | ICD     | Izmac Crash Damper                                    |
|               |                 | IDSP        | IPC     | Izmac Plastic type Crash damper                       |
|               |                 | IRAD        | IRD     | Izmac Rate control Double dial                        |
|               |                 | IASR        | ISR     | Izmac Short Range absorber                            |

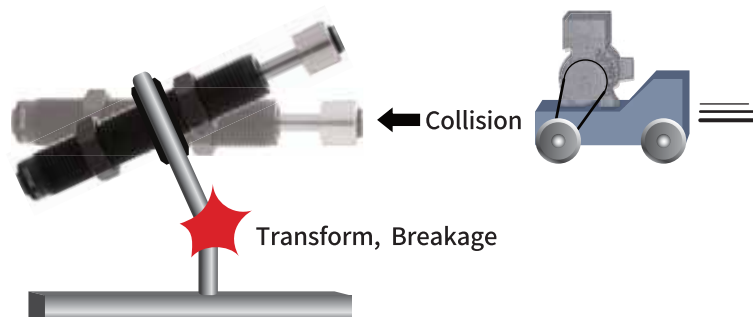


### Safety Instruction

#### 1. Shock absorber

IZMAC shock absorber has been developed to endure up to million cycles. To assure long life cycle please be aware of the following regulations.

- 1-1 Ensure correct energy capacity before operation of shock absorber.
- 1-2 Do not paint or weld to the body of shock absorber.
- 1-3 Piston rod not to be marked or painted. (Shock absorber may have malfunction with oil leakage.)
- 1-4 Adjustment dial knob is preset to "0" by factory. Once achieved the correct damping strength, tighten the lock screw.
- 1-5 Shock absorber shall be kept away from fire because can be ignited by oil.
- 1-6 Always check strength of installation site before using and it is recommended to installing shock absorber at strong base structure.  
Secure the strength as " Max shock energy x safety ratio. (2.5)"



- 1-7 Refer to below table of proper lock nut installation tightening torque. Tightening with excessive force may cause malfunction or damage to product.

| Screw<br>outdia<br>(mm) | M8x1.0 | M10x1.0 | M12x1.0 | M14x1.0<br>M14x1.5 | M16x1.5 | M20x1.5 | M25x1.5<br>M25x2.0 | M27x1.5<br>M27x2.0<br>M27x3.0 | M30x1.5 | M36x1.5 | M42x1.5 |
|-------------------------|--------|---------|---------|--------------------|---------|---------|--------------------|-------------------------------|---------|---------|---------|
| Nut<br>Torque<br>(Nm)   | 3.9    | 7.8     | 7.8     | 9.8                | 14.7    | 29.4    | 49                 | 58.8                          | 78.4    | 98      | 392     |

# It will be efficient when it apply additional anti-loosening adhesives.

- 1-8 Always check the operation temperature of shock absorber to get best performance.

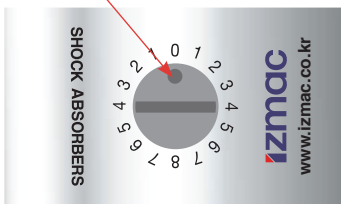


- 1-9 Impact mass should be perpendicular to the axis of shock absorber. In case of an side load impact especially angle of deviation over  $1^\circ$ , it is better to use an side load adapter. Otherwise failures can be occurred due to abrasion of inner sealings.



- 1-10 Shield the shock absorber to protect it from foreign material such as dust, acids, steam, solvents, cutting fluids, weld flash, and dust.
- 1-11 Check the operating temperature range of shock absorber ( $-10^\circ\text{C} \sim 80^\circ\text{C}$ ) to get best performance. If you need to use shock absorber in other temperature exceed standard operating temperature, please contact us.
- 1-12 Adjustable type of shock absorber is designed to get suitable resistance force according to impact velocity by controlling the area of orifice. Refer to below adjustment dial of IDA series, ISA Series and IHA Series.
- Lock the set screw bolt after adjust dial with fitting properly on your application conditions.

### IAA & ISA & IHA Series Adjustment Dial

| IAA Series                                                                                                | ISA Series                                                                                                | IHA Series                                                                                                  |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
|  <p>Set Screw bolt</p> |  <p>Set Screw bolt</p> |  <p>Set Screw bolt</p> |

# It will be efficient when it apply additional anti-loosening adhesives.



## 2. General Inspection

### 2-1 Regular Checking

1) Check the below points regularly.

It is impossible to disassemble and repair the shock absorber so change the shock absorber when it is broken.

2) Inspection Point

- Condition of lock nut fastened
- Condition of piston rod (Damage or Scratch)
- Checking the stroke
- Outer appearance of shock absorber
- Oil leakage
- Noise on process

If any trouble found after inspection, please do trouble shooting with considering "2-2. trouble & trouble shooting.

### 2-2 Trouble and trouble - shooting

| Trouble                           | Cause                                                         | Trouble - shooting                                                                             |
|-----------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Lock Nut is not fastened tightly. | Lack of fixing torque                                         | Fasten the nut or bolt tightly                                                                 |
|                                   | Vibration of machine                                          | Check the vibration of machine and adjust it                                                   |
| Not Suitable Shock absorption     | The required energy is over the capacity of shock absorber    | Change the shock absorber with higher capacity                                                 |
|                                   | The adjustable dial step is not suitable                      | Set adjustment dial to "0"                                                                     |
|                                   | Stop the stroke by stopper                                    | Adjust the stopper location to use long stroke                                                 |
|                                   | The shock absorber is not installed in line                   | Install the shock absorber in line                                                             |
|                                   | The moving load is hit the shock absorber over 1 degree       |                                                                                                |
|                                   | There is vibration in moving load                             | Install the guide to remove the vibration                                                      |
|                                   | Use the body of shock absorber as stopper                     | Install the stop collar or other stopper                                                       |
|                                   | Ambient temperature is too high                               | Change the shock absorber which is applicable for high temperature                             |
|                                   | Damaged seal by other material on the piston rod              | Clean the piston rod and protect it from the dust                                              |
|                                   | The rotating moving load is impacted direct to shock absorber | Install the protection object to prevent from the direct impact to shock absorber              |
|                                   | The lifespan of shock absorber is over                        | Change the shock absorber to new one                                                           |
| OIL Leakage                       | Damaged seal by other material on the piston rod              | Clean the piston rod and protect it from the dust                                              |
|                                   | Ambient temperature is too low                                | Warm the shock absorber as temp, over -10°C or change it to shock absorber for low temperature |
|                                   | The lifespan of shock absorber is over                        | Change the shock absorber to new one                                                           |
| Rod is not returned.              | The shock absorber is not installed in line                   | Install the shock absorber in line                                                             |
|                                   | The moving load is hit the shock absorber over 1 degree       |                                                                                                |
|                                   | There is vibration in moving load                             | Install the guide to remove the vibration                                                      |
|                                   | Use the body of shock absorber as stopper                     | Install the stop collar or other stopper                                                       |
|                                   | Ambient temperature is too low                                | Warm the shock absorber as temp, over -10°C or change it to shock absorber for low temperature |
|                                   | The lifespan of shock absorber is over                        | The lifespan of shock absorber is over                                                         |

※ When using the shock absorber inside the chamber, be sure to contact IZMAC before using it.

### ■ Symbols

| Symbol  | Unit       | Description                | Symbol   | Unit               | Description                        |
|---------|------------|----------------------------|----------|--------------------|------------------------------------|
| $E_K$   | Nm         | Kinetic energy             | T        | Nm                 | Torque                             |
| $E_W$   | Nm         | Work energy                | H        | m                  | Height                             |
| $E_T$   | Nm         | Total energy               | m        | kg                 | Weight                             |
| $E_T C$ | Nm / h     | Total energy per hour      | P        | bar                | Operating pressure                 |
| $F_S$   | N          | Impact force               | $\omega$ | rad / s            | Angular velocity                   |
| $F_P$   | N          | Propelling force           | $\alpha$ | °                  | Angle of incline                   |
| $m_e$   | kg         | Effective weight           | I        | kgm <sup>2</sup>   | Moment of inertia                  |
| C       | Cycles / h | Cycles per hour            | K        | m                  | Turning radius                     |
| S       | m          | Stroke                     | $R_S$    | m                  | Mounting distance from pivot point |
| V       | m / s      | Impact velocity            | $\mu$    |                    | Coefficient of friction            |
| $V_D$   | m / s      | Mass velocity              | g        | m / s <sup>2</sup> | Gravitational acceleration         |
| $P_M$   | kW         | Motor power                | a        | m / s <sup>2</sup> | Acceleration                       |
| D       | mm         | Inner diameter of Cylinder | t        | s                  | Time                               |
| $R_S$   | m          | Shock absorber location    | K        | m                  | Radius of gyration                 |
| $R_C$   | m          | Cylinder location          |          |                    |                                    |

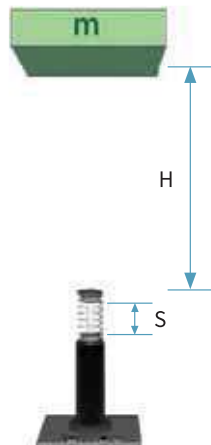
### ■ Useful formulas

| Description                  | Symbol | Formulas                       |
|------------------------------|--------|--------------------------------|
| Impact force                 | $F_S$  | $= E_T / S / 0.8$              |
| Motor power                  | $F_P$  | $= 3000 \times P_M / V$        |
| Cylinder power               | $F_P$  | $= 0.0785 \times D^2 \times P$ |
| Effective weight             | $m_e$  | $= 2 \times E_T / V^2$         |
| Velocity (free falling)      | V      | $= \sqrt{2 \times g \times H}$ |
| Velocity (with acceleration) | V      | $= 2 \times D / t$             |
| Velocity (w/o acceleration)  | V      | $= D / t$                      |
| Deceleration                 | a      | $= V^2 / 2 \times S$           |
| Stopping time                | t      | $= 2.5 \times S / V$           |

### ■ Examples

#### | A | Free falling mass

- Weight  $m = 600\text{kg}$
- Height  $H = 0.5\text{m}$
- Cycles per hour  $C = 60\text{Cycles/h}$
- Gravitational acceleration  $g = 9.81\text{m/s}^2$



$$E_K = m \times g \times H = 600 \times 9.81 \times 0.5 = 2,943 \text{ Nm}$$

Selected Model with  $E_K$  : ISA64 -100

$$E_W = m \times g \times S = 600 \times 9.81 \times 0.1 = 589 \text{ Nm}$$

$$E_T = E_K + E_W = 2,943 + 589 = 3,532 \text{ Nm}$$

$$E_T C = E_T \times C = 3,532 \times 60 = 211,896 \text{ Nm / h}$$

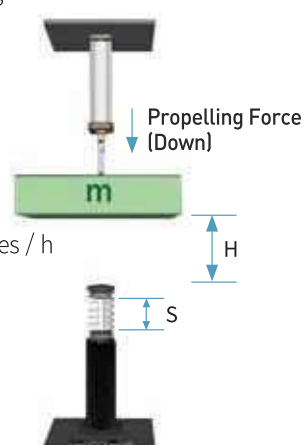
$$V = \sqrt{2 \times g \times H} = \sqrt{2 \times 9.81 \times 0.5} = 3.13 \text{ m / s}$$

$$m_e = \frac{2 \times E_T}{V^2} = \frac{2 \times 3,532}{3.13^2} = 720 \text{ kg}$$

Selected Model with  $E_T$ ,  $E_T C$  &  $m_e$  : ISA 64-100

#### | B | Down mass with propelling force

- Weight  $m = 1,200\text{kg}$
- Impact velocity  $V = 1.8\text{m / s}$
- Inner diameter of Cylinder  $D = 140\text{mm}$
- Operating pressure  $P = 5\text{bar}$
- Cycles per hour  $C = 125\text{Cycles / h}$



$$E_K = \frac{m \times V^2}{2} = \frac{1,200 \times 1.8^2}{2} = 1,944 \text{ Nm}$$

Selected Model with  $E_K$  : ISA 85-50

$$F_P = 0.0785 \times D^2 \times P + m \times g = 0.0785 \times 140^2 \times 5 + 1,200 \times 9.81 = 19,465 \text{ N}$$

$$E_W = F_P \times S = 19,465 \times 0.05 = 973 \text{ Nm}$$

$$E_T = E_K + E_W = 1,944 + 973 = 2,917 \text{ Nm}$$

$$E_T C = E_T \times C = 2,917 \times 125 = 364,656 \text{ Nm / h}$$

$$m_e = \frac{2 \times E_T}{V^2} = \frac{2 \times 2,917}{1.8^2} = 1,801 \text{ kg}$$

Selected Model with  $E_T$ ,  $E_T C$  &  $m_e$  : ISA 85-50

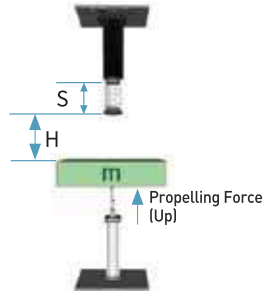


# Shock Absorber

## Formula and Calculation

### | C | Up mass with propelling force

- Weight  $m = 200\text{kg}$
- Impact velocity  $V = 2.5\text{m/s}$
- Inner diameter of Cylinder  $D = 150\text{mm}$
- Operating pressure  $P = 5\text{bar}$
- Cycles per hour  $C = 180\text{Cycles/h}$



$$E_k = \frac{m \times V^2}{2} = \frac{200 \times 2.5^2}{2} = 625 \text{ Nm}$$

Selected Model with  $E_k$ : ISA 45-50

$$F_p = 0.0785 \times D^2 \times P + m \times g = 0.0785 \times 150^2 \times 5 - 200 \times 9.81 = 6,869 \text{ N}$$

$$E_w = F_p \times S = 6,869 \times 0.05 = 343 \text{ Nm}$$

$$E_T = E_k + E_w = 625 + 343 = 968 \text{ Nm}$$

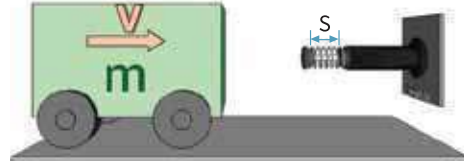
$$E_T C = E_T \times C = 968 \times 180 = 174,323 \text{ Nm / h}$$

$$m_e = \frac{2 \times E_T}{V^2} = \frac{2 \times 968}{2.5^2} = 310 \text{ kg}$$

Selected Model with  $E_T$ ,  $E_T C$  &  $m_e$ : ISA 45-50

### | D | Horizontal mass without propelling force

- Weight  $m = 100\text{kg}$
- Impact velocity  $V = 1.5\text{m/s}$
- Cycles per hour  $C = 600\text{Cycles/h}$



$$E_k = \frac{m \times V^2}{2} = \frac{100 \times 1.5^2}{2} = 113 \text{ Nm}$$

$$E_T = E_k + E_w = 113 + 0 = 113 \text{ Nm}$$

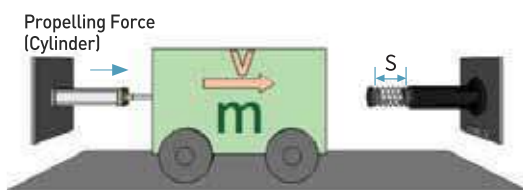
$$E_T C = E_T \times C = 113 \times 600 = 67,800 \text{ Nm / h}$$

$$m_e = \frac{2 \times E_T}{V^2} = \frac{2 \times 113}{1.5^2} = 100 \text{ kg}$$

Selected Model with  $E_T$ ,  $E_T C$  &  $m_e$ : IAA 25-25

### | E | Horizontal mass with propelling force [Cylinder]

- Weight  $m = 300\text{kg}$
- Impact velocity  $V = 1.2\text{m/s}$
- Inner diameter of Cylinder  $D = 50\text{mm}$
- Operating pressure  $P = 5\text{bar}$
- Cycles per hour  $C = 300\text{Cycles/h}$



$$E_k = \frac{m \times V^2}{2} = \frac{300 \times 1.2^2}{2} = 216 \text{ Nm}$$

Selected Model with  $E_k$ : IAA 36-25

$$F_p = 0.0785 \times D^2 \times P = 0.0785 \times 50^2 \times 5 = 981 \text{ N}$$

$$E_w = F_p \times S = 981 \times 0.025 = 25 \text{ Nm}$$

$$E_T = E_k + E_w = 216 + 25 = 241 \text{ Nm}$$

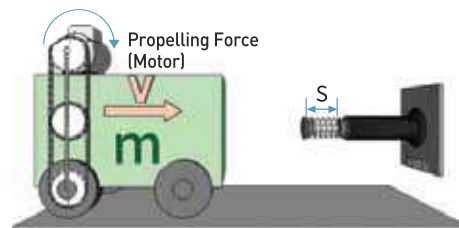
$$E_T C = E_T \times C = 241 \times 300 = 72,300 \text{ Nm / h}$$

$$m_e = \frac{2 \times E_T}{V^2} = \frac{2 \times 241}{1.2^2} = 335 \text{ kg}$$

Selected Model with  $E_T$ ,  $E_T C$  &  $m_e$ : IAA 36-25

### | F | Horizontal mass with propelling force [Motor]

- Weight  $m = 1,300\text{kg}$
- Impact velocity  $V = 1.6\text{m/s}$
- Motor power  $P = 3\text{kW}$
- Cycles per hour  $C = 100\text{Cycles/h}$



$$E_k = \frac{m \times V^2}{2} = \frac{1,300 \times 1.6^2}{2} = 1,664 \text{ Nm}$$

Selected Model with  $E_k$ : ISA 64-50

$$F_p = \frac{3,000 \times P_M}{V} = \frac{3,000 \times 3}{1.6} = 5,625 \text{ N}$$

$$E_w = F_p \times S = 5,625 \times 0.05 = 281 \text{ Nm}$$

$$E_T = E_k + E_w = 1,664 + 281 = 1,945 \text{ Nm}$$

$$E_T C = E_T \times C = 1,945 \times 100 = 194,500 \text{ Nm / h}$$

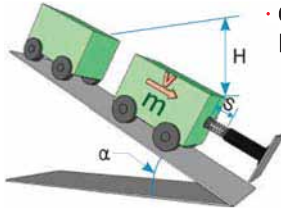
$$m_e = \frac{2 \times E_T}{V^2} = \frac{2 \times 1,945}{1.6^2} = 1,520 \text{ kg}$$

Selected Model with  $E_T$ ,  $E_T C$  &  $m_e$ : ISA 64-50



## | G | Sliding down mass without propelling force

- Weight  $m = 800\text{kg}$
- Height  $H = 0.7\text{m}$
- Angle of incline  $\alpha = 30^\circ$
- Cycles per hour  $C = 100\text{Cycles/h}$



$$E_k = m \times g \times H = 800 \times 0.7 \times 9.81 = 5,494 \text{ Nm}$$

Selected Model with  $E_k$ : ISA 85-100

$$F_s = m \times \sin(\alpha) \times g = 800 \times \sin(30) \times 9.81 = 3,922 \text{ N}$$

$$E_w = F_s \times S = 3,922 \times 0.1 = 392 \text{ Nm}$$

$$E_T = E_k + E_w = 5,494 + 392 = 5,886 \text{ Nm}$$

$$E_T C = E_T \times C = 5,886 \times 100 = 588,600 \text{ Nm}$$

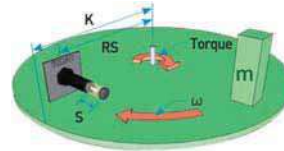
$$V = \sqrt{2 \times g \times H} = \sqrt{2 \times 9.81 \times 0.7} = 3.7 \text{ m/s}$$

$$m_e = \frac{2 \times E_T}{V^2} = \frac{2 \times 5,886}{3.7^2} = 860 \text{ Nm/h}$$

Selected Model with  $E_T$ ,  $E_T C$  &  $m_e$ : ISA 85-100

## | H | Rotary table mass with propelling force

- Weight  $m = 200\text{kg}$
- Angular Velocity  $\omega = 2\text{rad/s}$
- Mounting distance from pivot point  $R_s = 0.5\text{m}$
- Radius of gyration  $K = 0.8\text{m}$
- Torque  $T = 400\text{Nm}$
- Cycles per hour  $C = 350\text{Cycles/h}$



$$I = m \times K^2 = 200 \times 0.8^2 = 128 \text{ kgm}^2$$

$$E_k = \frac{I \times \omega^2}{2} = \frac{128 \times 2^2}{2} = 256 \text{ Nm}$$

Selected Model with  $E_k$ : IAA 36-25

$$F_p = \frac{T}{R_s} = \frac{400}{0.5} = 800 \text{ N}$$

$$E_w = F_p \times S = 800 \times 0.05 = 20 \text{ Nm}$$

$$E_T = E_k + E_w = 256 + 20 = 276 \text{ Nm}$$

$$E_T C = E_T \times C = 276 \times 350 = 96,600 \text{ Nm/h}$$

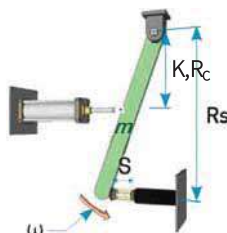
$$V = R_s \times \omega = 0.5 \times 2 = 1 \text{ m/s}$$

$$E_k = \frac{2 \times E_T}{V^2} = \frac{2 \times 276}{1^2} = 552 \text{ kg}$$

Selected Model with  $E_T$ ,  $E_T C$  &  $m_e$ : IAA 36-25

## | I | Swiveling mass with propelling force

- Weight  $m = 50\text{kg}$
- Angular Velocity  $\omega = 2.5\text{rad/s}$
- Mounting distance from pivot point  $R_s = 0.8\text{m}$
- Radius of gyration  $K = 0.5\text{m}$
- Inner diameter of Cylinder  $D = 40\text{mm}$
- Operating pressure  $P = 5\text{Bar}$
- Cycles per hour  $C = 1,000\text{Cycles/h}$



$$I = m \times K^2 = 50 \times 0.5^2 = 13 \text{ kgm}^2$$

$$E_k = \frac{I \times \omega^2}{2} = \frac{13 \times 2.5^2}{2} = 41 \text{ Nm}$$

Selected Model with  $E_k$ : IAA 20-15

$$F_s = \frac{0.0785 \times D^2 \times P \times R_c}{R_s} = \frac{0.0785 \times 40^2 \times 5 \times 0.5}{0.8} = 393 \text{ N}$$

$$E_w = F_p \times S = 393 \times 0.015 = 6 \text{ Nm}$$

$$E_T = E_k + E_w = 39 + 6 = 45 \text{ Nm}$$

$$E_T C = E_T \times C = 45 \times 1,000 = 45,000 \text{ Nm/h}$$

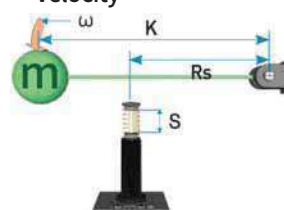
$$V = R_s \times \omega = 2.5 \times 0.8 = 2 \text{ m/s}$$

$$E_k = \frac{2 \times E_T}{V^2} = \frac{2 \times 45}{2^2} = 23 \text{ kg}$$

Selected Model with  $E_T$ ,  $E_T C$  &  $m_e$ : IAA 20-15

## | J | Swiveling mass without propelling force

- Weight  $m = 200\text{kg}$
- Mass velocity (from Mass Center)  $V_m = 2\text{m/s}$
- Angular Velocity  $\omega = 2.5\text{rad/s}$
- Mounting distance from pivot point  $R_s = 0.6\text{m}$
- Radius of gyration  $K = 0.8\text{m}$
- Cycles per hour  $C = 200\text{Cycles/h}$



$$I = m \times K^2 = 200 \times 0.8^2 = 128 \text{ kgm}^2$$

$$E_k = \frac{I \times \omega^2}{2} = \frac{128 \times 2.5^2}{2} = 400 \text{ Nm}$$

Selected Model with  $E_k$ : IAA 36-50

$$F_s = \frac{m \times g \times K}{R_s} = \frac{200 \times 9.81 \times 0.8}{0.6} = 2,616 \text{ N}$$

$$E_w = F_s \times S = 2,616 \times 0.05 = 131 \text{ Nm}$$

$$E_T = E_k + E_w = 400 + 131 = 531 \text{ Nm}$$

$$E_T C = E_T \times C = 531 \times 200 = 106,200 \text{ Nm/h}$$

$$V = R_s \times \omega = 0.6 \times 2.5 = 1.5 \text{ m/s}$$

$$m_e = \frac{2 \times E_T}{V^2} = \frac{2 \times 531}{1.5^2} = 472 \text{ kg}$$

Selected Model with  $E_T$ ,  $E_T C$  &  $m_e$ : IAA 36-50



# IAA Series Dial type

## IZMAC mini Absorber Adjustable type

### DESCRIPTION

IAA has dial which can set 12 steps of shock speed ranges and it is uparaded on wider damping ranges. Energy capacity was Improved over 200% and effective weight ranges become wider than before.

#### IAA series



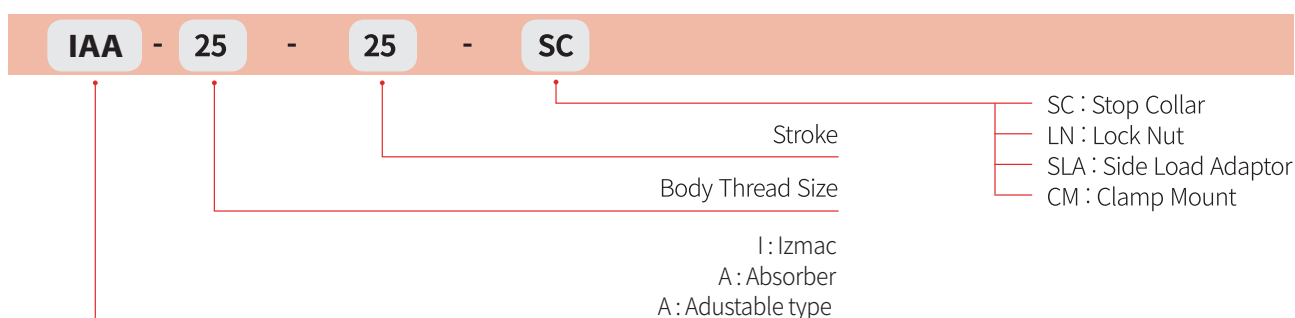
### FEATURES

- 1 All threaded body outside helps to install shock absorber and it can be fixed at the exact position. And it helps also to dissipate more shock energy by increased surface dimension.
- 2 Body surface treated as alloyed black colored steel or Zn-Ni and it develops anti corrosion features highly.
- 3 Stop collar is not necessary when shock absorber has steel head, and poly pad & urethane cap will help to decrease collision noises efficiently.
- 4 Rod cover is made of long lasting materials with function of seal protection and that enable to guarantee long life.
- 5 Speed ranges : 0.3 ~ 5m/s • Low Velocity (LV) : 0.08 ~ 1.3m/s
- 6 Temperature ranges : -10 ~ 80°C • Special : -40 ~ 120°C(Oil and Seal)

### APPLICATIONS

Pick N place robot, machining tools, automobile assembly line, tire assembly line, crane, safety devices and all other industrial ranges of multi purposes.

### IAA SERIES ORDERING INFORMATION



# IAA10 - 07(B) / 12 - 10(B)

Best engineered  
for energy absorption  
technology

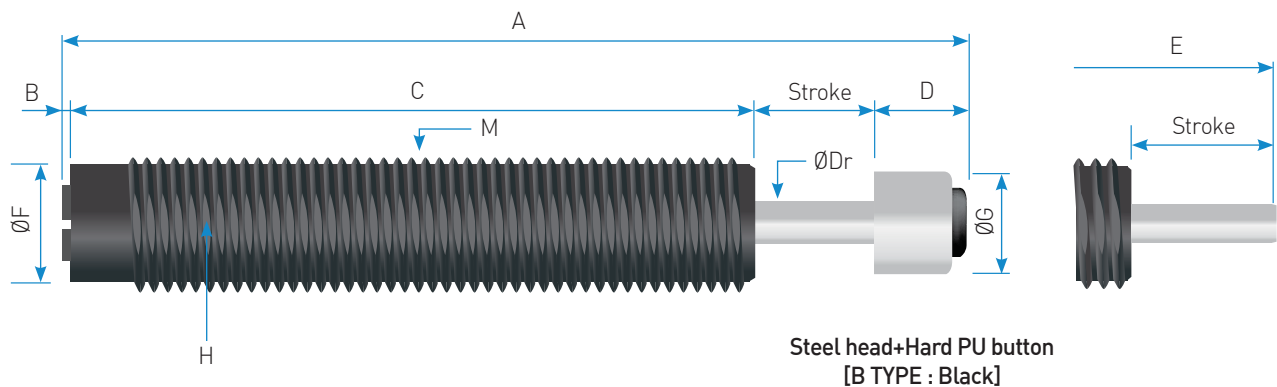


## Engineering Data

| Model         | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) E <sub>T</sub> | Max.Energy /<br>Hour(Nm/h)<br>E <sub>T</sub> C | Effective<br>Weight(kg) m <sub>e</sub> | Recoil Forec(N) |       | Weight(g) |
|---------------|----------------|------------------------------------------|------------------------------------------------|----------------------------------------|-----------------|-------|-----------|
|               |                |                                          |                                                |                                        | Ext.            | Comp. |           |
| IAA10 - 07(B) | 7              | 5.5                                      | 15,000                                         | 1 - 123                                | 2.4             | 5.4   | 21        |
| IAA12 - 10(B) | 10             | 16                                       | 30,500                                         | 1.5 - 333                              | 3.6             | 9.8   | 41        |

## Dimensions

| Model         | Stroke | A  | B   | C    | D  | E  | F     | G     | H      | M       | Dr |
|---------------|--------|----|-----|------|----|----|-------|-------|--------|---------|----|
| IAA10 - 07(B) | 7      | 68 | 1.2 | 52.8 | 7  | 61 | Ø8.6  | Ø8.6  | SW:8.8 | M10X1.0 | Ø3 |
| IAA12 - 10(B) | 10     | 84 | 1.2 | 62.8 | 10 | 74 | Ø10.5 | Ø10.5 | -      | M12X1.0 | Ø4 |

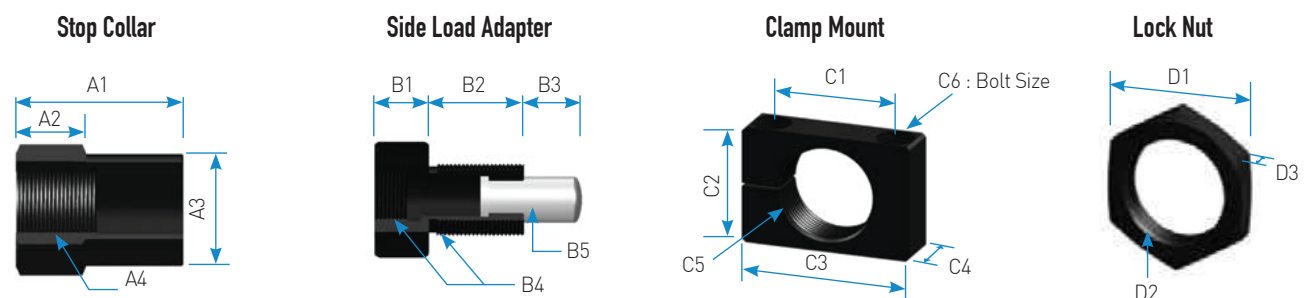


## Accessories

- Two Lock Nuts included for each absorber
- Other accessories to be ordered separately

(unit : mm)

| Stop Collar | Side Load Adapter | Clamp Mount | Lock Nut | A1 | A2 | A3    | A4        | B1   | B2   | B3        | B4        | B5 | C1 | C2 | C3 | C4        | C5        | C6       | D1        | D2        | D3 |
|-------------|-------------------|-------------|----------|----|----|-------|-----------|------|------|-----------|-----------|----|----|----|----|-----------|-----------|----------|-----------|-----------|----|
| SC10X1.0    | SLA10-07          | CM10X1.0    | LN10X1.0 |    |    | Ø12.8 | M10 X 1.0 | 12.5 | 7    | M10 X 1.0 | Ø4        | 16 | 14 | 25 | 10 | M10 X 1.0 | M4        | 13       | M10 X 1.0 |           |    |
| SC12X1.0    | SLA12-10          | CM12X1.0    | LN12X1.0 | 20 | 8  | Ø13.8 | M12 X 1.0 | 12   | 17.5 | 10        | M12 X 1.0 | Ø5 | 20 | 16 | 32 | 12        | M12 X 1.0 | M5 X 18L | 14        | M12 X 1.0 | 4  |



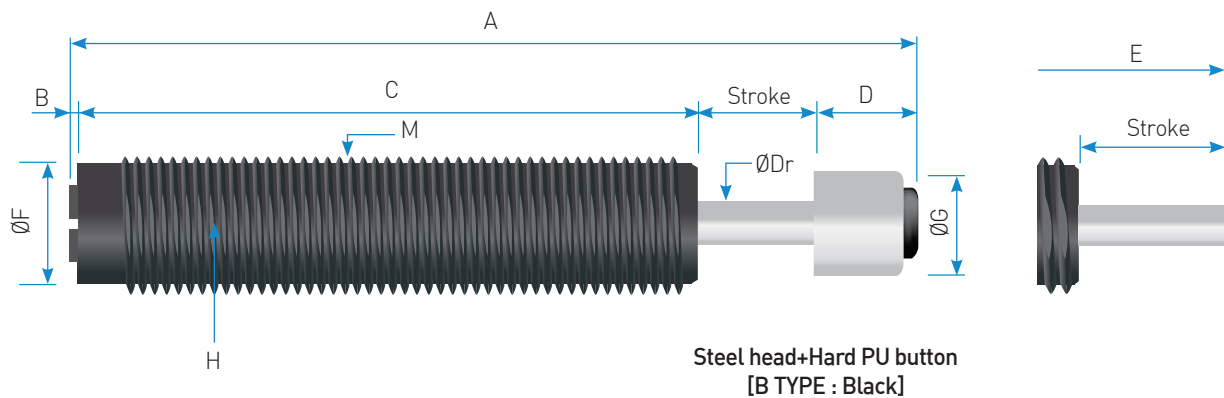
# IAA14 - 12(B) / 16 - 12(B)

## Engineering Data

| Model         | Stroke (mm) | Max.Energy / Cycle(Nm) $E_T$ | Max.Energy / Hour(Nm/h) $E_{TC}$ | Effective Weight(kg) $m_e$ | Recoil Forec(N) |       | Weight(g) |
|---------------|-------------|------------------------------|----------------------------------|----------------------------|-----------------|-------|-----------|
|               |             |                              |                                  |                            | Ext.            | Comp. |           |
| IAA14 - 12(B) | 12          | 18                           | 38,000                           | 1.8 - 400                  | 4.9             | 11.4  | 60        |
| IAA16 - 12(B) | 12          | 24                           | 47,200                           | 2-530                      | 5               | 11.5  | 84        |

## Dimensions

| Model         | Stroke | A  | B   | C    | D    | E    | F     | G     | H       | M       | Dr |
|---------------|--------|----|-----|------|------|------|-------|-------|---------|---------|----|
| IAA14 - 12(B) | 12     | 96 | 1.2 | 72.8 | 10   | 86   | Ø12   | Ø10.5 | SW:12.5 | M14X1.5 | Ø4 |
| IAA16 - 12(B) | 12     | 97 | 1.2 | 71.3 | 12.5 | 84.5 | Ø10.5 | Ø13.6 | SW:14.5 | M16X1.5 | Ø5 |



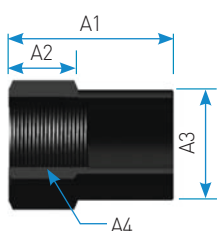
## Accessories

- Two Lock Nuts included for each absorber
- Other accessories to be ordered separately

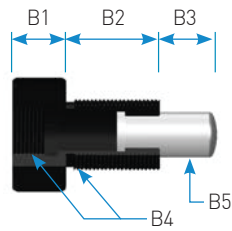
(unit : mm)

| Stop Collar | Side Load Adapter | Clamp Mount | Lock Nut | A1 | A2 | A3  | A4        | B1 | B2   | B3 | B4        | B5 | C1 | C2 | C3 | C4 | C5        | C6       | D1 | D2        | D3 |
|-------------|-------------------|-------------|----------|----|----|-----|-----------|----|------|----|-----------|----|----|----|----|----|-----------|----------|----|-----------|----|
| SC14X1.5    | SLA14-12          | CM14X1.5    | LN14X1.5 |    |    | Ø18 | M14 X 1.5 | 14 | 19.5 |    | M14 X 1.5 | Ø6 | 20 | 20 | 32 | 12 | M14 X 1.5 | M5 X 25L |    | M14 X 1.5 | 5  |
| SC16X1.5    | SLA16-12          | CM16X1.5    | LN16X1.5 | 27 | 12 | Ø19 | M16 X 1.5 | 19 | 22.8 | 12 | M16 X 1.5 | Ø8 | 24 | 22 | 36 | 16 | M16 X 1.5 | M6 X 25L | 19 | M16 X 1.5 | 6  |

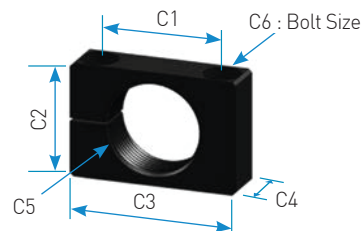
Stop Collar



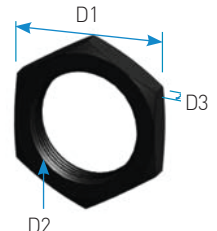
Side Load Adapter



Clamp Mount



Lock Nut





# IAA20 - 15(B) / 25 - 25(B)

Best engineered  
for energy absorption  
technology

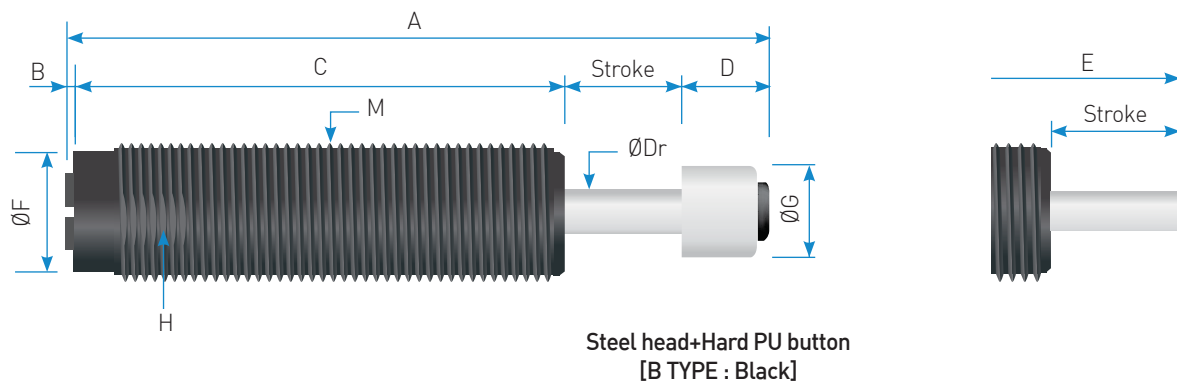


## Engineering Data

| Model         | Stroke (mm) | Max.Energy / Cycle(Nm) E <sub>T</sub> | Max.Energy / Hour(Nm/h) E <sub>T</sub> C | Effective Weight(kg) m <sub>e</sub> | Recoil Forec(N) |       | Weight(g) |
|---------------|-------------|---------------------------------------|------------------------------------------|-------------------------------------|-----------------|-------|-----------|
|               |             |                                       |                                          |                                     | Ext.            | Comp. |           |
| IAA20 - 15(B) | 15          | 56                                    | 58,000                                   | 4.5-1,240                           | 8               | 19,6  | 139       |
| IAA25 - 25(B) | 25          | 150                                   | 96,000                                   | 11.5-3,330                          | 10.2            | 29.5  | 285       |

## Dimensions

| Model         | Stroke | A   | B   | C   | D    | E     | F     | G     | H       | M       | Dr |
|---------------|--------|-----|-----|-----|------|-------|-------|-------|---------|---------|----|
| IAA20 - 15(B) | 15     | 109 | 1.5 | 79  | 13.5 | 95.5  | Ø17.6 | Ø13.5 | SW:18.5 | M20X1.5 | Ø6 |
| IAA25 - 25(B) | 25     | 144 | 1.5 | 101 | 16.5 | 127.5 | Ø22.6 | Ø16   | SW:23   | M25X1.5 | Ø8 |

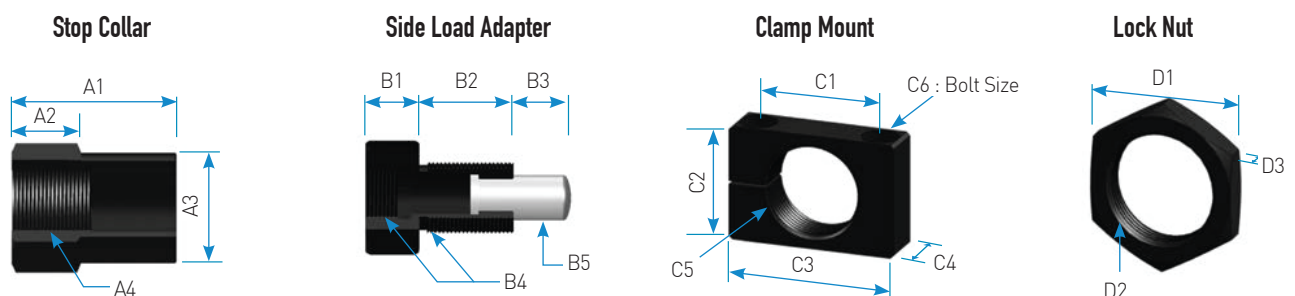


## Accessories

- Two Lock Nuts included for each absorber
- Other accessories to be ordered separately

(unit : mm)

| Stop Collar | Side Load Adapter | Clamp Mount | Lock Nut | A1 | A2 | A3    | A4        | B1 | B2 | B3 | B4        | B5  | C1 | C2 | C3 | C4 | C5        | C6       | D1 | D2        | D3 |
|-------------|-------------------|-------------|----------|----|----|-------|-----------|----|----|----|-----------|-----|----|----|----|----|-----------|----------|----|-----------|----|
| SC20X1.5    | SLA20-15          | CM20X1.5    | LN20X1.5 | 36 | 15 | Ø24   | M20 X 1.5 | 20 | 34 | 15 | M20 X 1.5 | Ø12 | 28 | 25 | 40 | 20 | M20 X 1.5 | M6 X 30L | 24 | M20 X 1.5 | 6  |
| SC25X1.5    | SLA25-25          | CM25X1.5    | LN25X1.5 | 42 | 18 | Ø31.5 | M25 X 1.5 |    | 42 | 25 | M25 X 1.5 | Ø16 | 32 | 32 | 46 | 25 | M25 X 1.5 | M6 X 35L | 32 | M25 X 1.5 | 8  |



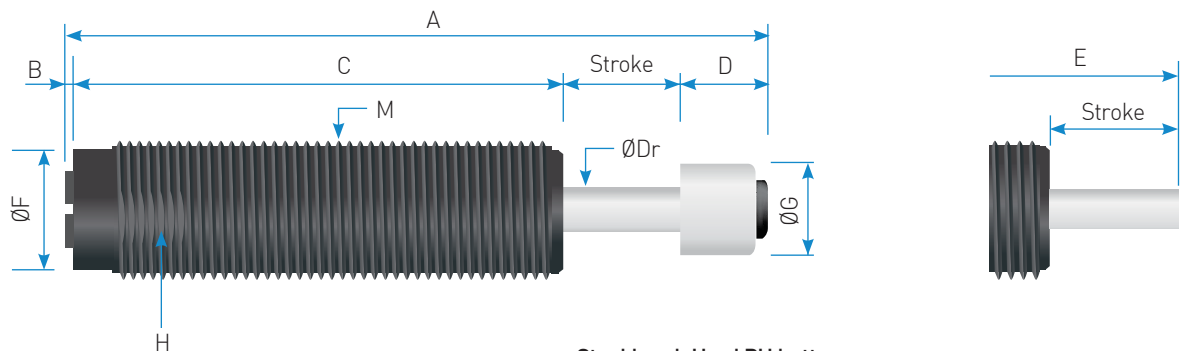
# IAA27 - 25(B) / 27 - 25F(B)

## Engineering Data

| Model         | Stroke (mm) | Max.Energy / Cycle(Nm) E <sub>T</sub> | Max.Energy / Hour(Nm/h) E <sub>T</sub> C | Effective Weight(kg) m <sub>e</sub> | Recoil Forec(N) |       | Weight(g) |
|---------------|-------------|---------------------------------------|------------------------------------------|-------------------------------------|-----------------|-------|-----------|
|               |             |                                       |                                          |                                     | Ext.            | Comp. |           |
| IAA27 - 25(B) | 25          | 150                                   | 96,000                                   | 11.5-3,330                          | 10.2            | 29.5  | 303       |
| - 25F(B)      |             |                                       |                                          |                                     |                 |       | 325       |

## Dimensions

| Model         | Stroke | A   | B   | C   | D    | E     | F     | G   | H         | M       | Dr |
|---------------|--------|-----|-----|-----|------|-------|-------|-----|-----------|---------|----|
| IAA27 - 25(B) | 25     | 144 | 1.5 | 101 | 16.5 | 127.5 | Ø22.6 | Ø16 | SW:25(23) | M27X3.0 | Ø8 |
| - 25F(B)      |        |     |     |     |      |       |       |     | SW:25     | M27X1.5 |    |



Steel head+Hard PU button  
[B TYPE : Black]

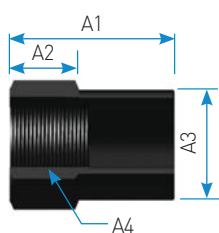
## Accessories

- Two Lock Nuts included for each absorber
- Other accessories to be ordered separately

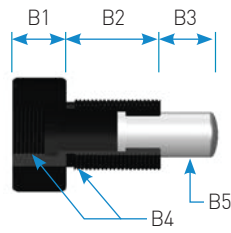
(unit : mm)

| Stop Collar | Side Load Adapter | Clamp Mount | Lock Nut | A1 | A2 | A3    | A4        | B1 | B2 | B3 | B4 | B5        | C1 | C2 | C3 | C4 | C5        | C6      | D1        | D2 | D3 |
|-------------|-------------------|-------------|----------|----|----|-------|-----------|----|----|----|----|-----------|----|----|----|----|-----------|---------|-----------|----|----|
| SC27X3.0    | SLA27-25          | CM27X3.0    | LN27X3.0 |    |    |       | M27 X 3.0 |    |    |    |    | M27 X 3.0 |    |    |    |    | M27 X 3.0 |         | M27 X 3.0 |    |    |
|             |                   |             |          | 42 | 18 | Ø31.5 |           | 20 | 42 | 25 |    | Ø16       | 35 | 32 | 48 | 25 |           | M6 X35L | 32        |    | 8  |
| SC27X1.5    | SLA27-25F         | CM27X1.5    | LN27X1.5 |    |    |       | M27 X 1.5 |    |    |    |    | M27 X 1.5 |    |    |    |    | M27 X 1.5 |         | M27 X 1.5 |    |    |

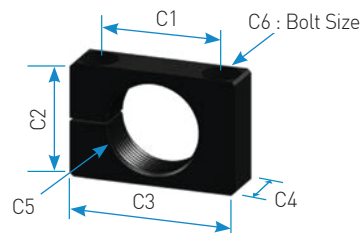
Stop Collar



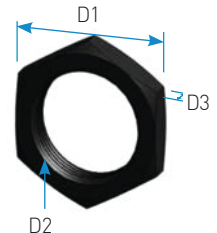
Side Load Adapter



Clamp Mount



Lock Nut



# IAA36 - 25(B) / 36 -50(B)

Best engineered  
for energy absorption  
technology

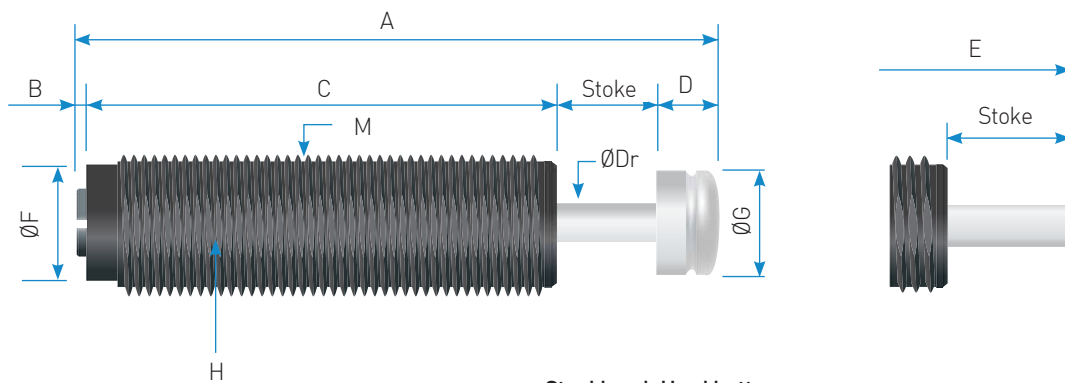


## Engineering Data

| Model         | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) E <sub>T</sub> | Max.Energy /<br>Hour(Nm/h)<br>E <sub>T</sub> C | Effective<br>Weight(kg) m <sub>e</sub> | Recoil Forec(N) |       | Weight(g) |
|---------------|----------------|------------------------------------------|------------------------------------------------|----------------------------------------|-----------------|-------|-----------|
|               |                |                                          |                                                |                                        | Ext.            | Comp. |           |
| IAA36 - 25(B) | 25             | 300                                      | 108,000                                        | 24 - 6,667                             | 25              | 56.2  | 700       |
| - 50(B)       | 50             | 615                                      | 142,000                                        | 49 - 13,667                            | 22.5            | 60    | 811       |

## Dimensions

| Model         | Stroke | A     | B | C     | D    | E     | F   | G   | H     | M         | Dr  |
|---------------|--------|-------|---|-------|------|-------|-----|-----|-------|-----------|-----|
| IAA36 - 25(B) | 25     | 158.5 | 2 | 116   | 15.5 | 143   | Ø31 | Ø28 | SW:34 | M36 X 1.5 | Ø10 |
| - 50(B)       | 50     | 218   | 2 | 150.5 | 15.5 | 202.5 | Ø31 | Ø28 | SW:34 | M36 X 1.5 | Ø10 |



Steel head+Hard button

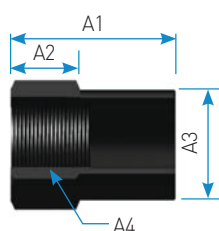
## Accessories

- Two Lock Nuts included for each absorber
- Other accessories to be ordered separately

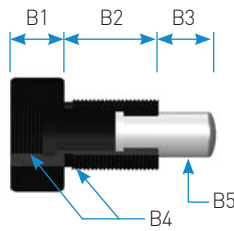
(unit : mm)

| Stop Collar | Side Load Adapter | Clamp Mount | Lock Nut | A1 | A2 | A3  | A4        | B1 | B2 | B3 | B4        | B5 | C1 | C2 | C3 | C4 | C5        | C6       | D1 | D2        | D3 |
|-------------|-------------------|-------------|----------|----|----|-----|-----------|----|----|----|-----------|----|----|----|----|----|-----------|----------|----|-----------|----|
| SC36X1.5    | SLA36-25          | CM36X1.5    | LN36X1.5 |    |    |     |           | 22 | 47 | 25 |           | 20 |    |    |    |    |           |          |    |           |    |
|             |                   |             |          | 62 | 26 | Ø45 | M36 X 1.5 |    |    |    | M36 X 1.5 |    | 45 | 42 | 61 | 25 | M36 X 1.5 | M6 X 45L | 46 | M36 X 1.5 | 10 |
| SC36X1.5    | SLA36-50          | CM36X1.5    | LN36X1.5 |    |    |     |           | 22 | 78 | 50 |           | 20 |    |    |    |    |           |          |    |           |    |

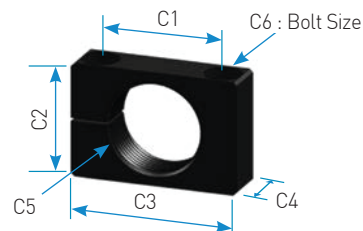
Stop Collar



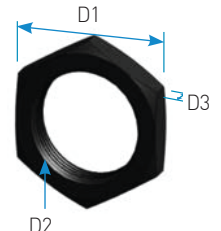
Side Load Adapter



Clamp Mount



Lock Nut





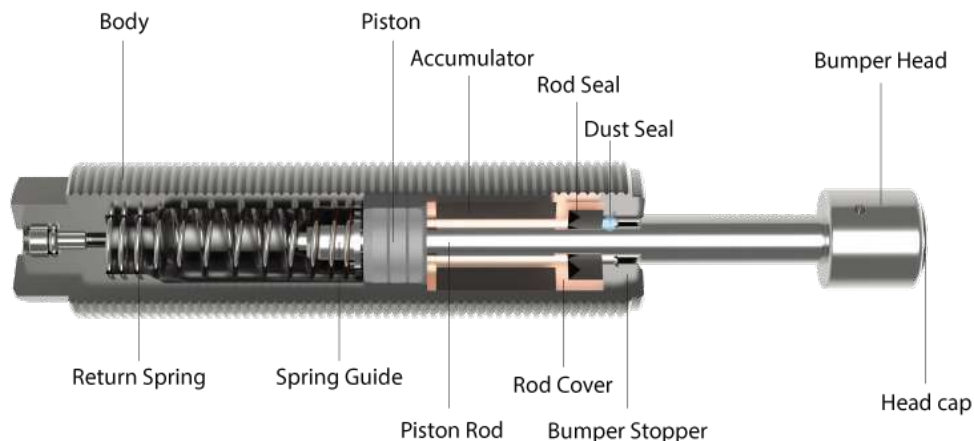
# IMS Series Self Compensating type

## IZMAC Mini absorber Self-compensating type

### DESCRIPTION

IMS is self compensating type and it controls the damping power itself against damping speed. Energy capacity was Improved over 200% and effective weight ranges become wider than before.

#### IMS series



### FEATURES

- 1 All threaded body outside helps to install shock absorber and it can be fixed at the exact position. And it helps also to dissipate more shock energy by increased surface dimension.
- 2 Body surface treated as alloyed black colored steel or Zn-Ni and it develops anti corrosion features highly.
- 3 Stop collar is not necessary when shock absorber has steel head, and poly pad & urethane cap will help to decrease collision noises efficiently.
- 4 Rod cover is made of long lasting materials with function of seal protection and that enable to guarantee long life.
- 5 Speed ranges : 0.3 ~ 5m/s
- 6 Temperature ranges : -10 ~ 80°C • Special : -40 ~ 120°C(Oil and Seal)

### APPLICATIONS

Pick N place robotic line, machining tools, automobile assembly line, tire assembly line, crane, safety devices and all other industrial ranges of multi purposes.

### IMS SERIES ORDERING INFORMATION

IMS - 25 - 25 - SC

Stroke  
Body Thread Size  
I : Izmec  
M : Mini & Middle absorber  
S : Self-compensating type  
SC : Stop Collar  
LN : Lock Nut  
CM : Clamp Mount

# IMS04 - 04(B) / 05 - 05(B) / 06 - 05(B)

Best engineered  
for energy absorption  
technology



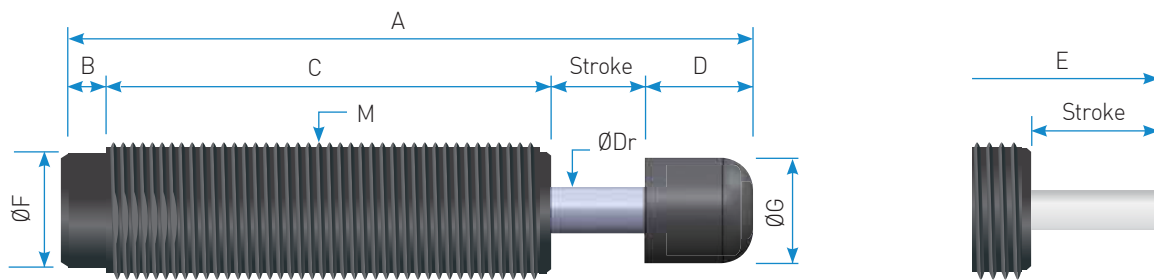
## Engineering Data

| Model         | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) E <sub>T</sub> | Max.Energy /<br>Hour(Nm/h)<br>E <sub>T</sub> C | Effective Weight(kg) m <sub>e</sub> |           |         | Recoil Force(N) |       | Weight(g) |
|---------------|----------------|------------------------------------------|------------------------------------------------|-------------------------------------|-----------|---------|-----------------|-------|-----------|
|               |                |                                          |                                                | 1                                   | 2         | 3       | Ext.            | Comp. |           |
| IMS04 - 04(B) | 4              | 0.4                                      | 1,500                                          | 0.1 - 1.0                           | 0.9 - 3.2 |         | 2               | 7     | 2.5       |
| IMS05 - 05(B) | 5              | 0.6                                      | 1,800                                          | 0.1 - 1.2                           | 1.0 - 5.0 |         | 2               | 7     | 2.5       |
| IMS06 - 05(B) | 5              | 1                                        | 2,500                                          | 0.05 - 1                            | 0.8 - 2.8 | 1.5 - 4 | 2               | 5     | 4.0       |

## Dimensions

(unit : mm)

| Model         | Stroke | A    | B   | C    | D | E    | F   | G  | M        | Dr   |
|---------------|--------|------|-----|------|---|------|-----|----|----------|------|
| IMS04 - 04(B) | 4      | 34.5 | 2.4 | 24   | 4 | 30.5 | 3.4 | Ø3 | M4 X 0.5 | Ø1.2 |
| IMS05 - 05(B) | 5      | 35   | 3.5 | 22.5 | 4 | 31   | 4.4 | Ø3 | M5 X 0.5 | Ø1.5 |
| IMS06 - 05(B) | 5      | 38.5 | 3.5 | 25   | 5 | 33.5 | 5.3 | Ø5 | M6 X 0.5 | Ø1.5 |



MC Nylon head

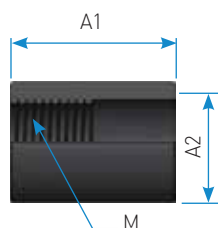
## Accessories

- Two Lock Nuts included for each absorber
- Other accessories to be ordered separately

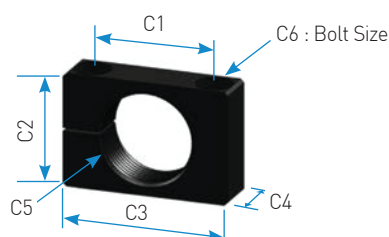
(unit : mm)

| Stop Collar | Clamp Mount | Lock Nut | A1 | A2 | A3     | C1 | C2 | C3 | C4 | C5     | C6 | D1 | D2     | D3 |
|-------------|-------------|----------|----|----|--------|----|----|----|----|--------|----|----|--------|----|
|             |             | LN04X0.5 |    |    | M4X0.5 |    |    |    |    |        |    | 8  | M4X0.5 | 5  |
| SC05X0.5    | CM05X0.5    | LN05X0.5 | 10 | 7  | M5X0.5 | 12 | 10 | 20 | 8  | M5X0.5 | M3 | 8  | M5X0.5 | 5  |
| SC06X0.5    | CM06X0.5    | LN06X0.5 | 12 | 8  | M6X0.5 |    |    |    |    | M6X0.5 |    |    | M6X0.5 |    |

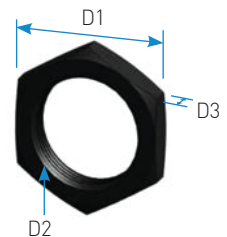
Stop Collar



Clamp Mount



Lock Nut



# IMS08 - 06(B) / 10 - 08(B)

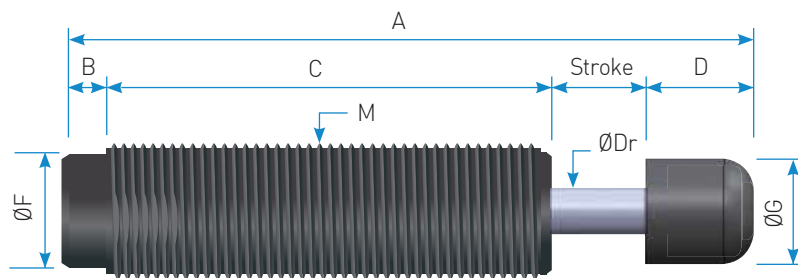
## ■ Engineering Data

| Model         | Stroke (mm) | Max.Energy / Cycle(Nm) $E_T$ | Max.Energy / Hour(Nm/h) $E_T C$ | Effective Weight(Kg) $m_e$ |            |          | Recoil Force(N) |       | Weight(g) |
|---------------|-------------|------------------------------|---------------------------------|----------------------------|------------|----------|-----------------|-------|-----------|
|               |             |                              |                                 | 1                          | 2          | 3        | Ext.            | Comp. |           |
| IMS08 - 06(B) | 6           | 5                            | 8,000                           | 0.8 - 2.8                  | 2.5 - 12.6 | 10 - 111 | 2.2             | 5.8   | 10        |
| IMS10 - 08(B) | 8           | 11                           | 14,500                          | 1.8 - 6.1                  | 5.5 - 27   | 22 - 244 | 2.5             | 6.9   | 15.5      |

## ■ Dimensions

(unit : mm)

| Model         | Stroke | A    | B   | C  | D | E    | F    | G    | M         | Dr |
|---------------|--------|------|-----|----|---|------|------|------|-----------|----|
| IMS08 - 06(B) | 6      | 54   | 7.5 | 35 | 7 | 47   | Ø6.8 | Ø6.8 | M8 X 1.0  | Ø2 |
| IMS10 - 08(B) | 8      | 64.5 | 7.5 | 42 | 7 | 57.5 | Ø7   | Ø8   | M10 X 1.0 | Ø3 |



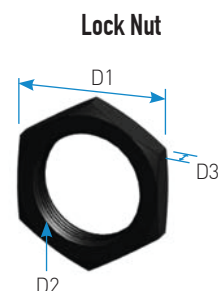
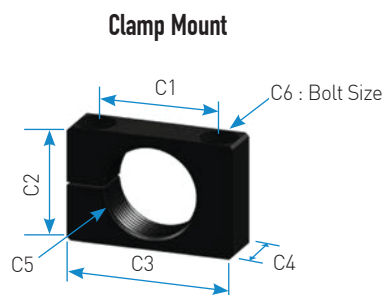
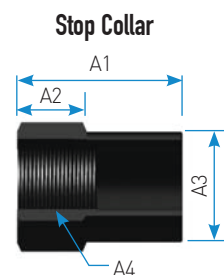
MC Nylon head

## ■ Accessories

- Two Lock Nuts included for each absorber
- Other accessories to be ordered separately

(unit : mm)

| Stop Collar | Clamp Mount | Lock Nut | A1 | A2 | A3 | A4    | C1        | C2 | C3 | C4 | C5        | C6       | D1 | D2        | D3 |
|-------------|-------------|----------|----|----|----|-------|-----------|----|----|----|-----------|----------|----|-----------|----|
| SC08X1.0    | CM08X1.0    | LN08X1.0 |    |    |    | Ø11.5 | M8X1.0    |    |    |    | M8X1.0    |          | 12 | M8X1.0    | 3  |
|             |             |          | 20 | 8  |    |       | 16        | 14 | 25 | 10 |           | M4 X 16L |    |           |    |
| SC10X1.0    | CM10X1.0    | LN10X1.0 |    |    |    | Ø12.8 | M10 X 1.0 |    |    |    | M10 X 1.0 |          | 13 | M10 X 1.0 | 4  |





# IMS12-10(B) / 14-15(B) / 14-15F(B)

Best engineered  
for energy absorption  
technology



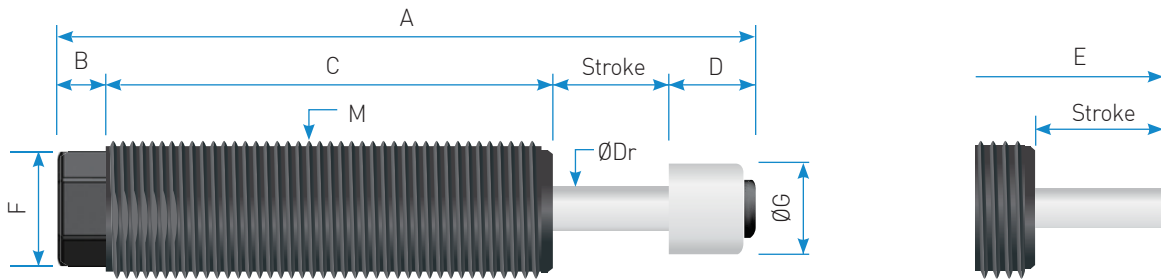
## Engineering Data

| Model       | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) $E_T$ | Max.Energy /<br>Hour(Nm/h)<br>$E_T \cdot C$ | Effective Weight(Kg) $m_e$ |        |         | Recoil Force(N) |       | Weight(g) |
|-------------|----------------|---------------------------------|---------------------------------------------|----------------------------|--------|---------|-----------------|-------|-----------|
|             |                |                                 |                                             | 1                          | 2      | 3       | Ext.            | Comp. |           |
| IMS12-10(B) | 10             | 16                              | 30,000                                      | 13~8                       | 6.6~22 | 16~200  | 3.8             | 9.8   | 39        |
| IMS14-15(B) | 15             | 30                              | 51,000                                      | 3.8~15                     | 12~42  | 31~1500 | 3.8             | 13    | 61        |
| - 15F(B)    | 15             | 30                              | 51,000                                      | 3.8~15                     | 12~42  | 31~1500 | 3.8             | 13    | 64        |

## Dimensions

(unit : mm)

| Model       | Stroke | A     | B   | C    | D  | E    | F       | G     | M         | Dr |
|-------------|--------|-------|-----|------|----|------|---------|-------|-----------|----|
| IMS12-10(B) | 10     | 79    | 6.5 | 52.5 | 10 | 69   | SW:10.8 | Ø10.5 | M12 X 1.0 | Ø4 |
| IMS14-15(B) | 15     | 100.5 | 6.5 | 69   | 10 | 90.5 | SW:11.8 | Ø10.5 | M14 X 1.5 | Ø4 |
| - 15F(B)    | 15     | 100.5 | 6.5 | 69   | 10 | 90.5 | SW:11.8 | Ø10.5 | M14 X 1.0 | Ø4 |



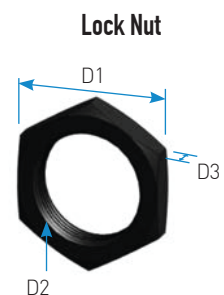
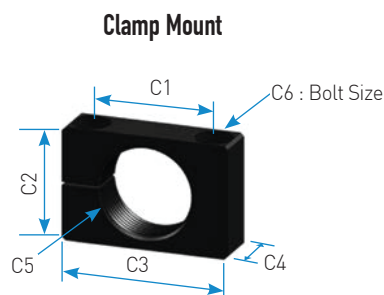
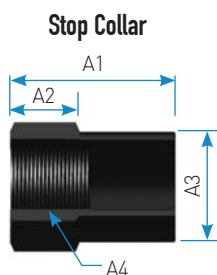
Steel head+Hard PU button  
[B TYPE : Black]

## Accessories

- Two Lock Nuts included for each absorber
- Other accessories to be ordered separately

(unit : mm)

| Stop Collar | Clamp Mount | Lock Nut | A1 | A2 | A3    | A4        | C1 | C2 | C3 | C4 | C5        | C6       | D1 | D2        | D3 |
|-------------|-------------|----------|----|----|-------|-----------|----|----|----|----|-----------|----------|----|-----------|----|
| SC12X1.0    | CM12X1.0    | LN12X1.0 | 20 | 8  | Ø13.8 | M12 X 1.0 | 20 | 16 | 32 | 12 | M12 X 1.0 | M5 X 18L | 14 | M12 X 1.0 | 4  |
| SC14X1.5    | CM14X1.5    | LN14X1.5 | 27 | 12 | Ø18   | M14 X 1.5 | 20 | 20 | 32 | 12 | M14 X 1.5 | M5 X 25L | 19 | M14 X 1.5 | 5  |
| SC14X1.0    | CM14X1.0    | LN14X1.0 |    |    |       | M14 X 1.0 |    |    |    |    | M14 X 1.0 |          |    | M14 X 1.0 |    |



# IMS20 - 20(B) / 25 - 25(B) / 25 - 25F(B)

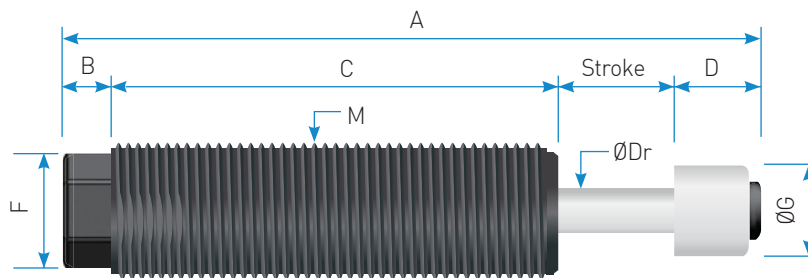
## Engineering Data

| Model         | Stroke (mm) | Max.Energy / Cycle(Nm) $E_T$ | Max.Energy / Hour(Nm/h) $E_T \cdot C$ | Effective Weight(Kg) $m_e$ |           |            | Recoil Force(N) |       | Weight(g) |
|---------------|-------------|------------------------------|---------------------------------------|----------------------------|-----------|------------|-----------------|-------|-----------|
|               |             |                              |                                       | 1                          | 2         | 3          | Ext.            | Comp. |           |
| IMS20 - 20(B) | 20          | 96                           | 55,000                                | 9~59                       | 48~192    | 133~4,800  | 8.1             | 23    | 138       |
| IMS25 - 25(B) | 25          | 205                          | 69,000                                | 33~209                     | 103~1,139 | 641~10,250 | 11              | 29    | 262       |
| - 25F(B)      | 25          | 205                          | 69,000                                | 33~209                     | 103~1,139 | 641~10,250 | 11              | 29    | 271       |

## Dimensions

(unit : mm)

| Model         | Stroke | A     | B | C  | D    | E   | F       | G     | M         | Dr |
|---------------|--------|-------|---|----|------|-----|---------|-------|-----------|----|
| IMS20 - 20(B) | 20     | 118.5 | 8 | 77 | 13.5 | 105 | SW : 15 | Ø13.5 | M20 X 1.5 | Ø6 |
| IMS25 - 25(B) | 25     | 141.5 | 9 | 91 | 16.5 | 125 | SW : 19 | Ø16   | M25 X 2.0 | Ø8 |
| - 25F(B)      | 25     | 141.5 | 9 | 91 | 16.5 | 125 | SW : 19 | Ø16   | M25 X 1.5 | Ø8 |



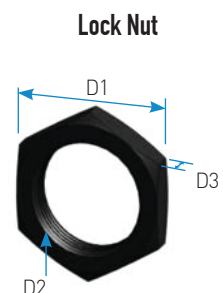
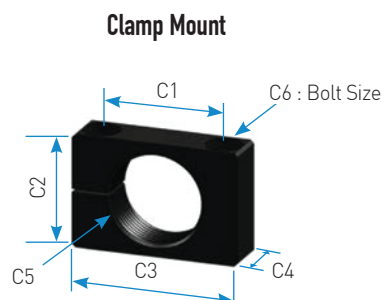
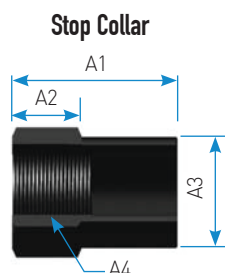
Steel head+Hard PU button  
[B TYPE : Black]

## Accessories

- Two Lock Nuts included for each absorber
- Other accessories to be ordered separately

(unit : mm)

| Stop Collar | Clamp Mount | Lock Nut | A1 | A2 | A3    | A4        | C1 | C2 | C3 | C4 | C5        | C6       | D1 | D2        | D3 |
|-------------|-------------|----------|----|----|-------|-----------|----|----|----|----|-----------|----------|----|-----------|----|
| SC20X1.5    | CM20X1.5    | LN20X1.5 | 36 | 15 | Ø24   | M20 X 1.5 | 28 | 25 | 40 | 20 | M20 X 1.5 | M6 X 30L | 24 | M20 X 1.5 | 6  |
| SC25X2.0    | CM25X2.0    | LN25X2.0 | 42 | 18 | Ø31.5 | M25 X 2.0 | 32 | 32 | 46 | 25 | M25 X 2.0 | M6 X 35L | 32 | M25 X 2.0 | 8  |
| SC25X1.5    | CM25X1.5    | LN25X1.5 |    |    |       | M25 X 1.5 |    |    |    |    | M25 X 1.5 |          |    | M25 X 1.5 |    |



# IMS36 - 25(B) / 36 - 50(B)

Best engineered  
for energy absorption  
technology



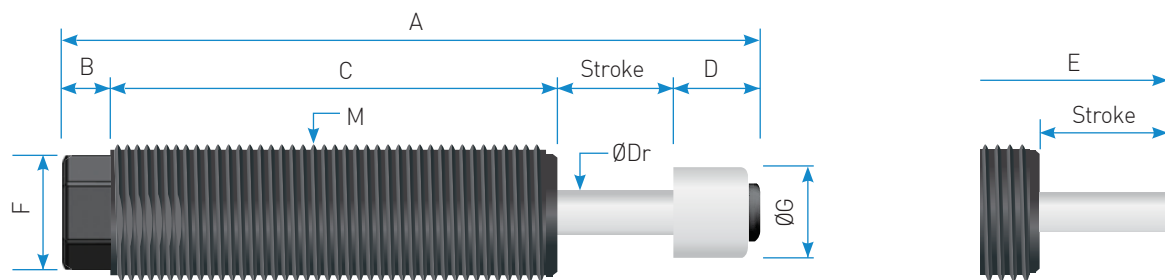
## Engineering Data

| Model         | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) $E_T$ | Max.Energy /<br>Hour(Nm/h)<br>$E_T \cdot C$ | Effective Weight(Kg) $m_e$ |         |            | Recoil Force(N) |       | Weight(g) |
|---------------|----------------|---------------------------------|---------------------------------------------|----------------------------|---------|------------|-----------------|-------|-----------|
|               |                |                                 |                                             | 1                          | 2       | 3          | Ext.            | Comp. |           |
| IMS36 - 25(B) | 25             | 490                             | 115,000                                     | 61-185                     | 170-500 | 435-10,888 | 25              | 64.3  | 758       |
| - 50(B)       | 50             | 220                             | 162,000                                     | 35-121                     | 110-543 | 440-4,888  | 24.4            | 44.6  | 994       |

## Dimensions

(unit : mm)

| Model         | Stroke | A   | B  | C   | D  | E   | F       | G   | M         | Dr  |
|---------------|--------|-----|----|-----|----|-----|---------|-----|-----------|-----|
| IMS36 - 25(B) | 25     | 164 | 14 | 101 | 24 | 140 | SW : 30 | Ø28 | M36 X 1.5 | Ø10 |
| - 50(B)       | 50     | 221 | 11 | 136 | 24 | 197 | SW : 30 | Ø28 | M36 X 1.5 | Ø10 |



Steel head+Hard PU button  
[B TYPE : Black]

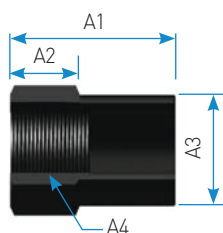
## Accessories

- Two Lock Nuts included for each absorber
- Other accessories to be ordered separately

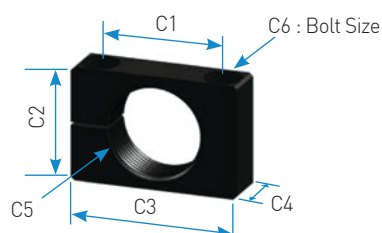
(unit : mm)

| Stop Collar | Clamp Mount | Lock Nut | A1 | A2 | A3  | A4        | C1 | C2 | C3 | C4 | C5        | C6 | D1 | D2        | D3 |
|-------------|-------------|----------|----|----|-----|-----------|----|----|----|----|-----------|----|----|-----------|----|
| SC36X1.5    | CM36X1.5    | LN36X1.5 | 62 | 26 | Ø45 | M36 X 1.5 | 45 | 42 | 61 | 25 | M36 X 1.5 | M6 | 46 | M36 X 1.5 | 10 |
| SC36X1.5    | CM36X1.5    | LN36X1.5 |    |    |     |           |    |    |    |    |           |    |    |           |    |

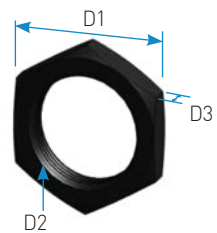
Stop Collar



Clamp Mount



Lock Nut

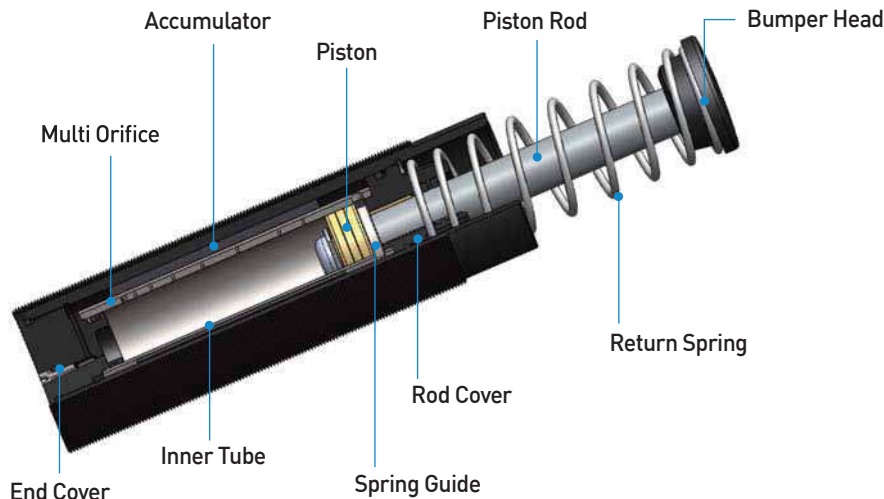




### IZMAC Mini absorber Self-compensating type

## DESCRIPTION

IMS is middle class shock absorber. Operation : when the collision comes piston compress oil chamber in inner tube according to the piston rod moving. Oil flows to accumulator position through outskirts line of tube. In this process, deceleration power comes by flow resistance. And kinetic energy is dissipated in the air by changed as heat energy. Recoil spring returns piston rod into its original position when the load is disappeared.

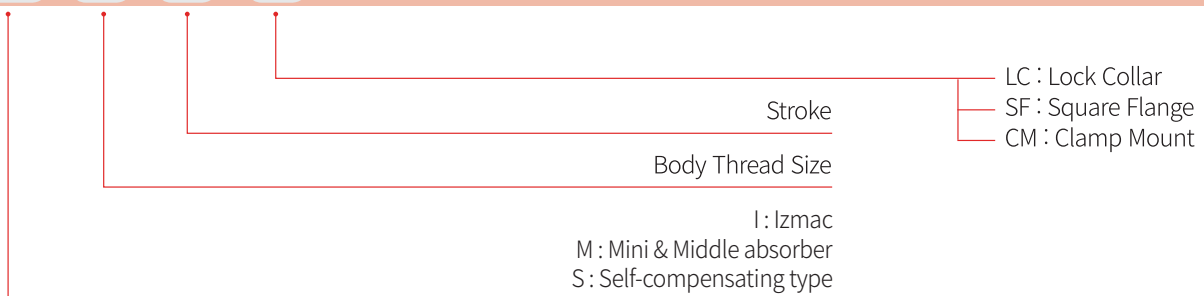


## FEATURES

- 1 Compact design and possible to absorb wide range of high shock energy.
- 2 The effective weight range is significantly expanded, resulting in a wide range of energy absorption capabilities.
- 3 Easy fastening and smooth heat dissipation due to the screw-type body.
- 4 Body surface strongly treated to prevent corrosion by nickel plating or alloy plating. (Black)
- 5 Body and mount : Black or Ni plating
- 6 Piston rod : Hard chrome (25μ or more)
- 7 Speed range : 0.15 ~ 5 m/s
- 8 Temperature range : -10 ~ 80°C • Special : -40 ~ 120°C (Oil and Seal)

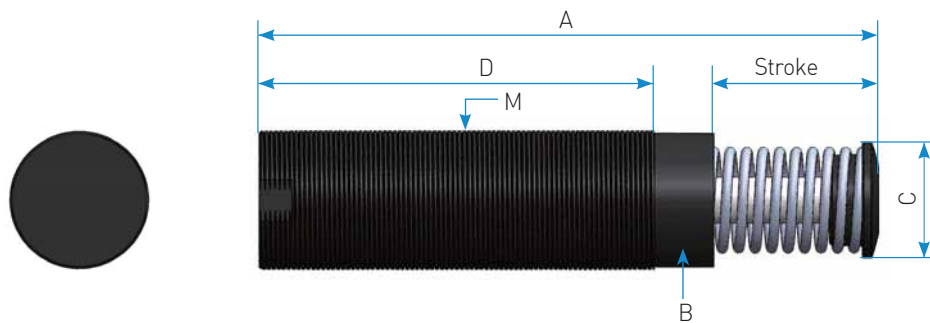
## IMS SERIES ORDERING INFORMATION

IMS - 45 - 25 - LC



## Engineering Data

| Model      | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) $E_T$ | Max.Energy /<br>Hour(Nm/h)<br>$E_T C$ | Effective<br>Weight(kg) $m_e$ | Recoil Forec(N) |       | Weight(kg) |
|------------|----------------|---------------------------------|---------------------------------------|-------------------------------|-----------------|-------|------------|
|            |                |                                 |                                       |                               | Ext.            | Comp. |            |
| IMS33-25-1 | 25             | 300                             | 120,000                               | 7-32                          | 45              | 90    | 0.53       |
| IMS33-25-2 | 25             | 300                             | 120,000                               | 28-130                        | 45              | 90    | 0.53       |
| IMS33-25-3 | 25             | 300                             | 120,000                               | 80-590                        | 45              | 90    | 0.53       |
| IMS33-25-4 | 25             | 300                             | 120,000                               | 440-2,050                     | 45              | 90    | 0.53       |
| IMS33-25-5 | 25             | 300                             | 120,000                               | 2,000-12,500                  | 45              | 90    | 0.53       |
| IMS33-50-1 | 50             | 500                             | 150,000                               | 13-60                         | 45              | 135   | 0.64       |
| IMS33-50-2 | 50             | 500                             | 150,000                               | 56-240                        | 45              | 135   | 0.64       |
| IMS33-50-3 | 50             | 500                             | 150,000                               | 160-1,200                     | 45              | 135   | 0.64       |
| IMS33-50-4 | 50             | 500                             | 150,000                               | 1,000-4,200                   | 45              | 135   | 0.64       |
| IMS33-50-5 | 50             | 500                             | 150,000                               | 4,000-25,000                  | 45              | 135   | 0.64       |



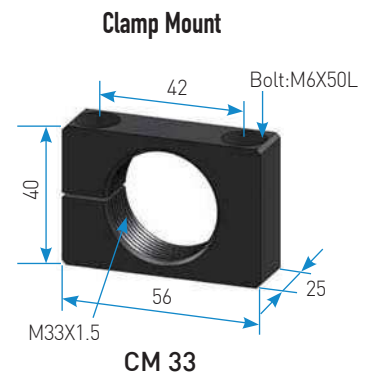
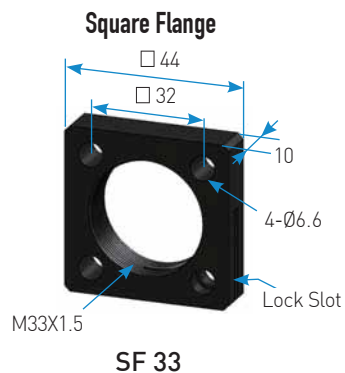
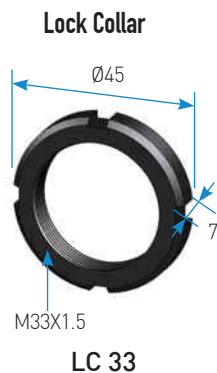
## Dimensions

| Model    | Stroke | A   | B   | C   | D   | M        |
|----------|--------|-----|-----|-----|-----|----------|
| IMS33-25 | 25     | 138 | Ø31 | Ø25 | 95  | M33*1.5P |
| IMS33-50 | 50     | 189 | Ø31 | Ø25 | 121 | M33*1.5P |

## Accessories

- Other accessories to be ordered separately

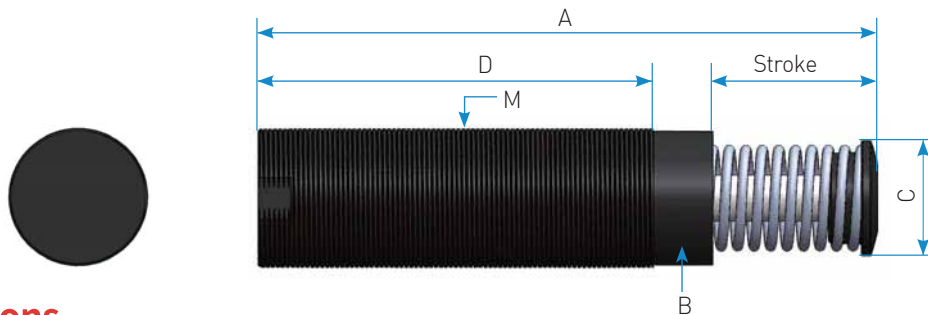
(unit : mm)



# IMS45 Series

## ■ Engineering Data

| Model          | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) $E_T$ | Max.Energy /<br>Hour(Nm/h)<br>$E_{TC}$ | Effective Weight(kg) $m_e$ |        | Recoil Forec(N) |       | Weight(kg) |
|----------------|----------------|---------------------------------|----------------------------------------|----------------------------|--------|-----------------|-------|------------|
|                |                |                                 |                                        |                            |        | Ext.            | Comp. |            |
| IMS45 - 25 - 1 | 25             | 850                             | 255,000                                | 24                         | 110    | 75              | 110   | 1.2        |
| - 25 - 2       | 25             | 850                             | 255,000                                | 100                        | 385    | 75              | 110   | 1.2        |
| - 25 - 3       | 25             | 850                             | 255,000                                | 350                        | 1,700  | 75              | 110   | 1.2        |
| - 25 - 4       | 25             | 850                             | 255,000                                | 1,400                      | 6,800  | 75              | 110   | 1.2        |
| - 25 - 5       | 25             | 850                             | 255,000                                | 6,000                      | 38,500 | 75              | 110   | 1.2        |
| IMS45 - 50 - 1 | 50             | 1,300                           | 325,000                                | 35                         | 170    | 75              | 150   | 1.4        |
| - 50 - 2       | 50             | 1,300                           | 325,000                                | 160                        | 590    | 75              | 150   | 1.4        |
| - 50 - 3       | 50             | 1,300                           | 325,000                                | 540                        | 2,600  | 75              | 150   | 1.4        |
| - 50 - 4       | 50             | 1,300                           | 325,000                                | 2,150                      | 10,400 | 75              | 150   | 1.4        |
| - 50 - 5       | 50             | 1,300                           | 325,000                                | 3,200                      | 58,900 | 75              | 150   | 1.4        |
| IMS45 - 75 - 1 | 75             | 2,100                           | 420,000                                | 50                         | 270    | 60              | 185   | 1.6        |
| - 75 - 2       | 75             | 2,100                           | 420,000                                | 250                        | 950    | 60              | 185   | 1.6        |
| - 75 - 3       | 75             | 2,100                           | 420,000                                | 870                        | 4,200  | 60              | 185   | 1.6        |
| - 75 - 4       | 75             | 2,100                           | 420,000                                | 3,500                      | 16,800 | 60              | 185   | 1.6        |
| - 75 - 5       | 75             | 2,100                           | 420,000                                | 14,900                     | 95,200 | 60              | 185   | 1.6        |



## ■ Dimensions

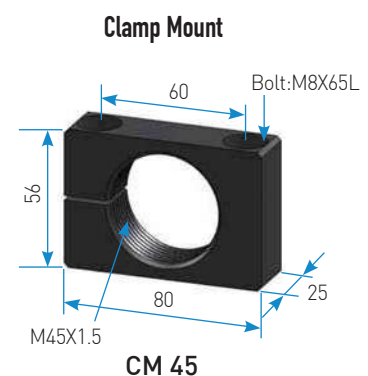
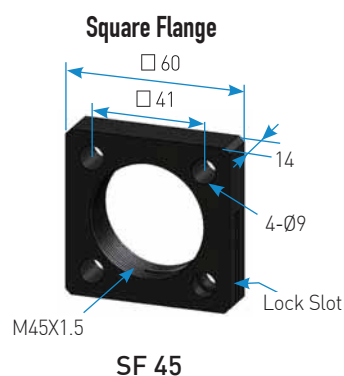
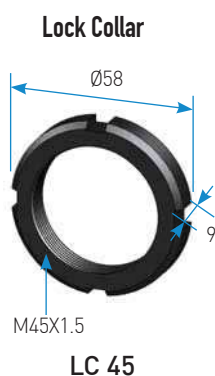
(unit : mm)

| Model      | Stroke | A   | B   | C   | D   | M          |
|------------|--------|-----|-----|-----|-----|------------|
| IMS45 - 25 | 25     | 145 | Ø42 | Ø35 | 95  | M45 X 1.5P |
| - 50       | 50     | 195 | Ø42 | Ø35 | 120 | M45 X 1.5P |
| - 75       | 75     | 246 | Ø42 | Ø35 | 145 | M45 X 1.5P |

## ■ Accessories

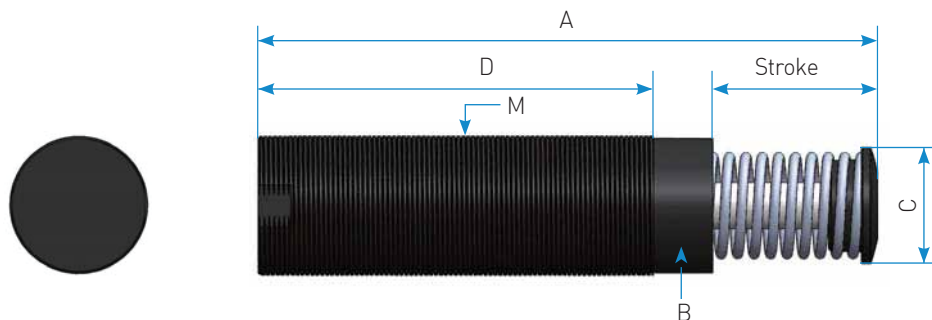
- Other accessories to be ordered separately

(unit : mm)



## Engineering Data

| Model           | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) $E_T$ | Max.Energy /<br>Hour(Nm/h)<br>$E_{TC}$ | Effective Weight(kg) $m_e$ |         | Recoil Forec(N) |       | Weight(kg) |
|-----------------|----------------|---------------------------------|----------------------------------------|----------------------------|---------|-----------------|-------|------------|
|                 |                |                                 |                                        |                            |         | Ext.            | Comp. |            |
| IMS64 - 50 - 1  | 50             | 2,400                           | 240,000                                | 60                         | 320     | 95              | 160   | 3          |
| - 50 - 2        | 50             | 2,400                           | 240,000                                | 280                        | 1,090   | 95              | 160   | 3          |
| - 50 - 3        | 50             | 2,400                           | 240,000                                | 990                        | 4,800   | 95              | 160   | 3          |
| - 50 - 4        | 50             | 2,400                           | 240,000                                | 4,000                      | 19,200  | 95              | 160   | 3          |
| - 50 - 5        | 50             | 2,400                           | 240,000                                | 17,000                     | 108,800 | 95              | 160   | 3          |
| IMS64 - 100 - 1 | 100            | 5,000                           | 350,000                                | 140                        | 650     | 110             | 275   | 3.7        |
| - 100 - 2       | 100            | 5,000                           | 350,000                                | 590                        | 2,270   | 110             | 275   | 3.7        |
| - 100 - 3       | 100            | 5,000                           | 350,000                                | 2,050                      | 10,000  | 110             | 275   | 3.7        |
| - 100 - 4       | 100            | 5,000                           | 350,000                                | 8,300                      | 40,000  | 110             | 275   | 3.7        |
| - 100 - 5       | 100            | 5,000                           | 350,000                                | 35,600                     | 226,700 | 110             | 275   | 3.7        |
| IMS64 - 150 - 1 | 150            | 8,000                           | 400,000                                | 220                        | 1,050   | 80              | 370   | 5          |
| - 150 - 2       | 150            | 8,000                           | 400,000                                | 950                        | 3,600   | 80              | 370   | 5          |
| - 150 - 3       | 150            | 8,000                           | 400,000                                | 3,300                      | 16,000  | 80              | 370   | 5          |
| - 150 - 4       | 150            | 8,000                           | 400,000                                | 13,200                     | 64,000  | 80              | 370   | 5          |
| - 150 - 5       | 150            | 8,000                           | 400,000                                | 16,000                     | 362,000 | 80              | 370   | 5          |



## Dimensions

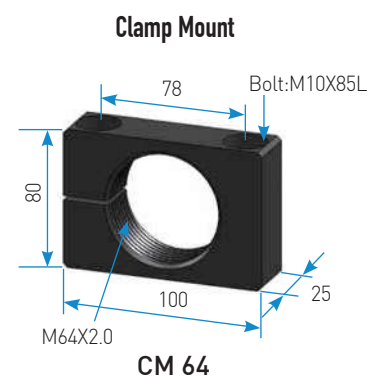
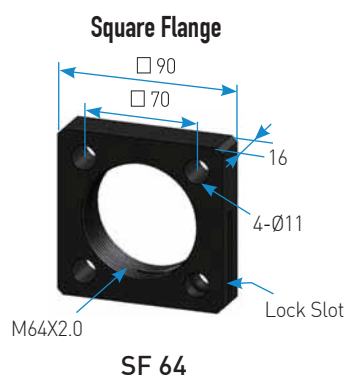
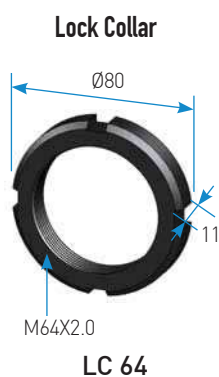
(unit : mm)

| Model      | Stroke | A   | B  | C  | D   | M          |
|------------|--------|-----|----|----|-----|------------|
| IMS64 - 50 | 50     | 225 | 60 | 48 | 140 | M64 X 2.0P |
| - 100      | 100    | 326 | 60 | 48 | 191 | M64 X 2.0P |
| - 150      | 150    | 450 | 60 | 48 | 241 | M64 X 2.0P |

## Accessories

- Other accessories to be ordered separately

(unit : mm)

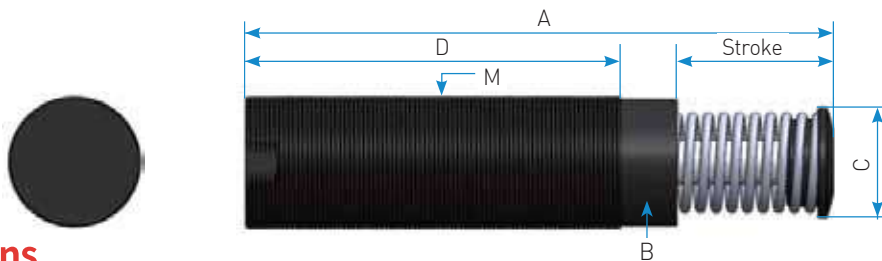




# IMS85 Series

## ■ Engineering Data

| Model           | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) E <sub>T</sub> | Max.Energy /<br>Hour(Nm/h)<br>E <sub>T</sub> C | Effective Weight(kg) m <sub>e</sub> |           | Recoil Forec(N) |       | Weight(kg) |
|-----------------|----------------|------------------------------------------|------------------------------------------------|-------------------------------------|-----------|-----------------|-------|------------|
|                 |                |                                          |                                                |                                     |           | Ext.            | Comp. |            |
| IMS85 - 50 - 1  | 50             | 4,000                                    | 1,200,000                                      | 110                                 | 520       | 140             | 330   | 7          |
| - 50 - 2        | 50             | 4,000                                    | 1,200,000                                      | 470                                 | 1,800     | 140             | 330   | 7          |
| - 50 - 3        | 50             | 4,000                                    | 1,200,000                                      | 1,600                               | 8,000     | 140             | 330   | 7          |
| - 50 - 4        | 50             | 4,000                                    | 1,200,000                                      | 6,600                               | 32,000    | 140             | 330   | 7          |
| - 50 - 5        | 50             | 4,000                                    | 1,200,000                                      | 28,400                              | 181,000   | 140             | 330   | 7          |
| IMS85 - 100 - 1 | 100            | 8,500                                    | 1,700,000                                      | 230                                 | 1,100     | 120             | 420   | 9          |
| - 100 - 2       | 100            | 8,500                                    | 1,700,000                                      | 1,000                               | 3,800     | 120             | 420   | 9          |
| - 100 - 3       | 100            | 8,500                                    | 1,700,000                                      | 3,500                               | 17,000    | 120             | 420   | 9          |
| - 100 - 4       | 100            | 8,500                                    | 1,700,000                                      | 14,100                              | 68,000    | 120             | 420   | 9          |
| - 100 - 5       | 100            | 8,500                                    | 1,700,000                                      | 60,500                              | 385,000   | 120             | 420   | 9          |
| IMS85 - 150 - 1 | 150            | 13,500                                   | 2,025,000                                      | 370                                 | 1,800     | 110             | 430   | 12         |
| - 150 - 2       | 150            | 13,500                                   | 2,025,000                                      | 1,600                               | 6,100     | 110             | 430   | 12         |
| - 150 - 3       | 150            | 13,500                                   | 2,025,000                                      | 5,600                               | 27,000    | 110             | 430   | 12         |
| - 150 - 4       | 150            | 13,500                                   | 2,025,000                                      | 22,300                              | 108,000   | 110             | 430   | 12         |
| - 150 - 5       | 150            | 13,500                                   | 2,025,000                                      | 96,100                              | 612,000   | 110             | 430   | 12         |
| IMS85 - 200 - 1 | 200            | 19,000                                   | 2,660,000                                      | 530                                 | 2,500     | 100             | 430   | 15         |
| - 200 - 2       | 200            | 19,000                                   | 2,660,000                                      | 2,200                               | 8,600     | 100             | 430   | 15         |
| - 200 - 3       | 200            | 19,000                                   | 2,660,000                                      | 7,800                               | 38,000    | 100             | 430   | 15         |
| - 200 - 4       | 200            | 19,000                                   | 2,660,000                                      | 31,400                              | 152,000   | 100             | 430   | 15         |
| - 200 - 5       | 200            | 19,000                                   | 2,660,000                                      | 135,000                             | 861,000   | 100             | 430   | 15         |
| IMS85 - 250 - 1 | 250            | 23,500                                   | 2,820,000                                      | 650                                 | 3,100     | 95              | 430   | 20         |
| - 250 - 2       | 250            | 23,500                                   | 2,820,000                                      | 2,800                               | 10,700    | 95              | 430   | 20         |
| - 250 - 3       | 250            | 23,500                                   | 2,820,000                                      | 9,700                               | 47,000    | 95              | 430   | 20         |
| - 250 - 4       | 250            | 23,500                                   | 2,820,000                                      | 38,800                              | 188,000   | 95              | 430   | 20         |
| - 250 - 5       | 250            | 23,500                                   | 2,820,000                                      | 167,000                             | 1,065,000 | 95              | 430   | 20         |



## ■ Dimensions

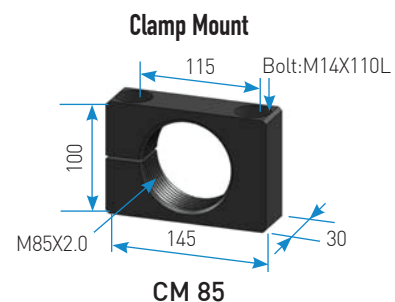
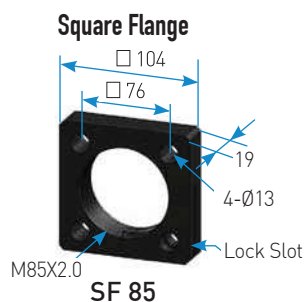
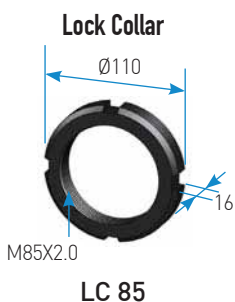
(unit : mm)

| Model      | Stroke | A   | B  | C  | D   | M          |
|------------|--------|-----|----|----|-----|------------|
| IMS85 - 50 | 50     | 258 | 81 | 68 | 165 | M85 X 2.0P |
| - 100      | 100    | 360 | 81 | 68 | 218 | M85 X 2.0P |
| - 150      | 150    | 485 | 81 | 68 | 285 | M85 X 2.0P |
| - 200      | 200    | 590 | 81 | 68 | 330 | M85 X 2.0P |
| - 250      | 250    | 700 | 81 | 68 | 390 | M85 X 2.0P |

## ■ Accessories

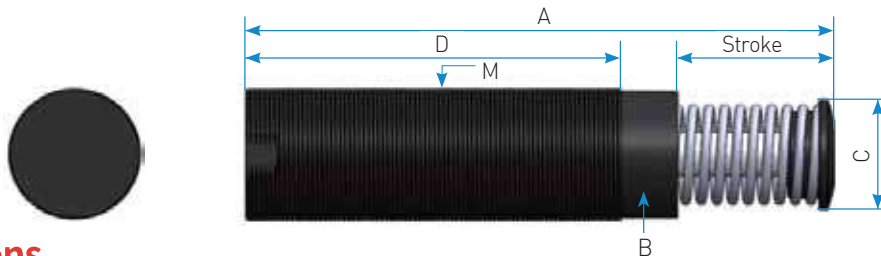
(unit : mm)

- Other accessories to be ordered separately



## Engineering Data

| Model            | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) $E_T$ | Max.Energy /<br>Hour(Nm/h)<br>$E_{TC}$ | Effective Weight(kg) $m_e$ |           | Recoil Forec(N) |       | Weight(kg) |
|------------------|----------------|---------------------------------|----------------------------------------|----------------------------|-----------|-----------------|-------|------------|
|                  |                |                                 |                                        |                            |           | Ext.            | Comp. |            |
| IMS115 - 50 - 1  | 50             | 11,000                          | 3,300,000                              | 300                        | 1,400     | 220             | 480   | 12         |
| - 50 - 2         | 50             | 11,000                          | 3,300,000                              | 1,300                      | 5,000     | 220             | 480   | 12         |
| - 50 - 3         | 50             | 11,000                          | 3,300,000                              | 4,500                      | 22,000    | 220             | 480   | 12         |
| - 50 - 4         | 50             | 11,000                          | 3,300,000                              | 18,100                     | 88,000    | 220             | 480   | 12         |
| - 50 - 5         | 50             | 11,000                          | 3,300,000                              | 78,300                     | 498,000   | 220             | 480   | 12         |
| IMS115 - 100 - 1 | 100            | 22,600                          | 4,520,000                              | 630                        | 3,000     | 220             | 550   | 14         |
| - 100 - 2        | 100            | 22,600                          | 4,520,000                              | 2,700                      | 10,300    | 220             | 550   | 14         |
| - 100 - 3        | 100            | 22,600                          | 4,520,000                              | 9,400                      | 45,200    | 220             | 550   | 14         |
| - 100 - 4        | 100            | 22,600                          | 4,520,000                              | 37,300                     | 180,800   | 220             | 550   | 14         |
| - 100 - 5        | 100            | 22,600                          | 4,520,000                              | 160,000                    | 1,024,000 | 220             | 550   | 14         |
| IMS115 - 150 - 1 | 150            | 34,000                          | 5,100,000                              | 950                        | 4,500     | 200             | 570   | 17         |
| - 150 - 2        | 150            | 34,000                          | 5,100,000                              | 4,100                      | 15,500    | 200             | 570   | 17         |
| - 150 - 3        | 150            | 34,000                          | 5,100,000                              | 14,000                     | 68,000    | 200             | 570   | 17         |
| - 150 - 4        | 150            | 34,000                          | 5,100,000                              | 56,000                     | 272,000   | 200             | 570   | 17         |
| - 150 - 5        | 150            | 34,000                          | 5,100,000                              | 242,000                    | 1,541,000 | 200             | 570   | 17         |
| IMS115 - 200 - 1 | 200            | 45,000                          | 6,300,000                              | 1,300                      | 6,000     | 180             | 570   | 20         |
| - 200 - 2        | 200            | 45,000                          | 6,300,000                              | 5,400                      | 20,500    | 180             | 570   | 20         |
| - 200 - 3        | 200            | 45,000                          | 6,300,000                              | 18,500                     | 90,000    | 180             | 570   | 20         |
| - 200 - 4        | 200            | 45,000                          | 6,300,000                              | 74,300                     | 360,000   | 180             | 570   | 20         |
| - 200 - 5        | 200            | 45,000                          | 6,300,000                              | 320,000                    | 2,040,000 | 180             | 570   | 20         |
| IMS115 - 250 - 1 | 250            | 56,000                          | 6,720,000                              | 1,500                      | 7,400     | 200             | 620   | 25         |
| - 250 - 2        | 250            | 56,000                          | 6,720,000                              | 6,600                      | 25,300    | 200             | 620   | 25         |
| - 250 - 3        | 250            | 56,000                          | 6,720,000                              | 23,100                     | 112,000   | 200             | 620   | 25         |
| - 250 - 4        | 250            | 56,000                          | 6,720,000                              | 92,500                     | 448,000   | 200             | 620   | 25         |
| - 250 - 5        | 250            | 56,000                          | 6,720,000                              | 398,000                    | 2,539,000 | 200             | 620   | 25         |



## Dimensions

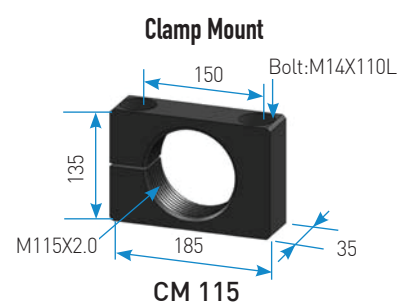
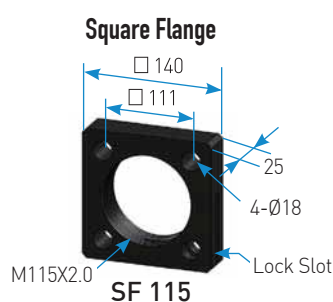
(unit : mm)

| Model       | Stroke | A   | B    | C   | D   | M           |
|-------------|--------|-----|------|-----|-----|-------------|
| IMS115 - 50 | 50     | 310 | Ø111 | Ø98 | 205 | M115 X 2.0P |
| - 100       | 100    | 420 | Ø111 | Ø98 | 265 | M115 X 2.0P |
| - 150       | 150    | 535 | Ø111 | Ø98 | 315 | M115 X 2.0P |
| - 200       | 200    | 655 | Ø111 | Ø98 | 380 | M115 X 2.0P |
| - 250       | 250    | 770 | Ø111 | Ø98 | 435 | M115 X 2.0P |

## Accessories

- Other accessories to be ordered separately

(unit : mm)

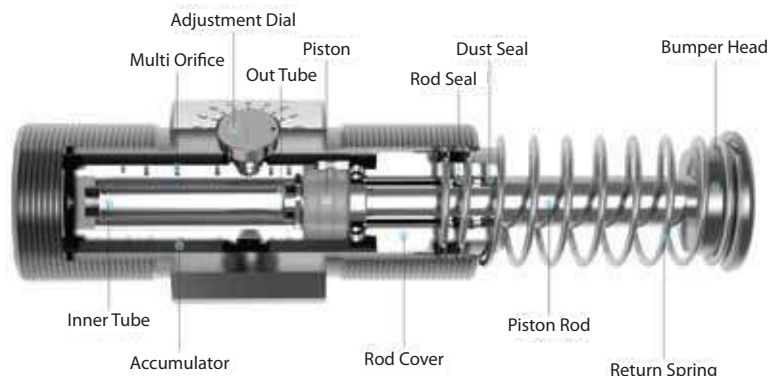


# IHA Series Mid dial type

## IZMAC Heavy Type Absorber

### DESCRIPTION

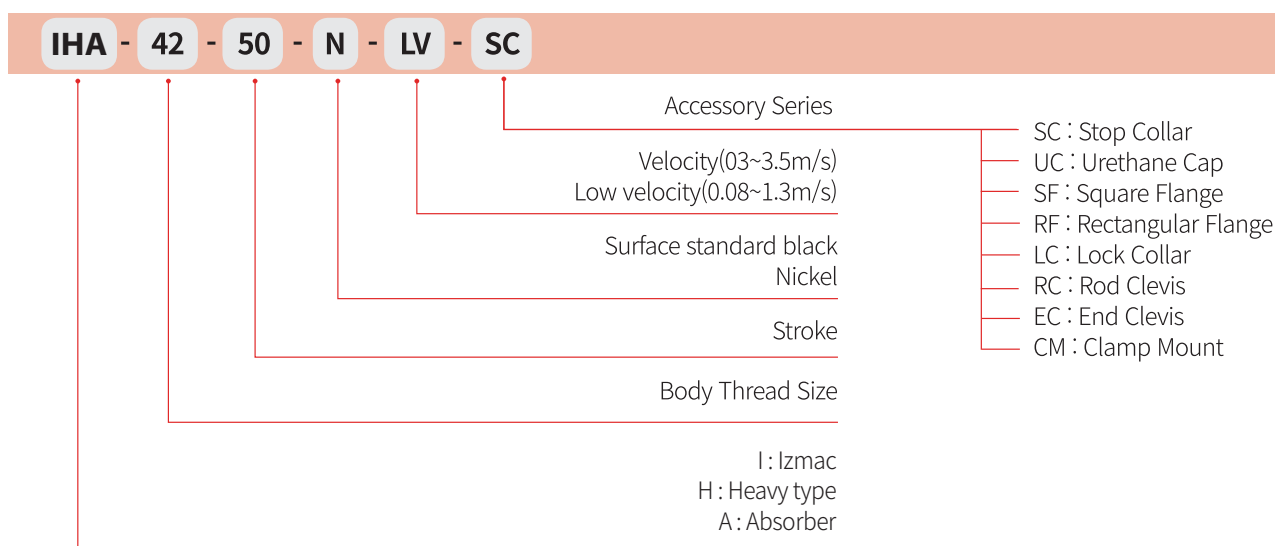
IHA is mid large bore series heavy type shock absorber. Operation : when the collision comes piston compress oil chamber in inner tube according to the piston rod moving. Oil flows to accumulator through outside grooves in this process, deceleration power comes by flow resistance. And kinetic energy is dissipated in the air by changed as heat energy. Recoil Spring returns piston rod into its original position when the load is disappeared.



### FEATURES

- 1 IHA is dial type and it can set the speed ranges as 8 steps.
- 2 Energy capacity was Improved over 200% and effective weight ranges become wider than before.
- 3 All threaded body outside helps to install shock absorber and it can be fixed at the exact position. And it helps also to dissipate more shock energy by increased surface dimension.
- 4 Body surface treated as alloyed black colored steel or Zn-Ni and it develops anti corrosion features highly.
- 5 Stop collar is not necessary when shock absorber has steel head, and poly pad & urethane cap will help to decrease collision noises efficiently.
- 6 Rod cover is made of long lasting materials with function of seal protection and that enable to guarantee long life.
- 7 Speed ranges : 0.3 ~ 3.5m/s • Low Velocity (LV) : 0.08 ~ 1.3m/s
- 8 Temperature ranges : -10 ~ 80°C • Special : -40 ~ 120°C(Oil and Seal)

### IHA SERIES ORDERING INFORMATION



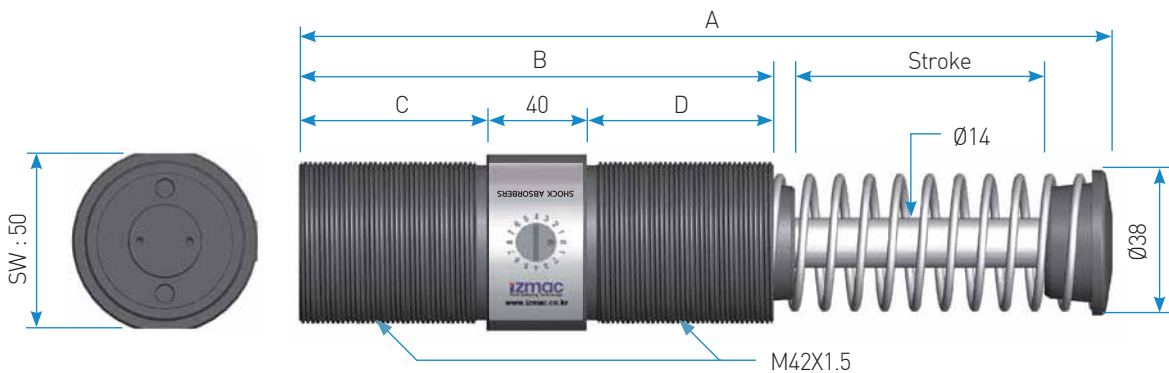
# IHA42 Series

Best engineered  
for energy absorption  
technology

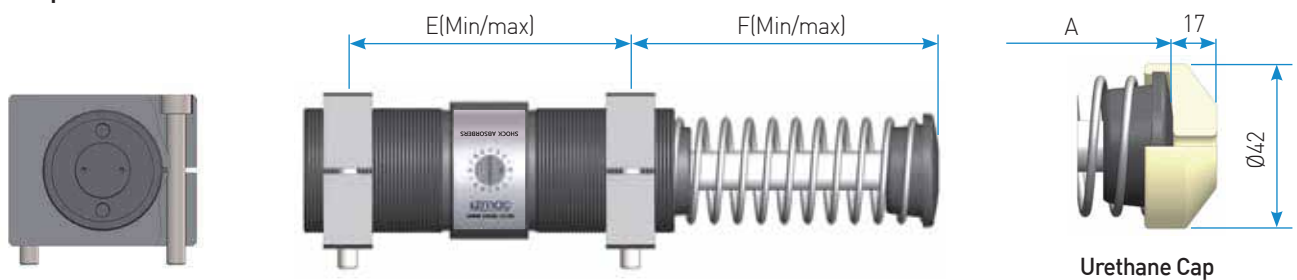


## Engineering Data

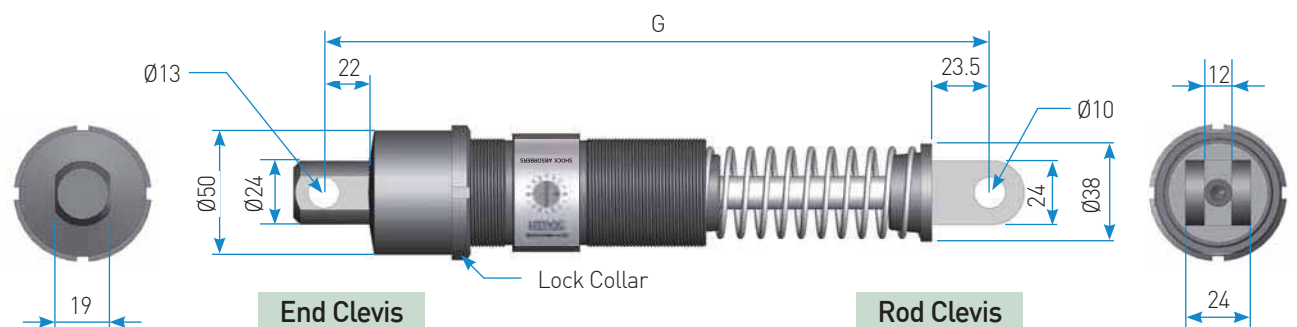
| Model      | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) E <sub>T</sub> | Max.Energy /<br>Hour(Nm/h)<br>E <sub>T</sub> C | Effective<br>Weight(kg) m <sub>e</sub> | Recoil Forec(N) |       | Weight(kg) |
|------------|----------------|------------------------------------------|------------------------------------------------|----------------------------------------|-----------------|-------|------------|
|            |                |                                          |                                                |                                        | Ext.            | Comp. |            |
| IHA42 - 25 | 25             | 250                                      | 125,000                                        | 36 - 3,600                             | 28              | 56    | 1.25       |
| - 50       | 50             | 500                                      | 166,000                                        | 45 - 6,150                             | 38              | 86    | 1.4        |
| - 75       | 75             | 750                                      | 200,000                                        | 54 - 9,500                             | 32              | 88    | 1.6        |



### Clamp Mount



### Clevis Mount



## Dimensions

(unit : mm)

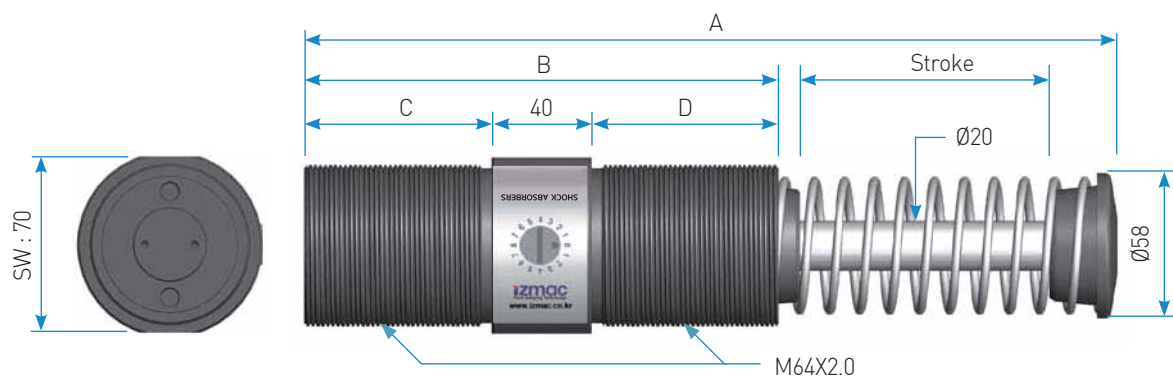
| Model      | Stroke | A   | B   | C  | D  | E(Min/Max) | F(Min/Max)  | G   |
|------------|--------|-----|-----|----|----|------------|-------------|-----|
| IHA42 - 25 | 25     | 145 | 95  | 28 | 27 | 65/70      | 62.5/64.5   | 200 |
| - 50       | 50     | 195 | 120 | 40 | 40 | 65/95      | 87.5/102.5  | 250 |
| - 75       | 75     | 245 | 145 | 52 | 53 | 65/120     | 117.6/145.5 | 300 |



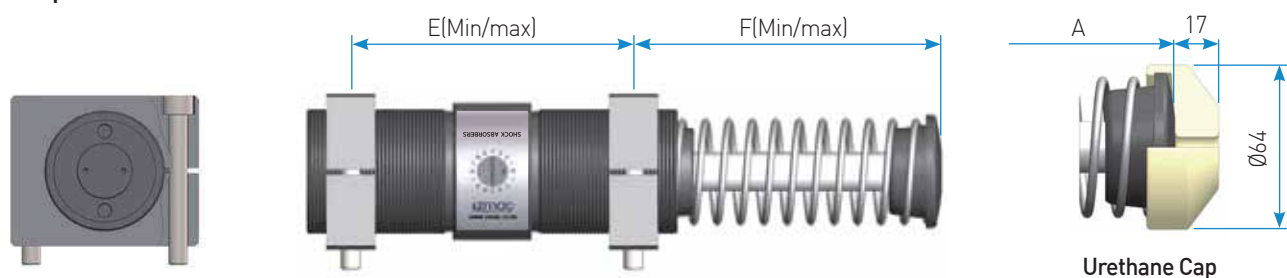
# IHA64 Series

## ■ Engineering Data

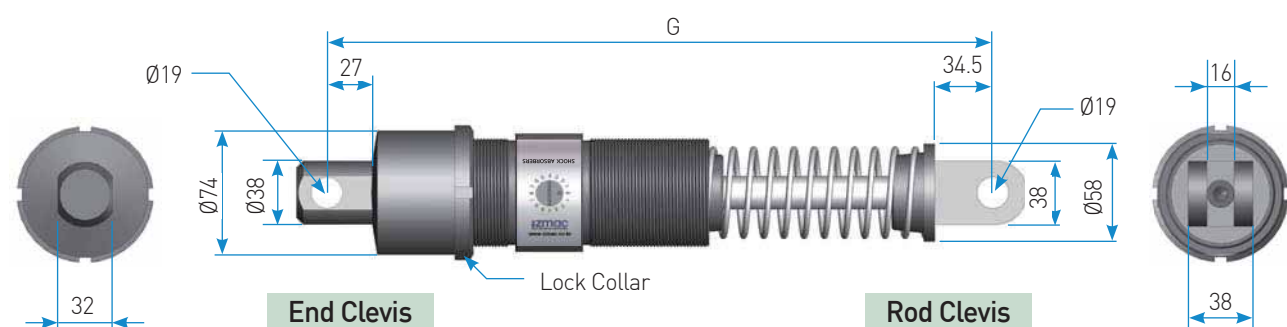
| Model      | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) E <sub>T</sub> | Max.Energy /<br>Hour(Nm/h)<br>E <sub>T</sub> C | Effective<br>Weight(kg) m <sub>e</sub> | Recoil Forec(N) |       | Weight(kg) |
|------------|----------------|------------------------------------------|------------------------------------------------|----------------------------------------|-----------------|-------|------------|
|            |                |                                          |                                                |                                        | Ext.            | Comp. |            |
| IHA64 - 50 | 50             | 1,300                                    | 270,000                                        | 82 - 12,600                            | 65              | 148   | 3.7        |
| - 100      | 100            | 2,600                                    | 360,000                                        | 115 - 17,000                           | 45              | 157   | 4.5        |
| - 150      | 150            | 3,900                                    | 420,000                                        | 140 - 22,000                           | 47              | 199   | 5.3        |



### Clamp Mount



### Clevis Mount



## ■ Dimensions

(unit : mm)

| Model      | Stroke | A   | B   | C   | D   | E(Min/Max) | F(Min/Max)  | G   |
|------------|--------|-----|-----|-----|-----|------------|-------------|-----|
| IHA64 - 50 | 50     | 225 | 140 | 50  | 50  | 65/115     | 97.5/122.5  | 305 |
| - 100      | 100    | 327 | 190 | 75  | 75  | 65/165     | 147.5/197.5 | 409 |
| - 150      | 150    | 455 | 240 | 100 | 100 | 65/215     | 197.5/272.5 | 534 |

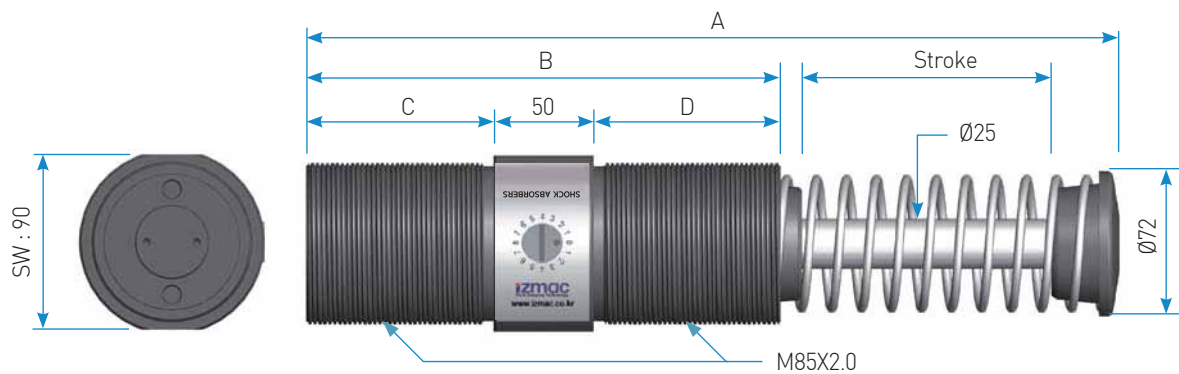
# IHA85 Series

Best engineered  
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technology

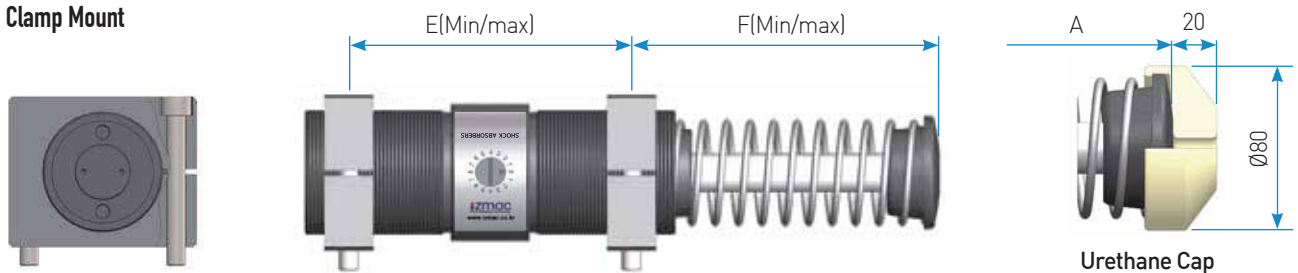


## ■ Engineering Data

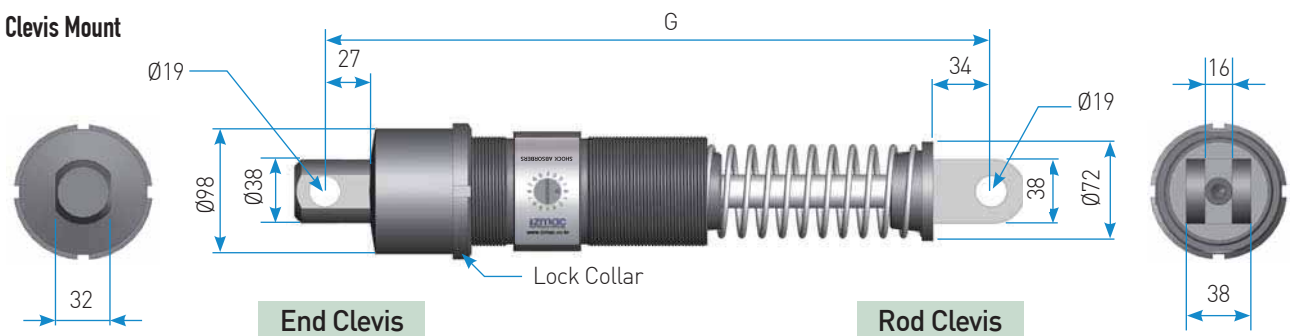
| Model      | Stroke<br>(mm) | Max. Energy /<br>Cycle(Nm) E <sub>T</sub> | Max. Energy /<br>Hour(Nm/h)<br>E <sub>T</sub> C | Effective<br>Weight(kg) m <sub>e</sub> | Recoil Fore(N) |       | Weight(kg) |
|------------|----------------|-------------------------------------------|-------------------------------------------------|----------------------------------------|----------------|-------|------------|
|            |                |                                           |                                                 |                                        | Ext.           | Comp. |            |
| IHA64 - 50 | 50             | 2,300                                     | 370,000                                         | 200 - 37,000                           | 148            | 315   | 6.4        |
| - 90       | 90             | 4,000                                     | 650,000                                         | 230 - 40,000                           | 121            | 365   | 7.6        |
| - 125      | 125            | 5,700                                     | 930,000                                         | 300 - 43,000                           | 114            | 365   | 8.6        |
| - 165      | 165            | 7,300                                     | 1,210,000                                       | 360 - 45,000                           | 98             | 429   | 9.8        |



### Clamp Mount



### Clevis Mount



## ■ Dimensions

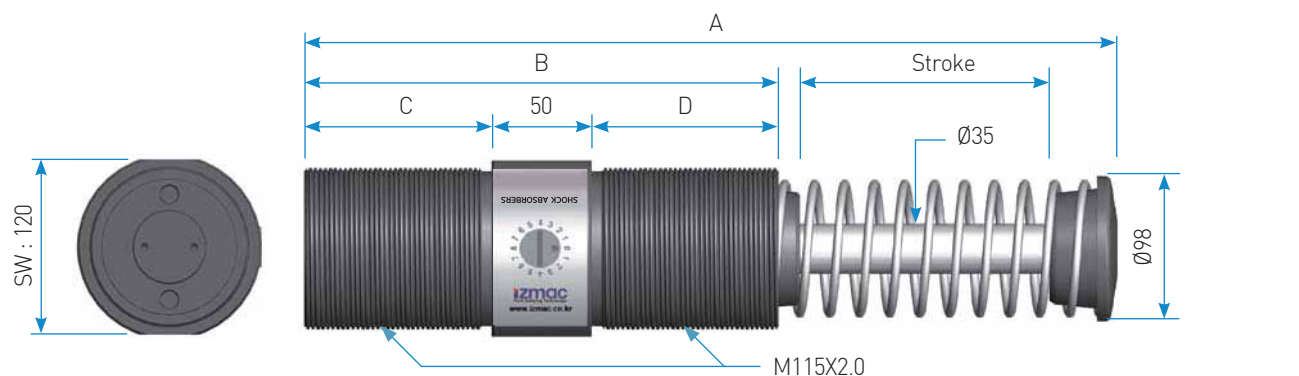
(unit : mm)

| Model      | Stroke | A   | B   | C   | D   | E(Min/Max) | F(Min/Max) | G   |
|------------|--------|-----|-----|-----|-----|------------|------------|-----|
| IHA64 - 50 | 50     | 245 | 140 | 42  | 48  | 80/110     | 120/138    | 325 |
| - 90       | 90     | 325 | 180 | 60  | 70  | 80/150     | 160/200    | 405 |
| - 125      | 125    | 400 | 216 | 82  | 84  | 80/186     | 195/249    | 480 |
| - 165      | 165    | 494 | 256 | 106 | 100 | 80/226     | 220/290    | 575 |

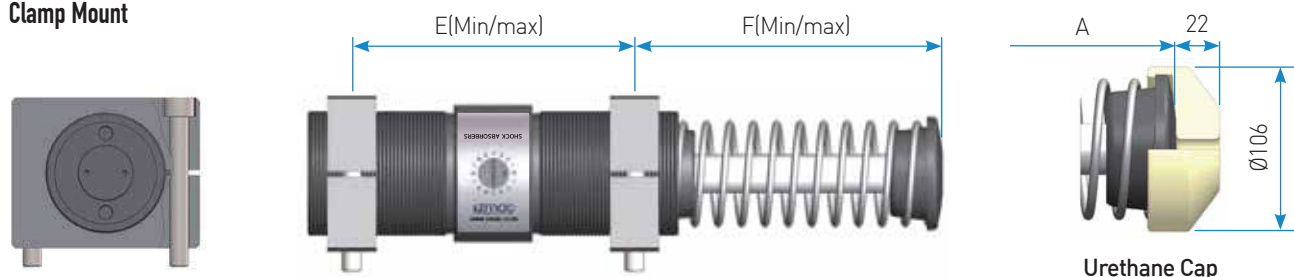
# IHA115 Series

## ■ Engineering Data

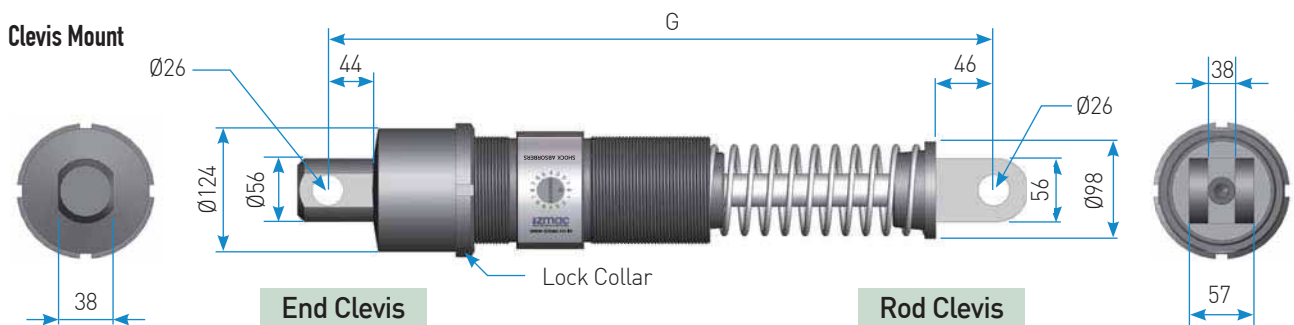
| Model       | Stroke (mm) | Max.Energy / Cycle(Nm) $E_T$ | Max.Energy / Hour(Nm/h) $E_{TC}$ | Effective Weight(kg) $m_e$ | Recoil Forec(N) |       | Weight(kg) |
|-------------|-------------|------------------------------|----------------------------------|----------------------------|-----------------|-------|------------|
|             |             |                              |                                  |                            | Ext.            | Comp. |            |
| IHA115 - 50 | 50          | 3,700                        | 1,501,000                        | 370 - 44,000               | 187             | 427   | 16         |
| - 100       | 100         | 7,400                        | 1,805,000                        | 370 - 57,000               | 178             | 482   | 19         |
| - 150       | 150         | 11,100                       | 2,100,000                        | 370 - 89,000               | 185             | 538   | 21         |
| - 200       | 200         | 14,800                       | 2,405,000                        | 390 - 118,000              | 185             | 665   | 31         |
| - 250       | 250         | 18,500                       | 2,710,000                        | 440 - 145,000              | 184             | 649   | 34         |



### Clamp Mount



### Clevis Mount

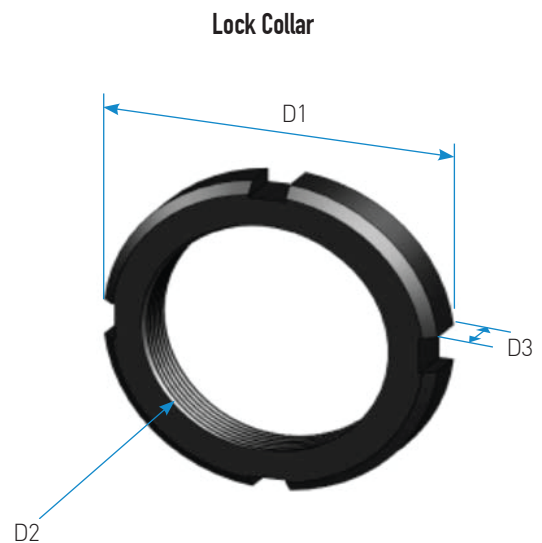
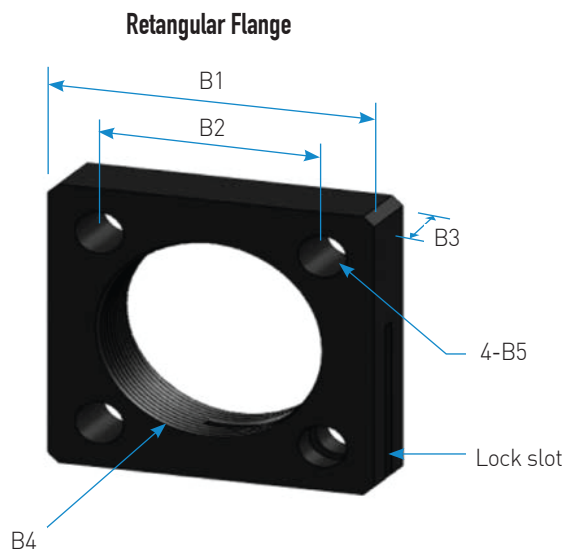
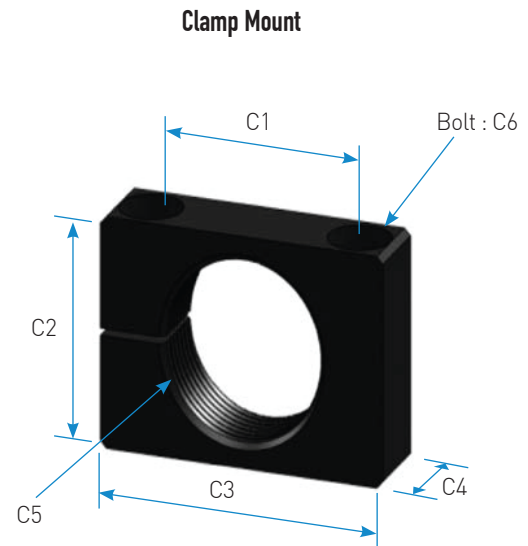
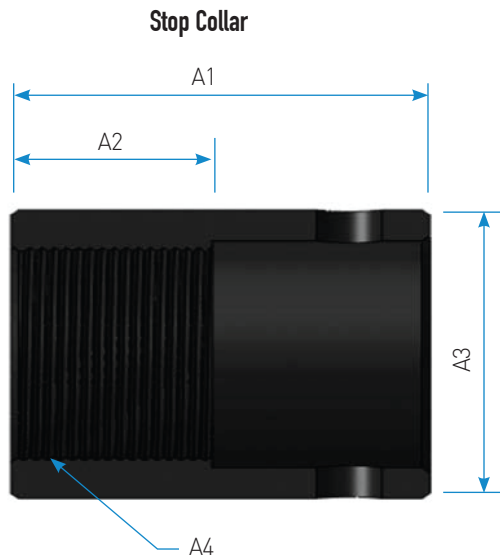


## ■ Dimensions

(unit : mm)

| Model       | Stroke | A   | B   | C   | D   | E(Min/Max) | F(Min/Max) | G   |
|-------------|--------|-----|-----|-----|-----|------------|------------|-----|
| IHA115 - 50 | 50     | 314 | 200 | 75  | 75  | 80/170     | 129/174    | 430 |
| - 100       | 100    | 414 | 250 | 100 | 100 | 80/220     | 179/249    | 535 |
| - 150       | 150    | 514 | 300 | 125 | 125 | 80/270     | 229/324    | 635 |
| - 200       | 200    | 644 | 350 | 150 | 150 | 80/320     | 279/399    | 760 |
| - 250       | 250    | 744 | 400 | 180 | 170 | 80/370     | 329/479    | 865 |

## ■ Accessories



## ■ Dimensions

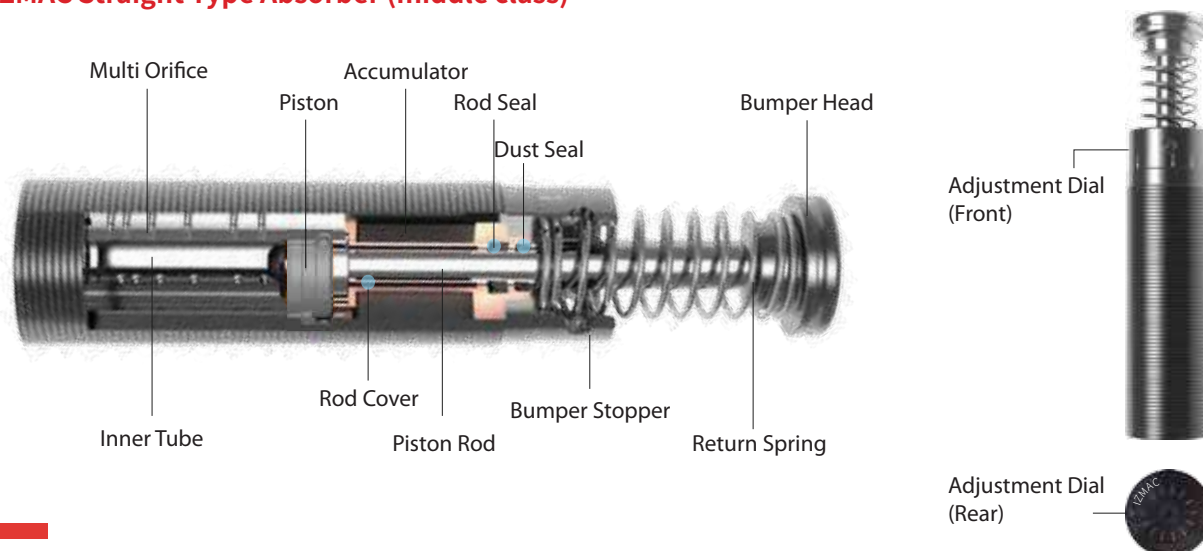
(unit : mm)

| Stop Collar | Rectangular Flange | Clamp Mount | Lock Nut  | A1  | A2 | A3   | A4         | B1   | B2   | B3 | B4         | B5      | C1  | C2  | C3  | C4 | C5         | C6         | D1   | D2         | D3 |
|-------------|--------------------|-------------|-----------|-----|----|------|------------|------|------|----|------------|---------|-----|-----|-----|----|------------|------------|------|------------|----|
| SC42X1.5    | RF42X1.5           | CM42X1.5    | LN42X1.5  | 52  | 25 | Ø56  | M42 X 1.5P | □60  | □41  | 14 | M42 X 1.5P | 4 - Ø9  | 60  | 56  | 80  | 25 | M42 X 1.5P | M8 X 60L   | Ø58  | M42 X 1.5P | 9  |
| SC64X2.0    | RF64X2.0           | CM64X2.0    | LN64X2.0  | 85  | 45 | Ø75  | M64 X 2.0P | □90  | □70  | 16 | M64 X 2.0P | 4 - Ø11 | 78  | 80  | 100 | 25 | M64 X 2.0P | M10 X 80L  | Ø80  | M64 X 2.0P | 11 |
| SC85X2.0    | RF85X2.0           | CM85X2.0    | LN85X2.0  | 95  | 45 | Ø98  | M85 X 2.0P | □104 | □76  | 19 | M85 X 2.0P | 4 - Ø13 | 115 | 100 | 145 | 30 | M85 X 2.0P | M14 X 100L | Ø110 | M85 X 2.0P | 16 |
| SC115X2.0   | RF115X2.0          | CM115X2.0   | LN115X2.0 | 130 | 70 | Ø138 | M115 X 2.0 | □140 | □111 | 25 | M115 X 2.0 | 4 - Ø18 | 150 | 135 | 185 | 35 | M115 X 2.0 | M16 X 150L | Ø150 | M115 X 2.0 | 19 |



# ISA Series Mid dial type

## IZMAC Straight Type Absorber (middle class)



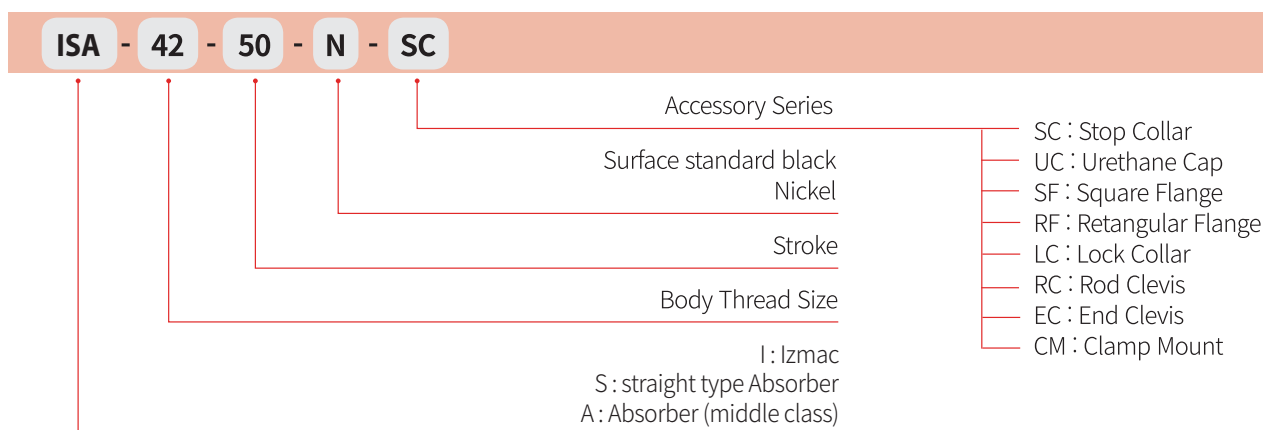
## FEATURES

- 1 According to the collision speed, you can fit the buffer dial from 0 to 300 degrees in 12 steps.
- 2 Adjustment is easy in various installation environment by adopting front and rear adjustment dial.
- 3 Shock absorber body is built as one body, not only robust also fundamentally solves the shoot out trouble so called bottom out problem.
- 4 The entire body is threaded and easy to assemble. Also, due to the surface area is increased, the heat energy which comes from shock absorption can be emitted to the outside faster.
- 5 Piston rod is hard chrome-plated. Rod cover is a special material that can be used even for a long time, it protects seal and guarantees life.
- 6 Body surface treatment is resistant to corrosion by nickel plating or alloy plating (Black).
- 7 By using urethane cap, noise can be reduced.
- 8 Speed ranges : 0.3 ~ 5m/s
- 9 Temperature ranges : -10 ~ 80°C • Special : -40 ~ 120°C(Oil and Seal)

## APPLICATIONS

Pick n place robot, packing machine, loom, machine tool, automobile manufacturing facility, crane etc.

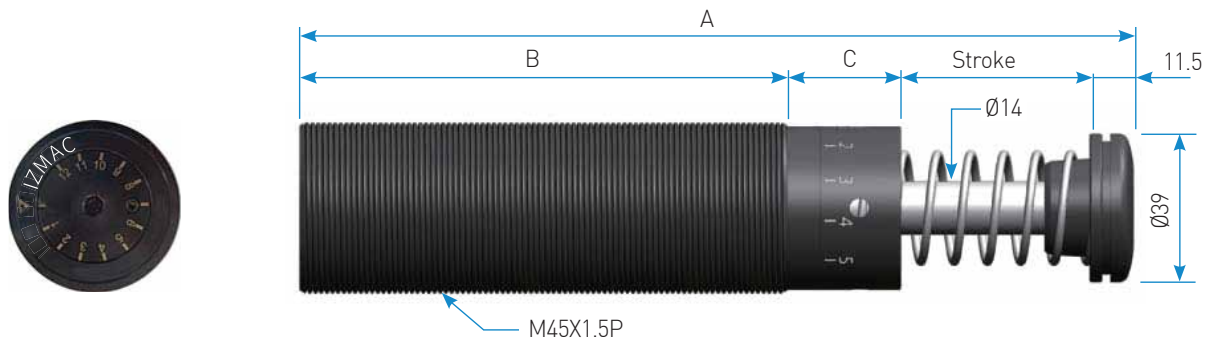
## ISA SERIES ORDERING INFORMATION





## Engineering Data

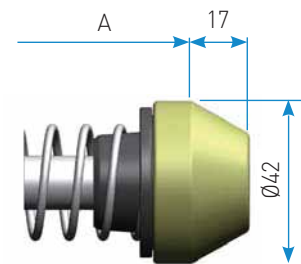
| Model      | Stroke<br>(mm) St | Max.Energy /<br>Cycle(Nm) E <sub>T</sub> | Max.Energy /<br>Hour(Nm/h)<br>E <sub>T</sub> C | Effective<br>Weight(kg) m <sub>e</sub> | Recoil Forec(N) |       | Weight(kg) |
|------------|-------------------|------------------------------------------|------------------------------------------------|----------------------------------------|-----------------|-------|------------|
|            |                   |                                          |                                                |                                        | Ext.            | Comp. |            |
| ISA45 - 25 | 25                | 650                                      | 195,000                                        | 50-13,354                              | 49.7            | 82.8  | 1.13       |
| - 50       | 50                | 1,300                                    | 260,000                                        | 99-26,700                              | 45.6            | 54    | 1.3        |
| - 75       | 75                | 2,000                                    | 300,000                                        | 148-39,060                             | 44.3            | 86.3  | 1.52       |



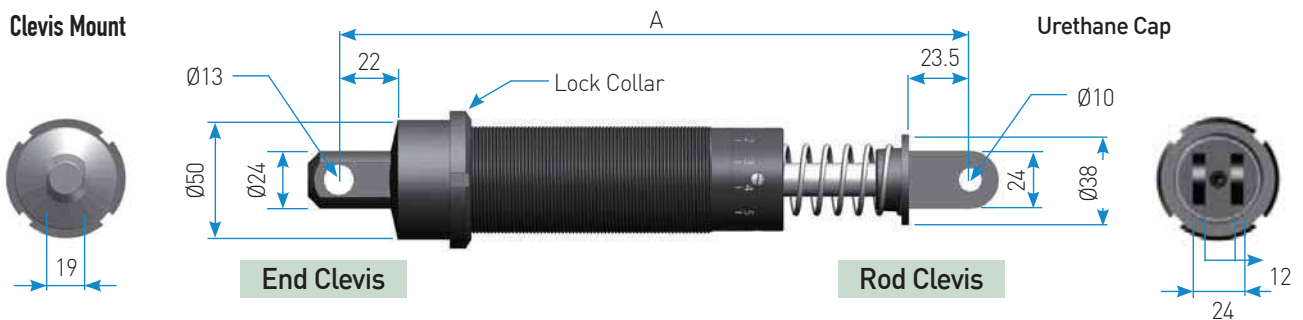
## Dimensions

(unit : mm)

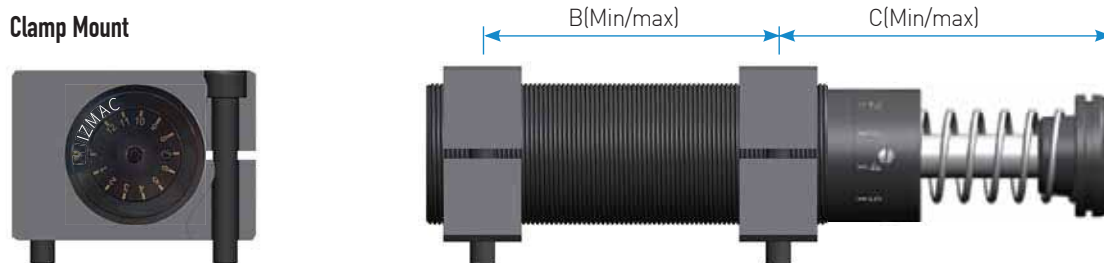
| Model      | Stroke | A     | B     | C    |
|------------|--------|-------|-------|------|
| ISA45 - 25 | 25     | 159.5 | 93    | 30   |
| - 50       | 50     | 220.5 | 129   | 30   |
| - 75       | 75     | 292.5 | 168.5 | 37.5 |



### Clevis Mount



### Clamp Mount



## Accessories

- Other accessories to be ordered separately

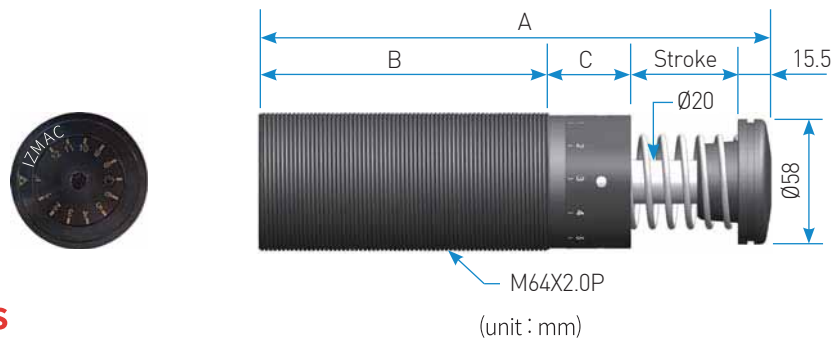
(unit : mm)

| Model      | ISA45-25   | ISA45-50  | ISA45-75        |
|------------|------------|-----------|-----------------|
| A          | 210        | 271       | 343             |
| B(Min/Max) | 25/68      | 25/104    | 25/143.5        |
| C(Min/Max) | 79.5/100.5 | 104/143.5 | 37.5/129/170.75 |

# ISA64 Series

## Engineering Data

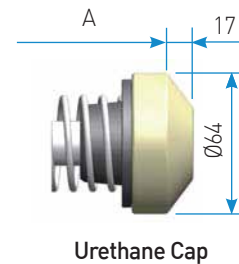
| Model      | Stroke (mm) | Max.Energy / Cycle(Nm) $E_T$ | Max.Energy / Hour(Nm/h) $E_{TC}$ | Effective Weight(kg) $m_e$ | Recoil Forec(N) |       | Weight(kg) |
|------------|-------------|------------------------------|----------------------------------|----------------------------|-----------------|-------|------------|
|            |             |                              |                                  |                            | Ext.            | Comp. |            |
| ISA64 - 25 | 25          | 1,250                        | 152,000                          | 92~24,400                  | 61.8            | 110   | 2.9        |
| - 50       | 50          | 2,500                        | 248,000                          | 185-48,800                 | 60.8            | 133.2 | 3.3        |
| - 75       | 75          | 3,750                        | 265,000                          | 277-73,240                 | 61.4            | 148.3 | 3.8        |
| - 100      | 100         | 5,000                        | 360,000                          | 370-97,650                 | 59.4            | 140.1 | 4          |
| - 125      | 125         | 6,250                        | 413,000                          | 462-122,000                | 57.1            | 160.4 | 7.7        |
| - 150      | 150         | 7,500                        | 450,000                          | 555-146,480                | 51              | 166.8 | 8.9        |



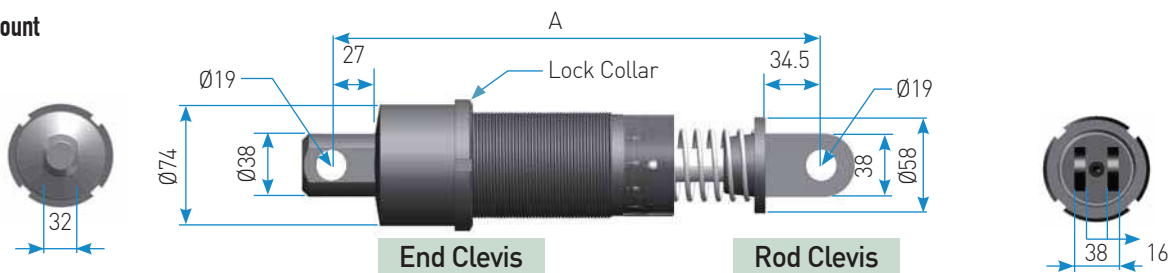
## Dimensions

(unit : mm)

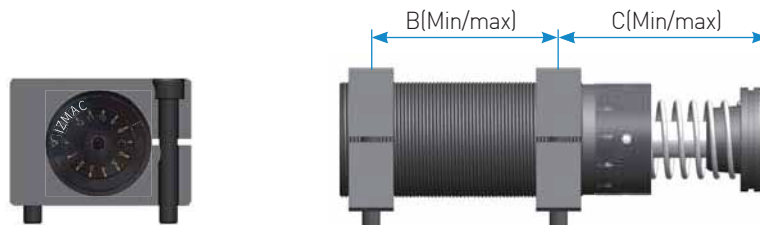
| Model      | Stroke | A     | B     | C    |
|------------|--------|-------|-------|------|
| ISA64 - 25 | 25     | 176   | 97.5  | 38   |
| - 50       | 50     | 239   | 135.5 | 38   |
| - 75       | 75     | 309   | 173.5 | 45   |
| - 100      | 100    | 375   | 214.5 | 45   |
| - 125      | 125    | 452   | 256.5 | 55   |
| - 150      | 150    | 518.5 | 294.5 | 58.5 |



### Clevis Mount



### Clamp Mount



## Accessories

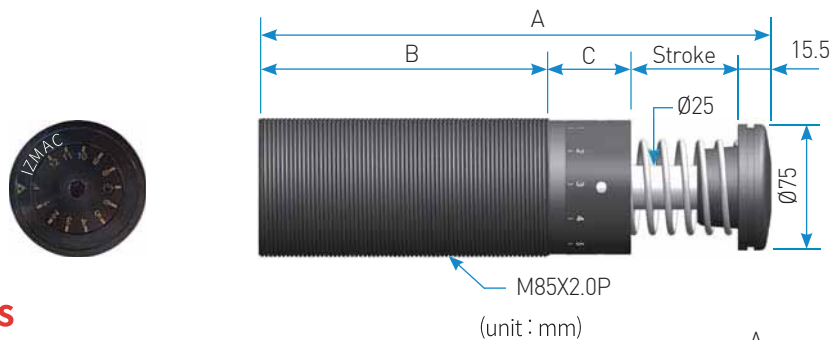
- Other accessories to be ordered separately

(unit : mm)

| Model      | ISA64 -25   | ISA64 -50    | ISA64 -75    | ISA64 -100   | ISA64 -125   | ISA64 -150 |
|------------|-------------|--------------|--------------|--------------|--------------|------------|
| A          | 244         | 307          | 377          | 442          | 520          | -          |
| B(Min/Max) | 25/72.5     | 25/110.5     | 25/148.5     | 25/189.5     | 25/231.5     | 25/269.5   |
| C(Min/Max) | 92.5/116.25 | 117.5/160.25 | 149.5/211.25 | 174.5/256.75 | 209.5/312.75 | 238/360.25 |

## Engineering Data

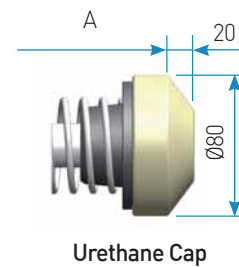
| Model      | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) E <sub>T</sub> | Max.Energy /<br>Hour(Nm/h)<br>E <sub>T</sub> C | Effective<br>Weight(kg) m <sub>e</sub> | Recoil Forec(N) |       | Weight(kg) |
|------------|----------------|------------------------------------------|------------------------------------------------|----------------------------------------|-----------------|-------|------------|
|            |                |                                          |                                                |                                        | Ext.            | Comp. |            |
| ISA85 - 25 | 25             | 2,000                                    | 330,000                                        | 148-39,600                             | 132             | 245.8 | 5.4        |
| - 50       | 50             | 4,000                                    | 462,000                                        | 296-78,100                             | 131.6           | 271.7 | 6.5        |
| - 75       | 75             | 6,000                                    | 680,000                                        | 444-117,200                            | 130             | 325.1 | 7.9        |
| - 100      | 100            | 8,000                                    | 825,000                                        | 591-156,300                            | 125.4           | 327.7 | 9.2        |
| - 125      | 125            | 10,000                                   | 859,000                                        | 740-195,300                            | 126.6           | 343.3 | 10.4       |
| - 150      | 150            | 12,000                                   | 901,000                                        | 930-220,300                            | 126.6           | 386.7 | 12         |



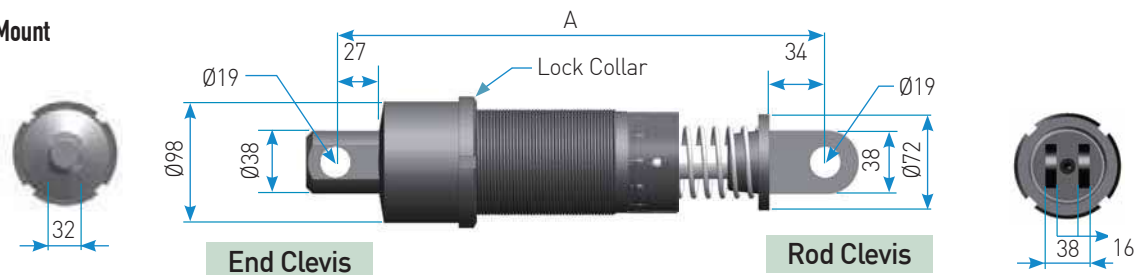
## Dimensions

(unit : mm)

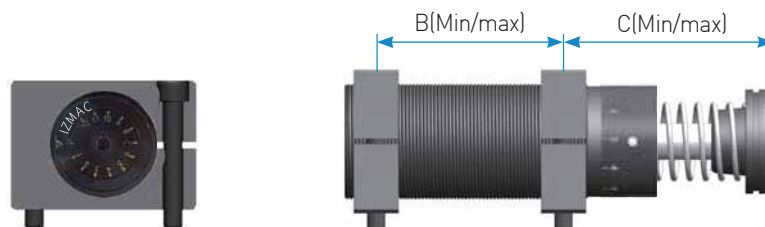
| Model      | Stroke | A     | B     | C  |
|------------|--------|-------|-------|----|
| ISA85 - 25 | 25     | 189.5 | 109   | 40 |
| - 50       | 50     | 251   | 145.5 | 40 |
| - 75       | 75     | 314.5 | 180   | 44 |
| - 100      | 100    | 384.5 | 217   | 52 |
| - 125      | 125    | 452.5 | 256   | 56 |
| - 150      | 150    | 513.5 | 292   | 56 |



### Clevis Mount



### Clamp Mount



## Accessories

- Other accessories to be ordered separately

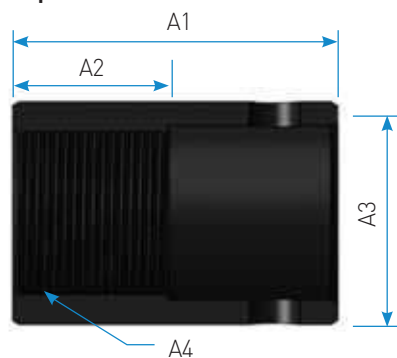
(unit : mm)

| Model      | ISA85 -25 | ISA85 -50   | ISA85 -75 | ISA85 -100 | ISA85 -125  | ISA85 -150  |
|------------|-----------|-------------|-----------|------------|-------------|-------------|
| A          | 256       | 316         | 381       | 451        | 519         | -           |
| B(Min/Max) | 25/79     | 25/114      | 25/150    | 25/187     | 25/226      | 25/262      |
| C(Min/Max) | 95.5/120  | 120.5/162.5 | 149/209.5 | 182.5/261  | 211.5/309.5 | 236.5/352.5 |

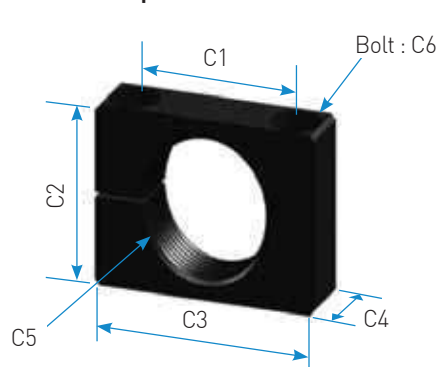


## ■ Accessories

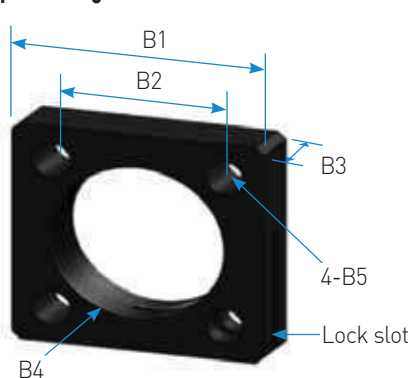
**Stop Collar**



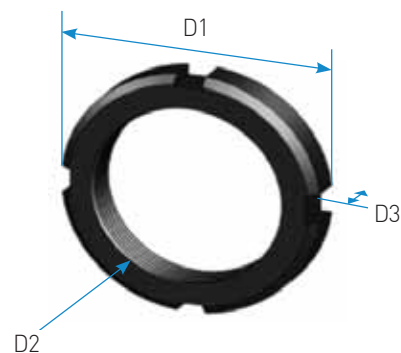
**Clamp Mount**



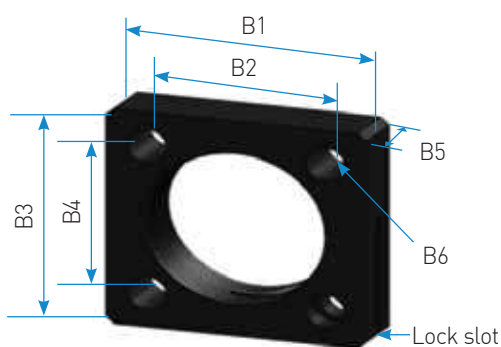
**Square Flange**



**Lock Collar**



**Rectangular Flange**



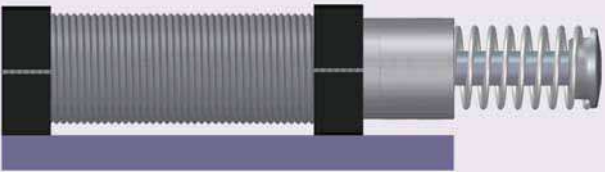
| Rectangular Flange | B1  | B2   | B3 | B4 | B5 | B6  |
|--------------------|-----|------|----|----|----|-----|
| RF45X1.5           | 78  | 60   | 60 | 41 | 14 | Ø9  |
| RF64X2.0           | 115 | 87.6 | 90 | 70 | 16 | Ø11 |

## ■ Dimensions

(unit : mm)

| Stop Collar | Square Flange | Clamp Mount | Lock Nut | A1 | A2 | A3  | A4         | B1   | B2  | B3 | B4         | B5  | C1  | C2  | C3  | C4 | C5         | C6         | D1   | D2         | D3 |
|-------------|---------------|-------------|----------|----|----|-----|------------|------|-----|----|------------|-----|-----|-----|-----|----|------------|------------|------|------------|----|
| SC45X1.5    | SF45X1.5      | CM45X1.5    | LN45X1.5 | 52 | 25 | Ø56 | M45 X 1.5P | □60  | □41 | 14 | M45 X 1.5P | Ø9  | 60  | 56  | 80  | 25 | M45 X 1.5P | M8 X 60L   | Ø58  | M45 X 1.5P | 9  |
| SC64X2.0    | SF64X2.0      | CM64X2.0    | LN64X2.0 | 85 | 45 | Ø75 | M64 X 2.0P | □90  | □70 | 16 | M64 X 2.0P | Ø11 | 78  | 80  | 100 | 25 | M64 X 2.0P | M10 X 80L  | Ø80  | M64 X 2.0P | 11 |
| SC85X2.0    | SF85X2.0      | CM85X2.0    | LN85X2.0 | 95 | 45 | Ø98 | M85 X 2.0P | □104 | □76 | 19 | M85 X 2.0P | Ø13 | 115 | 100 | 145 | 30 | M85 X 2.0P | M14 X 100L | Ø110 | M85 X 2.0P | 16 |



| NAME                      | Installation Drawing                                                                 |                                                                                                                                                     |
|---------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Lock Collar               |    | As a basic mounting way by using lock collar, shock absorber can be installed easily.                                                               |
| Stop Collar + Lock Collar |   | Stop collar makes it easy to stop precisely and easy to select location to set and it prevents troubles so called bottom-out problem in the piston. |
| Flange Mount              |  | Flange or rectangular flange can be used to fix the shock absorber conveniently.                                                                    |
| Clamp Mount               |  | Clamp mount is mainly used for horizontal surfaces and is useful when the length of the shock absorber is long.                                     |
| Clevis Mount              |  | It is mainly used in rotary motion and it is used to absorb shock indirectly by using clevis. Shock absorber can be protected from side loads.      |



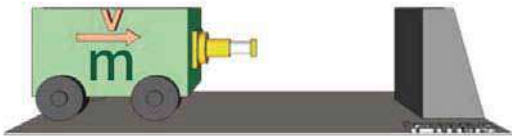
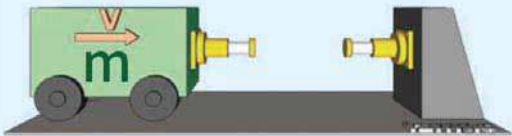
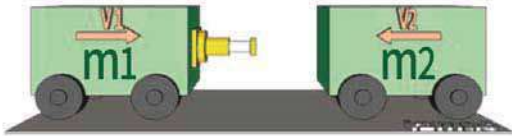
### Symbols

| Symbol         | Unit             | Description       | Symbol         | Unit               | Description                |
|----------------|------------------|-------------------|----------------|--------------------|----------------------------|
| m              | kg               | Weight            | r              | m                  | Radius of rotation         |
| m <sub>e</sub> | kg               | Designed/weight   | g              | m / s <sup>2</sup> | Gravitational acceleration |
| H              | m                | Height            | d              | m / s <sup>2</sup> | Deceleration               |
| S              | m                | Stroke            | E <sub>k</sub> | Nm                 | Kinetic energy             |
| V              | m / s            | Impact velocity   | E <sub>w</sub> | Nm                 | Work energy                |
| V <sub>E</sub> | m / s            | Designed velocity | E <sub>T</sub> | Nm                 | Total energy               |
| ω              | rad / s          | Angular velocity  | P              | kW                 | Motor power                |
| I              | Nms <sup>2</sup> | Moment of inertia | F <sub>s</sub> | N                  | Impact force               |
| T              | Nm               | Torque            | η              | s                  | Efficiency                 |

### Useful Formulas

|                     |                             |
|---------------------|-----------------------------|
| Maximum Shock Force | $F_s = E_T / S / 0.8 + F_D$ |
| Stroke              | $S = V_E^2 / 2 / d / 0.8$   |
| Deceleration        | $d = V_E^2 / 2 / S / 0.8$   |
| Deceleration Time   | $t = 2.5 \times S / V$      |

### Application

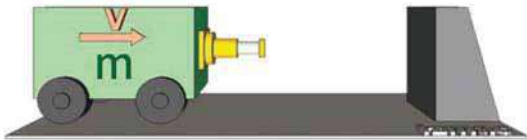
| Arrangement                                                                         | Design Speed (V <sub>E</sub> )  | Design Weight (M <sub>E</sub> )             |
|-------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------|
|  | V                               | m                                           |
|  | $\frac{V}{2}$                   | 2m                                          |
|  | V <sub>1</sub> + V <sub>2</sub> | $\frac{m_1 \times m_2}{m_1 + m_2}$          |
|  | $\frac{V_1 + V_2}{2}$           | $\frac{2 \times m_1 \times m_2}{m_1 + m_2}$ |



### Examples

#### A | Horizontal Mass without Propelling Force

- Weight  $m = 100\text{ton}$
- Impact velocity  $V = 0.5\text{m/s}$



$$E_k = \frac{m \times V^2}{2} = \frac{100 \times 0.5^2}{2} = 12.5 \text{ kNm}$$

Selected Model with  $E_k$  : IHG65-200

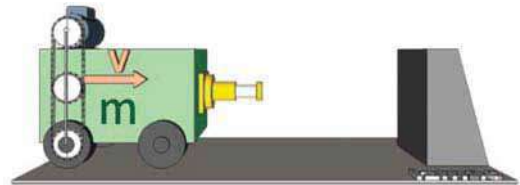
$$E_T = E_k + E_w = 12.5 + 0 = 12.5 \text{ kNm}$$

$$F_s = \frac{E_T}{S \times \eta} = \frac{12.5}{0.2 \times 0.8} = 78.13 \text{ kN}$$

Selected Model with  $E_T$  &  $F_s$  : IHG65-200

#### B | Horizontal Mass with Propelling Force [Motor]

- Weight  $m = 100\text{ton}$
- Impact velocity  $V = 1.5\text{m/s}$
- Motor Power  $P = 12\text{kW}$



$$E_k = \frac{m \times V^2}{2} = \frac{100 \times 1.5^2}{2} = 112.5 \text{ kNm}$$

$$F_p = \frac{2.5P}{V} = \frac{2.5 \times 12}{1.5} = 20 \text{ kN}$$

Selected Model with  $E_k$  : IHG120-400

$$E_w = F_p \times S = 20 \times 0.4 = 8 \text{ kNm}$$

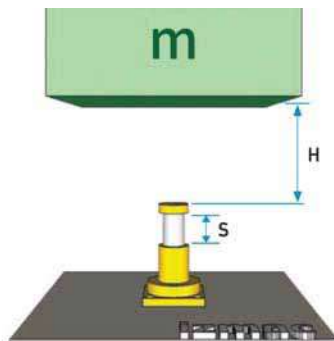
$$E_T = E_k + E_w = 112.5 + 8 = 120.5 \text{ kNm}$$

$$F_s = \frac{E_T}{S \times \eta} = \frac{120.5}{0.4 \times 0.8} = 376.6 \text{ kN}$$

Selected Model with  $E_T$  &  $F_s$  : IHG120-400

#### C | Free Falling Mass

- Weight  $m = 4\text{ton}$
- Height  $H = 0.3\text{m}$



$$E_k = m \times g \times H = 4 \times 9.81 \times 0.3 = 11.772 \text{ kNm}$$

Selected Model with  $E_k$  : IHG100-10

$$E_w = m \times g \times S = 4 \times 9.81 \times 0.1 = 3.924 \text{ kNm}$$

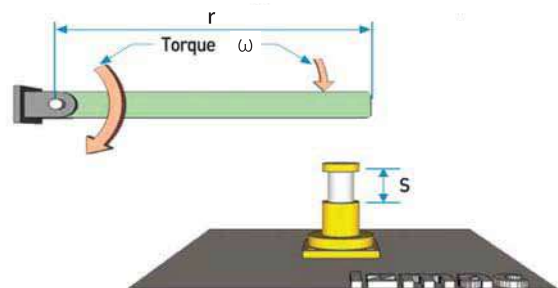
$$E_T = E_k + E_w = 11.772 + 3.924 = 15.696 \text{ kNm}$$

$$F_s = \frac{E_T}{S \times \eta} = \frac{15.696}{0.1 \times 0.8} = 196.2 \text{ kN}$$

Selected Model with  $E_T$  &  $F_s$  : IHG100-100

#### D | Swiveling Mass with Propelling Force

- Torque  $T = 200\text{Nm}$
- Impact velocity  $\omega = 2\text{rad/s}$
- Radius of gyration  $r = 8\text{m}$
- Mass moment of inertia  $I = 35\text{Nms}^2$
- Stroke  $S = 0.2\text{m}$



$$E_k = \frac{I \times \omega^2}{2} = \frac{35 \times 2^2}{2} = 70 \text{ kNm}$$

Selected Model with  $E_k$  : IHG120-200

$$E_w = \frac{T \times S}{r} = \frac{200 \times 0.2}{8} = 5 \text{ kNm}$$

$$E_T = E_k + E_w = 70 + 5 = 75 \text{ kNm}$$

$$F_s = \frac{E_T}{S \times \eta} = \frac{75}{0.2 \times 0.8} = 468.75 \text{ kN}$$

Selected Model with  $E_T$  &  $F_s$  : IHG120-200

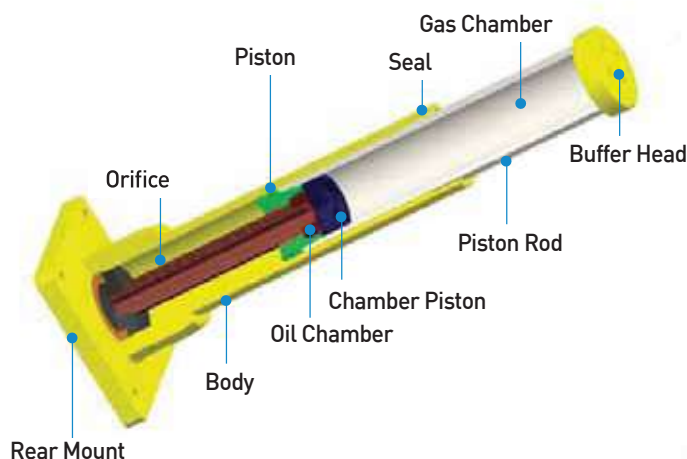


# IHG Series

## IZMAC Hydraulic Gas & Oil type Buffer

### DESCRIPTION

IHG is a heavy-duty application product that has a high impact energy absorption capacity compared to its size. Operation : When an object collides, the piston enters the oil chamber by the tube shape piston rod, and the oil flows through the orifice of the throttle port and finally oil is pushing floating piston up to compress gas. In this process, flow resistance power arouse and gas pressure takes a role of accumulator which compensating piston rod space and compressed gas has the role of returning piston rod to its original position when the load is released.



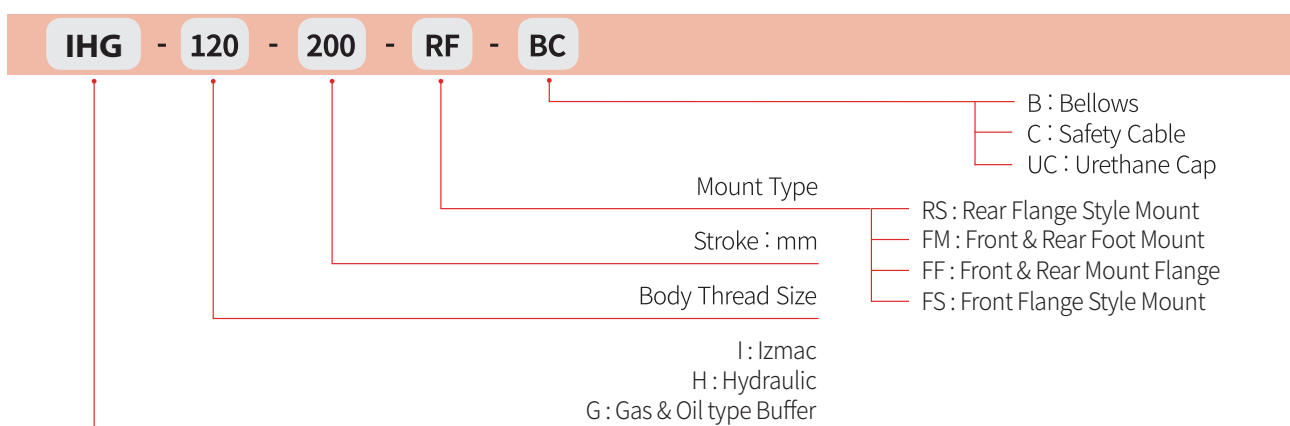
### FEATURES

- 1 IHG is high capable efficient energy hydraulic buffer.
- 2 Customizing orifice
- 3 Fulfilled international standards : OSHA, AIST, CMMA, DIN, FEM etc.
- 4 Operation temperature : -10 ~ 80°C • Special : -40 ~ 120°C
- 5 Piton rod surface treatment : Hard Chrome Plated
- 6 Body surface treatment : Epoxy Painting

### APPLICATION

Container crane, overhead crane, stacker crane, transfer car, rail end stop, heavy industry, steel mill

### IHG SERIES ORDERING INFORMATION

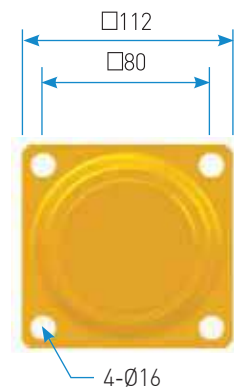
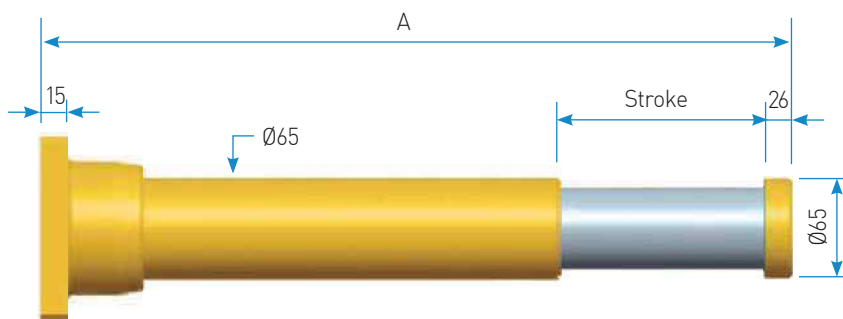




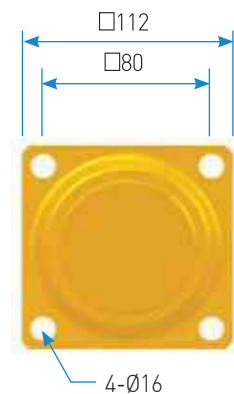
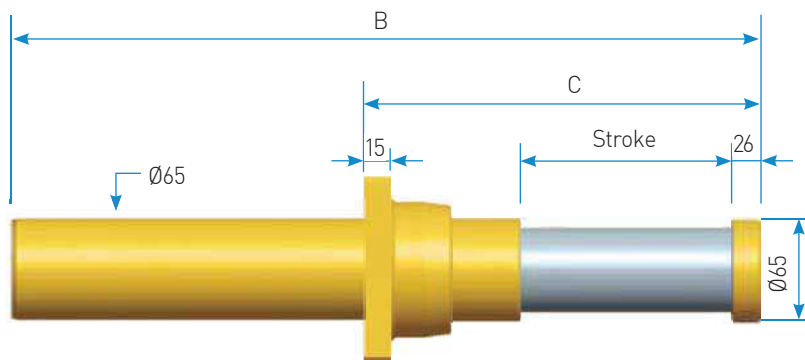
## ■ Engineering Data

| Model      | Stroke<br>(mm) | Max.Energy /<br>Cycle(kJ) $E_T$ | Max.Buffer Force<br>(kN) $F_s$ | Recoil Force (kN) |       | Weight(kg) |
|------------|----------------|---------------------------------|--------------------------------|-------------------|-------|------------|
|            |                |                                 |                                | Ext.              | Comp. |            |
| IHG65 - 50 | 50             | 3.8                             | 95                             | 1.0               | 3.5   | 6          |
| - 75       | 75             | 5.7                             | 95                             | 1.0               | 3.5   | 8          |
| - 100      | 100            | 7.5                             | 95                             | 1.0               | 3.5   | 9          |
| - 150      | 150            | 11                              | 95                             | 1.0               | 3.5   | 11         |
| - 200      | 200            | 14.7                            | 95                             | 1.0               | 3.5   | 12         |

### Rear Mount



### Front Mount



## ■ Dimensions

(unit : mm)

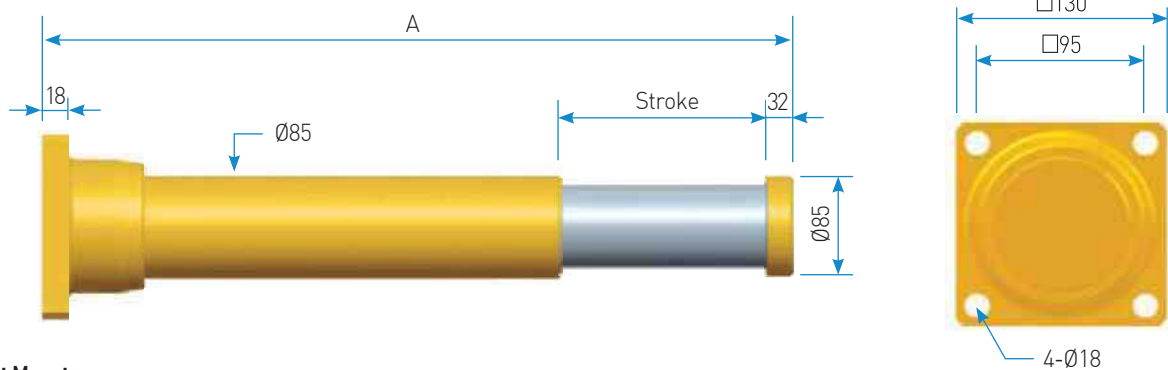
| Model      | Stroke (mm) | Rear Type (mm) | Front Type (mm) |     | Mounting Bolt Size |
|------------|-------------|----------------|-----------------|-----|--------------------|
|            |             | A              | B               | C   |                    |
| IHG65 - 50 | 50          | 312            | 300             | 151 | 14                 |
| - 75       | 75          | 372            | 360             | 176 |                    |
| - 100      | 100         | 432            | 420             | 231 |                    |
| - 150      | 150         | 552            | 540             | 281 |                    |
| - 200      | 200         | 682            | 670             | 371 |                    |

# IHG85 Series

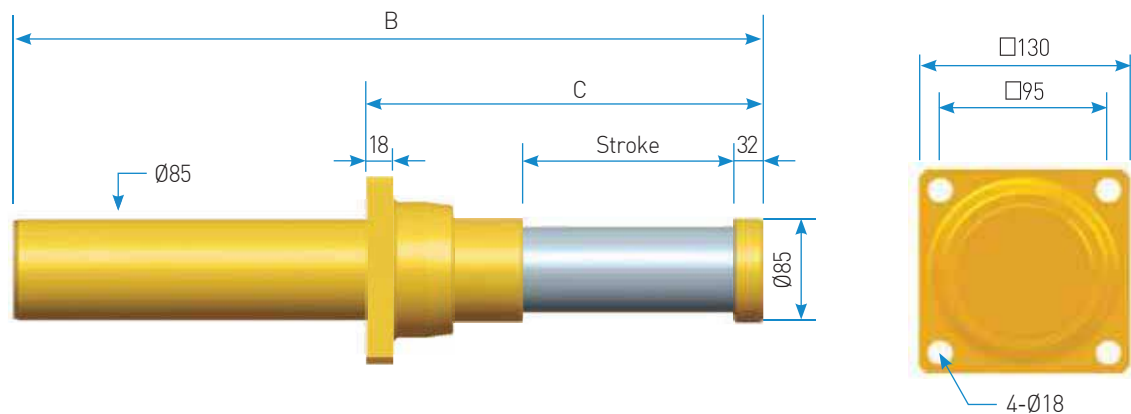
## ■ Engineering Data

| Model      | Stroke (mm) | Max.Energy / Cycle(kJ) $E_T$ | Max.Buffer Force (kN) $F_s$ | Recoil Force (kN) |       | Weight(kg) |
|------------|-------------|------------------------------|-----------------------------|-------------------|-------|------------|
|            |             |                              |                             | Ext.              | Comp. |            |
| IHG85 - 50 | 50          | 7.5                          | 188                         | 1.5               | 13    | 12         |
| - 100      | 100         | 15                           | 188                         | 1.5               | 13    | 14         |
| - 150      | 150         | 22.5                         | 188                         | 1.5               | 20    | 17         |
| - 200      | 200         | 30                           | 188                         | 1.5               | 20    | 20         |
| - 250      | 250         | 37.5                         | 188                         | 1.5               | 20    | 22         |

### Rear Mount



### Front Mount



## ■ Dimensions

(unit : mm)

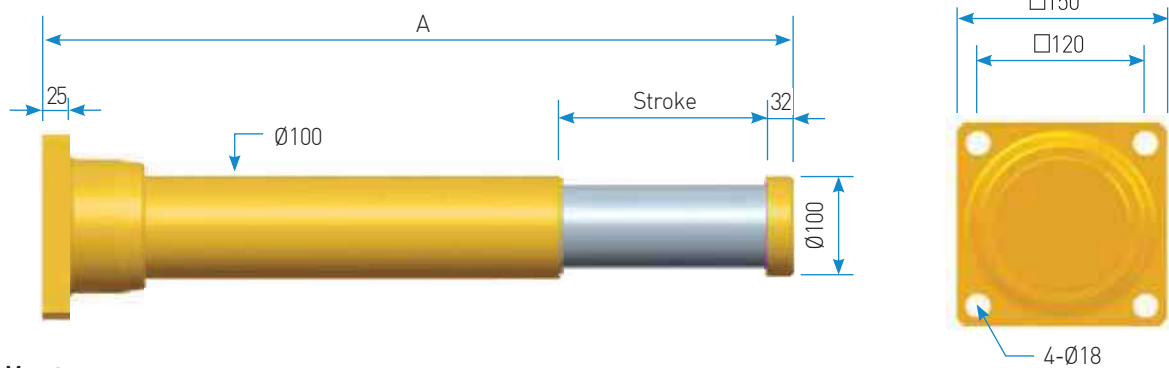
| Model      | Stroke (mm) | Rear Type (mm) | Front Type (mm) |     | Mounting Bolt Size |
|------------|-------------|----------------|-----------------|-----|--------------------|
|            |             | A              | B               | C   |                    |
| IHG85 - 50 | 50          | 323            | 310             | 183 | 16                 |
| - 100      | 100         | 463            | 450             | 242 |                    |
| - 150      | 150         | 603            | 590             | 305 |                    |
| - 200      | 200         | 743            | 730             | 367 |                    |
| - 250      | 250         | 883            | 870             | 430 |                    |



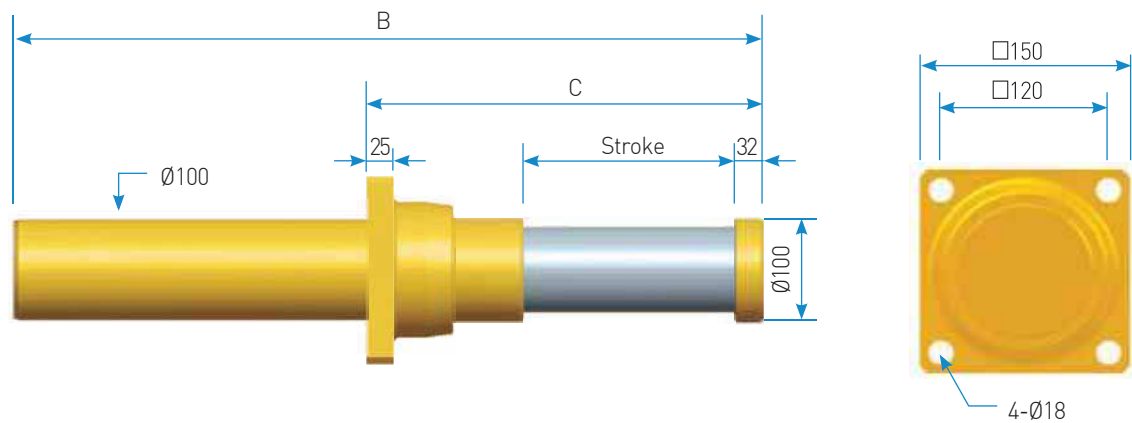
## Engineering Data

| Model       | Stroke (mm) | Max.Energy / Cycle(kJ) $E_T$ | Max.Buffer Force (kN) $F_s$ | Recoil Force (kN) |       | Weight(kg) |
|-------------|-------------|------------------------------|-----------------------------|-------------------|-------|------------|
|             |             |                              |                             | Ext.              | Comp. |            |
| IHG100 - 80 | 80          | 16                           | 250                         | 2.4               | 16    | 20         |
| - 100       | 100         | 20                           | 250                         | 2.4               | 16    | 25         |
| - 150       | 150         | 30                           | 250                         | 2.4               | 20    | 28         |
| - 200       | 200         | 40                           | 250                         | 2.4               | 20    | 34         |
| - 250       | 250         | 50                           | 250                         | 2.4               | 25    | 39         |
| - 300       | 300         | 60                           | 250                         | 2.4               | 25    | 43         |

### Rear Mount



### Front Mount



## Dimensions

(unit : mm)

| Model       | Stroke (mm) | Rear Type (mm) | Front Type (mm) |     | Mounting Bolt Size |
|-------------|-------------|----------------|-----------------|-----|--------------------|
|             |             | A              | B               | C   |                    |
| IHG100 - 80 | 80          | 423            | 403             | 215 | 16                 |
| - 100       | 100         | 450            | 430             | 252 |                    |
| - 150       | 150         | 580            | 560             | 315 |                    |
| - 200       | 200         | 720            | 700             | 377 |                    |
| - 250       | 250         | 865            | 845             | 440 |                    |
| - 300       | 300         | 1010           | 990             | 502 |                    |

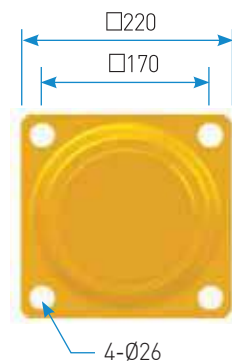
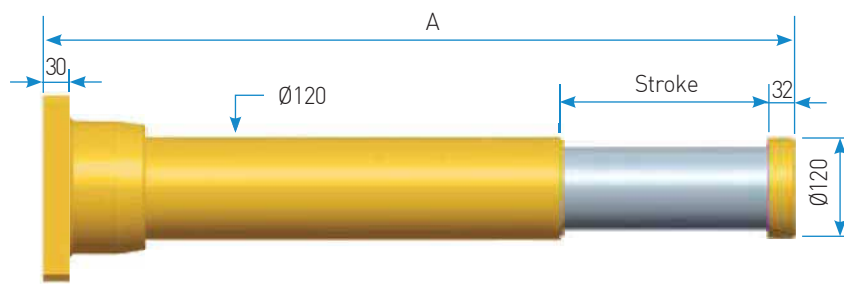


# IHG120 Series

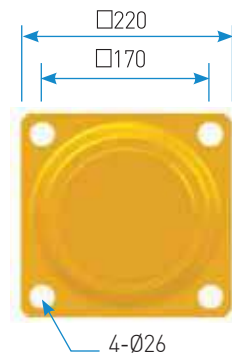
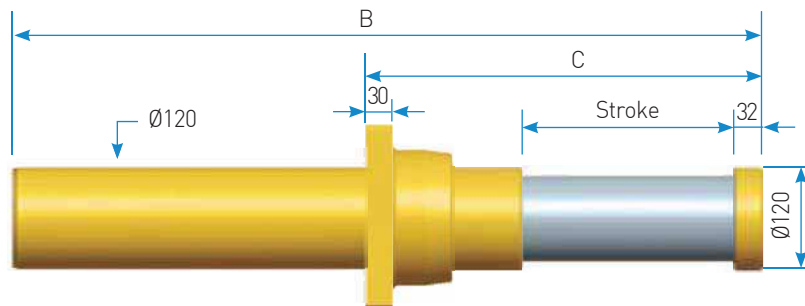
## ■ Engineering Data

| Model        | Stroke (mm) | Max.Energy / Cycle(kJ) $E_T$ | Max.Buffer Force (kN) $F_s$ | Recoil Force (kN) |       | Weight(kg) |
|--------------|-------------|------------------------------|-----------------------------|-------------------|-------|------------|
|              |             |                              |                             | Ext.              | Comp. |            |
| IHG120 - 100 | 100         | 40                           | 500                         | 3.5               | 40    | 41         |
| - 150        | 150         | 60                           | 500                         | 3.5               | 40    | 48         |
| - 200        | 200         | 80                           | 500                         | 3.5               | 40    | 58         |
| - 250        | 250         | 100                          | 500                         | 3.5               | 40    | 65         |
| - 300        | 300         | 120                          | 400                         | 3.5               | 40    | 72         |
| - 400        | 400         | 160                          | 400                         | 3.5               | 40    | 78         |
| - 500        | 500         | 180                          | 400                         | 3.5               | 40    | 86         |
| - 600        | 600         | 200                          | 400                         | 3.5               | 40    | 95         |
| - 800        | 800         | 240                          | 375                         | 3.5               | 40    | 112        |
| - 1000       | 1000        | 280                          | 350                         | 3.5               | 40    | 118        |

Rear Mount



Front Mount



## ■ Dimensions

(unit : mm)

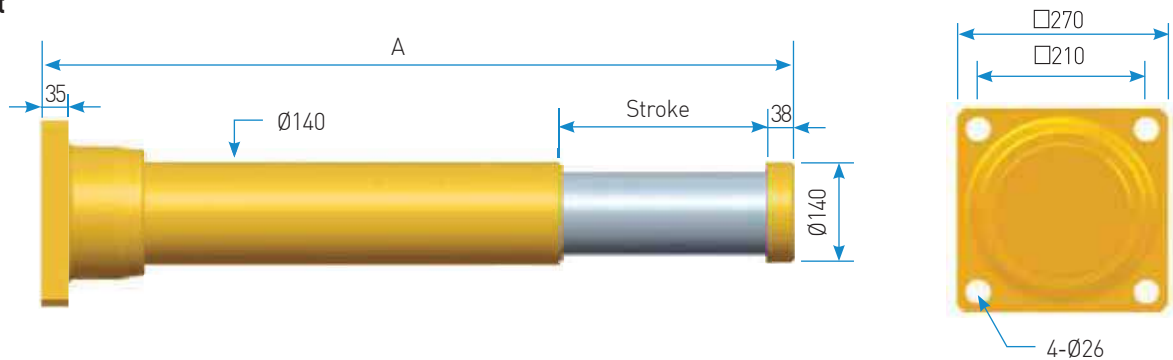
| Model        | Stroke (mm) | Rear Type (mm) | Front Type (mm) |      | Mounting Bolt Size |
|--------------|-------------|----------------|-----------------|------|--------------------|
|              |             | A              | B               | C    |                    |
| IHG120 - 100 | 100         | 470            | 450             | 277  | 24                 |
| - 150        | 150         | 610            | 590             | 340  |                    |
| - 200        | 200         | 760            | 740             | 402  |                    |
| - 250        | 250         | 900            | 880             | 465  |                    |
| - 300        | 300         | 1050           | 1030            | 527  |                    |
| - 400        | 400         | 1340           | 1320            | 680  |                    |
| - 500        | 500         | 1620           | 1600            | 815  |                    |
| - 600        | 600         | 1920           | 1900            | 950  |                    |
| - 800        | 800         | -              | 2400            | 1290 |                    |
| - 1000       | 1000        | -              | 2960            | 1360 |                    |



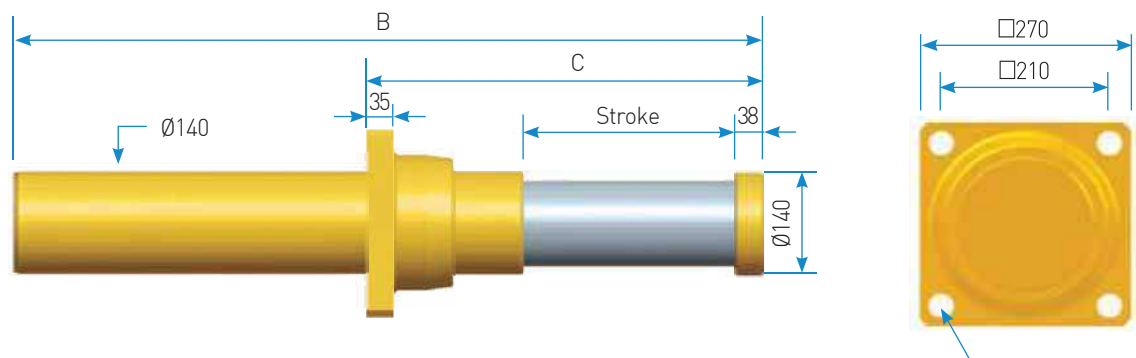
## Engineering Data

| Model        | Stroke (mm) | Max.Energy / Cycle(kJ) $E_T$ | Max.Buffer Force (kN) $F_S$ | Recoil Force (kN) |       | Weight(kg) |
|--------------|-------------|------------------------------|-----------------------------|-------------------|-------|------------|
|              |             |                              |                             | Ext.              | Comp. |            |
| IHG140 - 100 | 100         | 55                           | 688                         | 6                 | 70    | 60         |
| - 200        | 200         | 110                          | 688                         | 6                 | 70    | 85         |
| - 300        | 300         | 165                          | 688                         | 6                 | 70    | 110        |
| - 400        | 400         | 220                          | 688                         | 6                 | 78    | 135        |
| - 500        | 500         | 270                          | 680                         | 6                 | 78    | 150        |
| - 600        | 600         | 300                          | 630                         | 6                 | 78    | 160        |
| - 800        | 800         | 325                          | 510                         | 6                 | 78    | 185        |
| - 1000       | 1000        | 360                          | 450                         | 6                 | 78    | 200        |

### Rear Mount



### Front Mount



## Dimensions

(unit : mm)

| Model        | Stroke (mm) | Rear Type (mm) | Front Type (mm) |      | Mounting Bolt Size |
|--------------|-------------|----------------|-----------------|------|--------------------|
|              |             | A              | B               | C    |                    |
| IHG140 - 100 | 100         | 480            | 460             | 297  | 24                 |
| - 200        | 200         | 770            | 750             | 422  |                    |
| - 300        | 300         | 1060           | 1040            | 547  |                    |
| - 400        | 400         | 1350           | 1330            | 712  |                    |
| - 500        | 500         | 1630           | 1610            | 847  |                    |
| - 600        | 600         | 1930           | 1910            | 982  |                    |
| - 800        | 800         | 2350           | 2330            | 1252 |                    |
| - 1000       | 1000        | -              | 2880            | 1595 |                    |

# ICA Series Stacker Crane Buffers

NEW

## IZMAC Hydraulic & Oil Stacker Crane Adjustable Buffer

### DESCRIPTION

ICA is newly launched state-of-the-art dial type buffer which is engineered to select damping forces against wide application conditions. Engineered to maintain rather low peak figures & rebounding forces. Therefore it can be operated by lowest rebounding force and decelerate softly in emergency stop conditions. Used for mainly stacker crane application in automated logistic system, dial type has models upto 300mm stroke and maximum energy 376 kJ, there are front dial control type & back dial control type according to the dial location.

### FEATURES

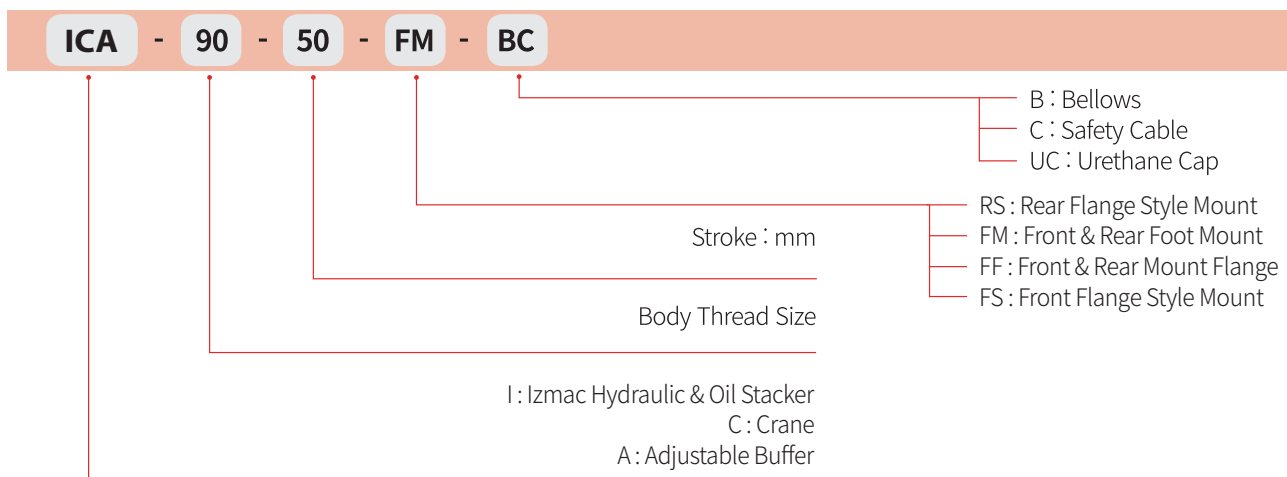
- 1 Customized orifice
- 2 Piston rod surface treatment : Hardened, Hard chrome plated
- 3 Body surface treatment : White zinc plated
- 4 Operation temperature : -10 ~ 80°C • Special : -40 ~ 120°C
- 5 Fullfilled international standards : OSHA, AIST, CMAA, DIN, FEM etc.
- 6 Option : Urethane cap, Safety cable, Mounting plates, Adjustment dial position

### APPLICATION

Automatic warehouse stacker crane, logistic automation system, amusement park, STS crane, production facilities.



### ICA SERIES ORDERING INFORMATION



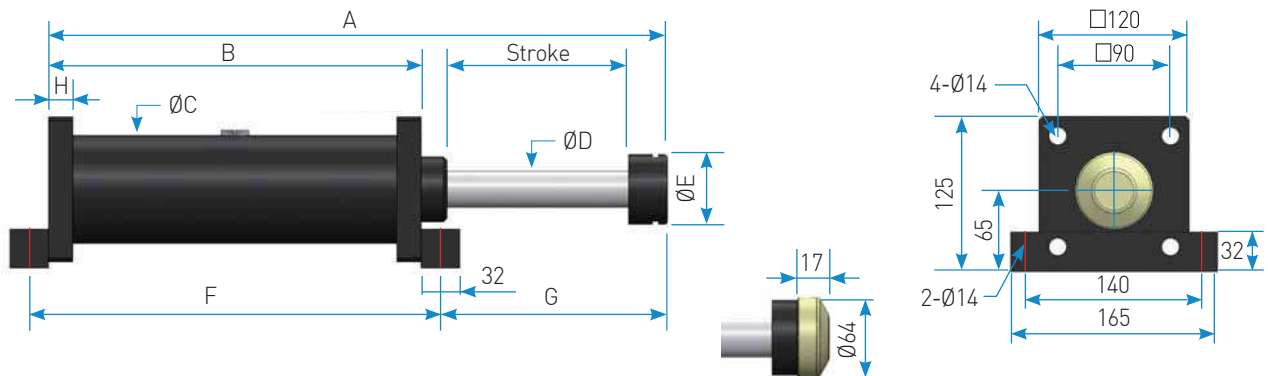
# ICA90 Series

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technology



## Engineering Data

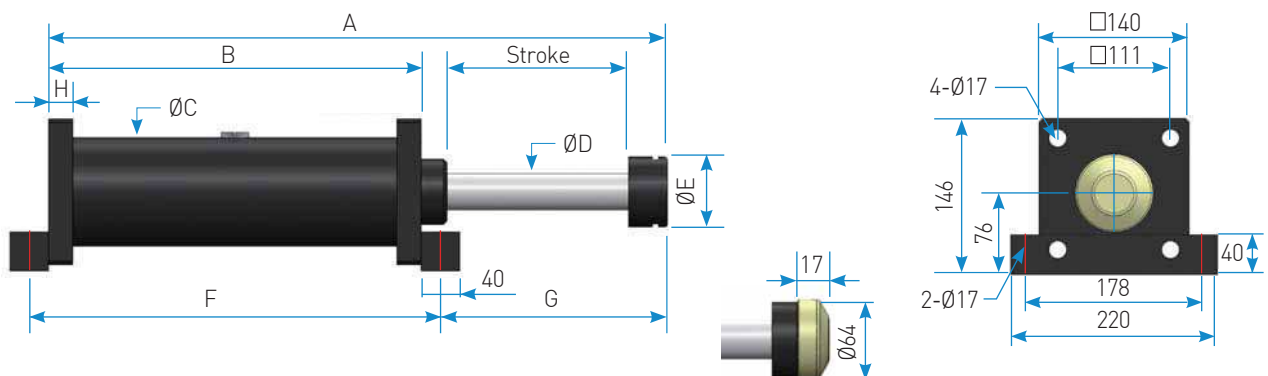
| Model      | Stroke (mm) | Max. Energy / Cycle(kJ) $E_T$ | Max.Buffer Force (kN) $F_s$ | Dimension [unit:mm] |     |    |    |    |     |     |    |
|------------|-------------|-------------------------------|-----------------------------|---------------------|-----|----|----|----|-----|-----|----|
|            |             |                               |                             | A                   | B   | C  | D  | E  | F   | G   | H  |
| ICA90 - 50 | 50          | 3                             | 77                          | 300                 | 218 | 90 | 30 | 58 | 250 | 66  | 20 |
| - 100      | 100         | 6                             | 77                          | 400                 | 268 | 90 | 30 | 58 | 300 | 116 | 20 |
| - 150      | 150         | 9                             | 77                          | 500                 | 318 | 90 | 30 | 58 | 350 | 166 | 20 |
| - 200      | 200         | 12                            | 77                          | 600                 | 368 | 90 | 30 | 58 | 400 | 216 | 20 |
| - 250      | 250         | 15                            | 77                          | 700                 | 418 | 90 | 30 | 58 | 450 | 266 | 20 |
| - 300      | 300         | 18                            | 77                          | 800                 | 468 | 90 | 30 | 58 | 500 | 316 | 20 |



# ICA110 Series

## Engineering Data

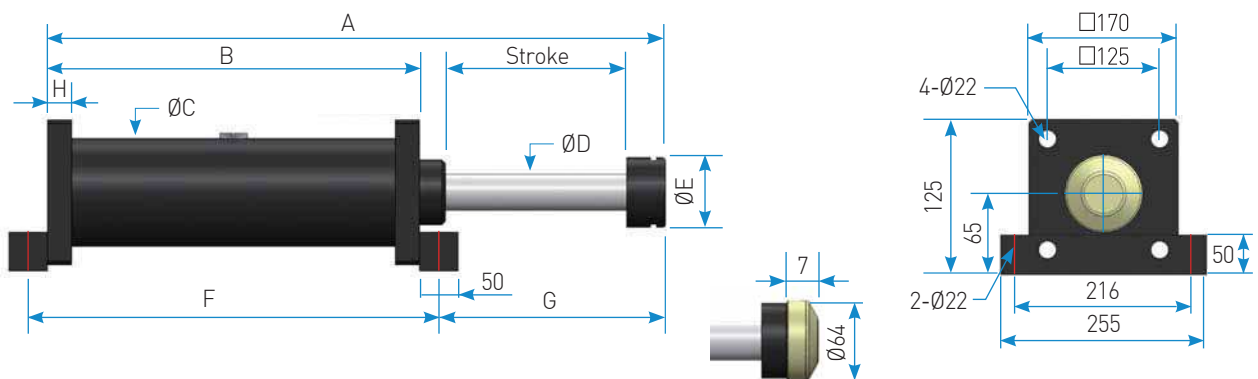
| Model       | Stroke (mm) | Max. Energy / Cycle(kJ) $E_T$ | Max.Buffer Force (kN) $F_s$ | Dimension [unit:mm] |     |     |    |    |     |     |    |
|-------------|-------------|-------------------------------|-----------------------------|---------------------|-----|-----|----|----|-----|-----|----|
|             |             |                               |                             | A                   | B   | C   | D  | E  | F   | G   | H  |
| ICA110 - 50 | 50          | 4                             | 114                         | 320                 | 230 | 110 | 35 | 58 | 270 | 70  | 25 |
| - 100       | 100         | 9                             | 114                         | 420                 | 280 | 110 | 35 | 58 | 320 | 120 | 25 |
| - 150       | 150         | 13                            | 114                         | 520                 | 330 | 110 | 35 | 58 | 370 | 170 | 25 |
| - 200       | 200         | 18                            | 114                         | 620                 | 380 | 110 | 35 | 58 | 420 | 220 | 25 |
| - 250       | 250         | 23                            | 114                         | 720                 | 430 | 110 | 35 | 58 | 470 | 270 | 25 |
| - 300       | 300         | 27                            | 114                         | 820                 | 480 | 110 | 35 | 58 | 520 | 320 | 25 |



# ICA130 Series

## Engineering Data

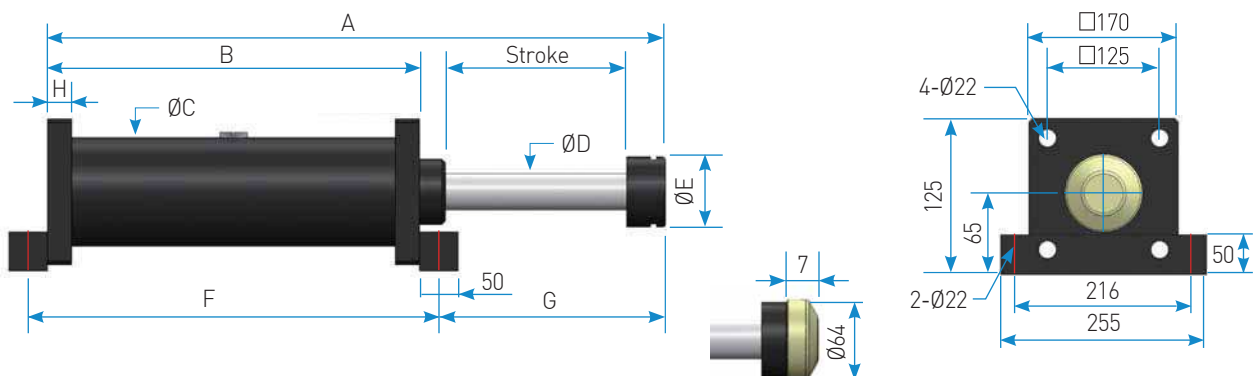
| Model       | Stroke (mm) | Max. Energy / Cycle(kJ) $E_T$ | Max.Buffer Force (kN) $F_s$ | Dimension [unit:mm] |     |     |    |    |     |     |    |
|-------------|-------------|-------------------------------|-----------------------------|---------------------|-----|-----|----|----|-----|-----|----|
|             |             |                               |                             | A                   | B   | C   | D  | E  | F   | G   | H  |
| ICA130 - 50 | 50          | 8                             | 210                         | 350                 | 250 | 138 | 45 | 58 | 300 | 75  | 25 |
| - 75        | 75          | 16                            | 210                         | 450                 | 300 | 138 | 45 | 58 | 350 | 125 | 25 |
| - 125       | 125         | 25                            | 210                         | 550                 | 350 | 138 | 45 | 58 | 400 | 175 | 25 |
| - 200       | 200         | 33                            | 210                         | 650                 | 400 | 138 | 45 | 58 | 450 | 225 | 25 |
| - 250       | 250         | 42                            | 210                         | 750                 | 450 | 138 | 45 | 58 | 500 | 275 | 25 |
| - 300       | 300         | 50                            | 210                         | 850                 | 500 | 138 | 45 | 58 | 550 | 325 | 25 |



# ICA160 Series

## Engineering Data

| Model       | Stroke (mm) | Max. Energy / Cycle(kJ) $E_T$ | Max.Buffer Force (kN) $F_s$ | Dimension [unit:mm] |     |     |    |    |     |     |    |
|-------------|-------------|-------------------------------|-----------------------------|---------------------|-----|-----|----|----|-----|-----|----|
|             |             |                               |                             | A                   | B   | C   | D  | E  | F   | G   | H  |
| ICA160 - 50 | 50          | 10                            | 267                         | 380                 | 280 | 160 | 50 | 98 | 330 | 75  | 25 |
| - 100       | 100         | 21                            | 267                         | 480                 | 330 | 160 | 50 | 98 | 380 | 125 | 25 |
| - 150       | 150         | 32                            | 267                         | 580                 | 380 | 160 | 50 | 98 | 430 | 175 | 25 |
| - 200       | 200         | 42                            | 267                         | 680                 | 430 | 160 | 50 | 98 | 480 | 225 | 25 |
| - 250       | 250         | 53                            | 267                         | 780                 | 480 | 160 | 50 | 98 | 530 | 275 | 25 |
| - 300       | 300         | 64                            | 267                         | 880                 | 530 | 160 | 50 | 98 | 580 | 325 | 25 |





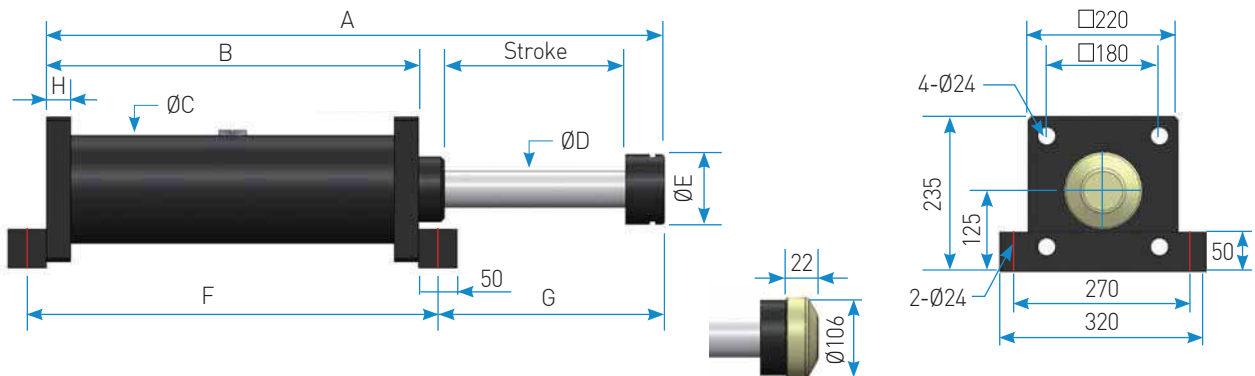
# ICA180 Series

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## Engineering Data

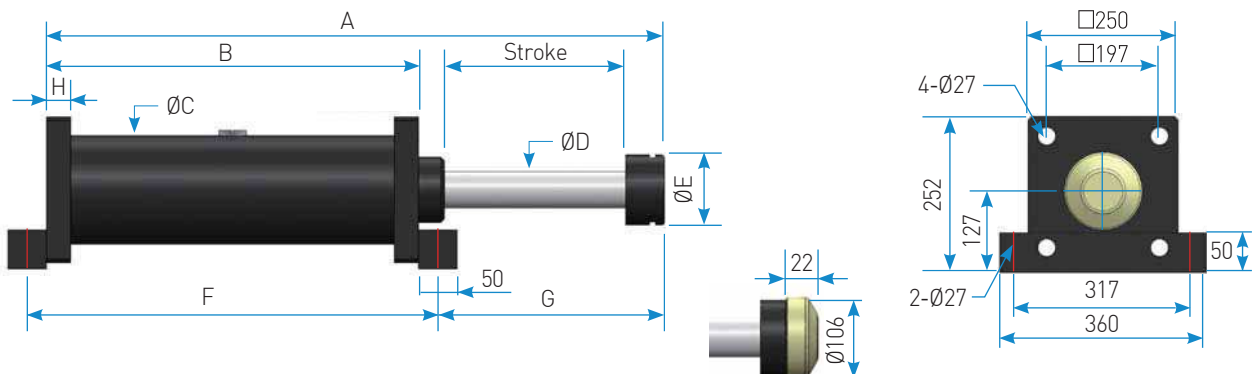
| Model       | Stroke (mm) | Max. Energy / Cycle(kJ) $E_T$ | Max.Buffer Force (kN) $F_s$ | Dimension [unit:mm] |     |     |    |    |     |     |    |
|-------------|-------------|-------------------------------|-----------------------------|---------------------|-----|-----|----|----|-----|-----|----|
|             |             |                               |                             | A                   | B   | C   | D  | E  | F   | G   | H  |
| ICA180 - 50 | 50          | 13                            | 338                         | 440                 | 340 | 180 | 55 | 98 | 390 | 75  | 40 |
| - 100       | 100         | 27                            | 338                         | 540                 | 390 | 180 | 55 | 98 | 440 | 125 | 40 |
| - 150       | 150         | 40                            | 338                         | 640                 | 440 | 180 | 55 | 98 | 490 | 175 | 40 |
| - 200       | 200         | 54                            | 338                         | 740                 | 490 | 180 | 55 | 98 | 540 | 225 | 40 |
| - 250       | 250         | 67                            | 338                         | 840                 | 540 | 180 | 55 | 98 | 590 | 275 | 40 |
| - 300       | 300         | 81                            | 338                         | 940                 | 590 | 180 | 55 | 98 | 640 | 325 | 40 |



# ICA200 Series

## Engineering Data

| Model       | Stroke (mm) | Max. Energy / Cycle(kJ) $E_T$ | Max.Buffer Force (kN) $F_s$ | Dimension [unit:mm] |     |     |    |    |     |     |    |
|-------------|-------------|-------------------------------|-----------------------------|---------------------|-----|-----|----|----|-----|-----|----|
|             |             |                               |                             | A                   | B   | C   | D  | E  | F   | G   | H  |
| ICA200 - 50 | 50          | 15                            | 376                         | 450                 | 350 | 200 | 65 | 98 | 400 | 75  | 40 |
| - 100       | 100         | 30                            | 376                         | 550                 | 400 | 200 | 65 | 98 | 450 | 125 | 40 |
| - 150       | 150         | 45                            | 376                         | 650                 | 450 | 200 | 65 | 98 | 500 | 175 | 40 |
| - 200       | 200         | 60                            | 376                         | 750                 | 500 | 200 | 65 | 98 | 550 | 225 | 40 |
| - 250       | 250         | 75                            | 376                         | 850                 | 550 | 200 | 65 | 98 | 600 | 275 | 40 |
| - 300       | 300         | 90                            | 376                         | 950                 | 600 | 200 | 65 | 98 | 650 | 325 | 40 |



# ISC Series Stacker Crane Buffers

## IZMAC Hydraulic & Oil Stacker Crane Buffer

### DESCRIPTION

ISC model max energy capacity is upto 915kJm, and it's longest stroke is 1,200mm and it is mainly applied for automatic logistic warehouse system(AS/RS). Engineered to maintain rather Low Peak figures & Low Recoil Force figures. Therefore it can be operated by lowest rebounding force and decelerate softly in emergency stop conditions.

### FEATURES

- 1 Custom orifice
- 2 Piston rod : Hardened, hard chrome plated
- 3 Cylinder : Zinc plated
- 4 Operation temperature : -10 ~ 80°C • Special : -40 ~ 120°C
- 5 Fullfilled international standards : OSHA, AIST, CMAA, DIN, FEM etc.
- 6 Option : Urethane cap, Safety cable, Mounting plates

### APPLICATION

Automatic warehouse system(AS/RS), Theme park, Stacker crane, Automobile assembly line, Overhead crane



### ISC SERIES ORDERING INFORMATION

ISC - 90 - 250 - FM - BC

B : Bellows  
C : Safety Cable  
UC : Urethane Cap

RS : Rear Flange Style Mount  
FM : Front & Rear Foot Mount  
FF : Front & Rear Mount Flange  
FS : Front Flange Style Mount

Stroke(mm)

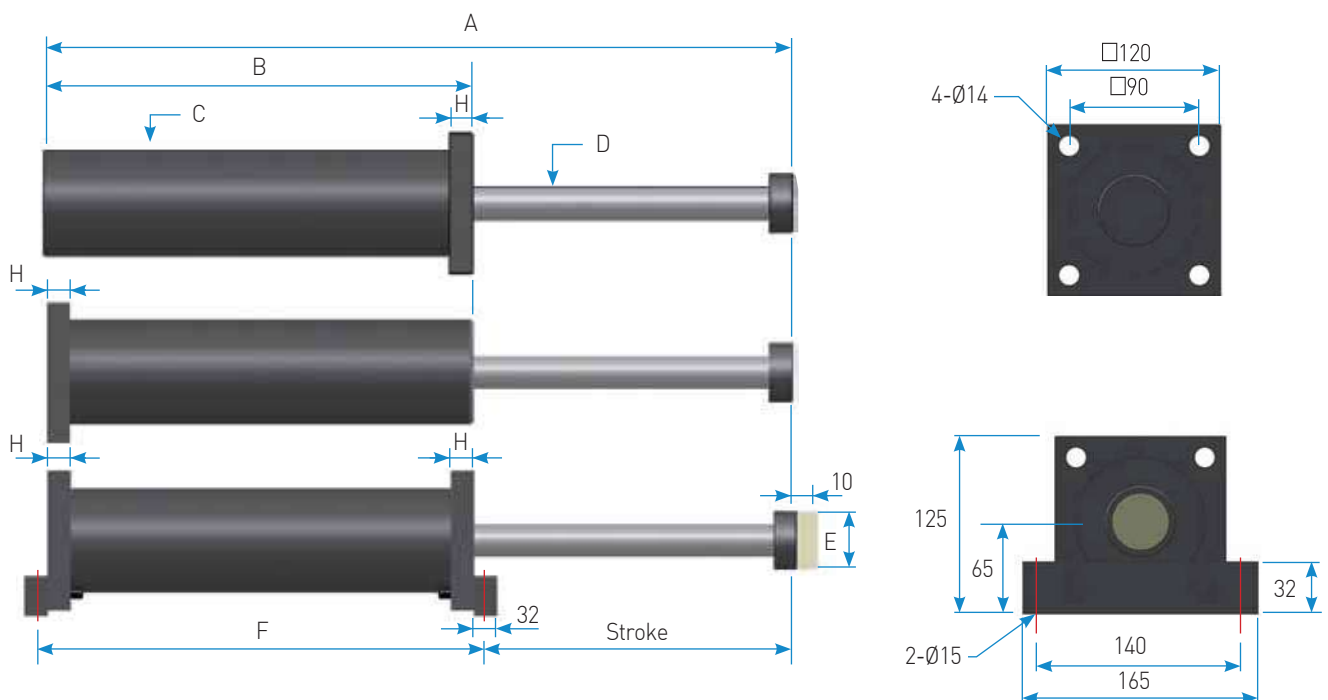
Body Thread Size

I : Izmac Hydraulic & Oil  
S : Stacker  
C : Crane Buffer



## ■ Engineering Data & Dimensions

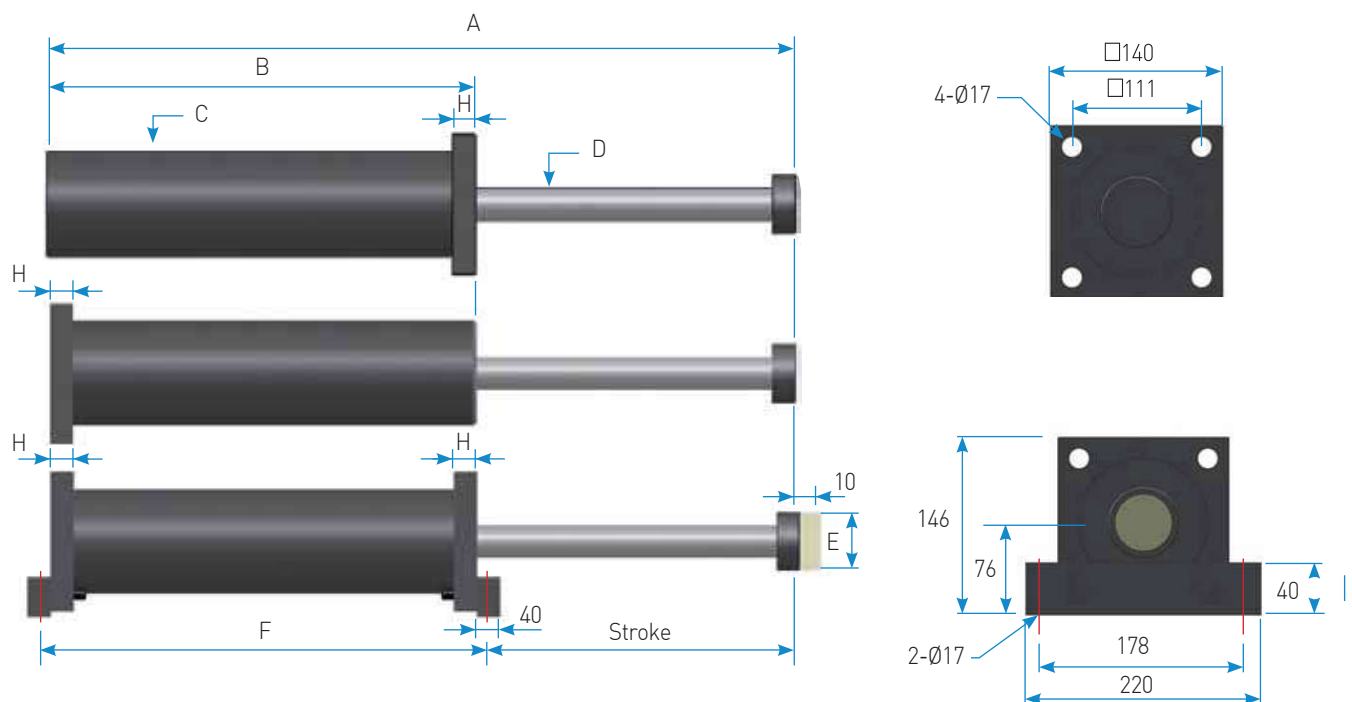
| Model      | Stroke<br>(mm) | Max. Energy /<br>Cycle(kJ)<br>$E_T$ | Max. Energy /<br>Hour(kJ/hr)<br>$E_{TC}$ | Max.Buffer<br>Force(kN)<br>$F_s$ | Dimension [unit:mm] |       |    |    |    |       |     |    |
|------------|----------------|-------------------------------------|------------------------------------------|----------------------------------|---------------------|-------|----|----|----|-------|-----|----|
|            |                |                                     |                                          |                                  | A                   | B     | C  | D  | E  | F     | G   | H  |
| ISC90 - 50 | 50             | 4                                   | 190                                      | 75                               | 310                 | 208   | 90 | 30 | 50 | 240   | 86  | 20 |
| - 100      | 100            | 7                                   | 390                                      | 75                               | 410                 | 258   | 90 | 30 | 50 | 290   | 136 | 20 |
| - 150      | 150            | 10                                  | 580                                      | 75                               | 510                 | 308   | 90 | 30 | 50 | 340   | 186 | 20 |
| - 200      | 200            | 13                                  | 780                                      | 75                               | 613                 | 360   | 90 | 30 | 50 | 392   | 237 | 20 |
| - 250      | 250            | 16                                  | 830                                      | 75                               | 715                 | 411   | 90 | 30 | 50 | 443   | 288 | 20 |
| - 300      | 300            | 20                                  | 940                                      | 75                               | 817                 | 462   | 90 | 30 | 50 | 496   | 339 | 20 |
| - 350      | 350            | 23                                  | 1,260                                    | 75                               | 918                 | 512   | 90 | 30 | 50 | 544   | 390 | 20 |
| - 400      | 400            | 21                                  | 1,150                                    | 67                               | 1,019               | 563   | 90 | 30 | 50 | 595   | 440 | 20 |
| - 450      | 450            | 20                                  | 1,090                                    | 55                               | 1,121               | 614   | 90 | 30 | 50 | 646   | 491 | 20 |
| - 500      | 500            | 19                                  | 1,060                                    | 47                               | 1,223               | 665   | 90 | 30 | 50 | 697   | 542 | 20 |
| - 600      | 600            | 15                                  | 880                                      | 31                               | 1,427               | 767   | 90 | 30 | 50 | 799   | 644 | 20 |
| - 700      | 700            | 13                                  | 610                                      | 24                               | 1,668               | 910   | 90 | 30 | 50 | 956   | 742 | 20 |
| - 800      | 800            | 12                                  | 530                                      | 19                               | 1,888               | 1,030 | 90 | 30 | 50 | 1,076 | 842 | 20 |



# ISC110 Series

## ■ Engineering Data & Dimensions

| Model     | Stroke<br>(mm) | Max. Energy /<br>Cycle (kJ)<br>$E_T$ | Max. Energy /<br>Hour (kJ/hr)<br>$E_{TC}$ | Max. Buffer<br>Force (kN)<br>$F_s$ | Dimension [unit:mm] |       |      |    |    |       |       |    |
|-----------|----------------|--------------------------------------|-------------------------------------------|------------------------------------|---------------------|-------|------|----|----|-------|-------|----|
|           |                |                                      |                                           |                                    | A                   | B     | C    | D  | E  | F     | G     | H  |
| ISC110-50 | 50             | 5                                    | 290                                       | 115                                | 370                 | 203   | Ø110 | 40 | 60 | 270   | 120   | 25 |
| - 100     | 100            | 10                                   | 560                                       | 115                                | 470                 | 280   | Ø110 | 40 | 60 | 320   | 170   | 25 |
| - 150     | 150            | 15                                   | 880                                       | 115                                | 553                 | 339   | Ø110 | 40 | 60 | 379   | 194   | 25 |
| - 200     | 200            | 20                                   | 930                                       | 115                                | 655                 | 390   | Ø110 | 40 | 60 | 430   | 245   | 25 |
| - 250     | 250            | 25                                   | 1,050                                     | 115                                | 757                 | 441   | Ø110 | 40 | 60 | 481   | 296   | 25 |
| - 300     | 300            | 29                                   | 1,180                                     | 115                                | 859                 | 492   | Ø110 | 40 | 60 | 532   | 347   | 25 |
| - 350     | 350            | 34                                   | 1,350                                     | 115                                | 960                 | 543   | Ø110 | 40 | 60 | 583   | 397   | 25 |
| - 400     | 400            | 39                                   | 1,510                                     | 115                                | 1,062               | 594   | Ø110 | 40 | 60 | 634   | 448   | 25 |
| - 450     | 450            | 44                                   | 1,680                                     | 115                                | 1,164               | 645   | Ø110 | 40 | 60 | 685   | 499   | 25 |
| - 500     | 500            | 49                                   | 1,840                                     | 115                                | 1,265               | 695   | Ø110 | 40 | 60 | 735   | 550   | 25 |
| - 600     | 600            | 59                                   | 2,160                                     | 115                                | 1,469               | 797   | Ø110 | 40 | 60 | 837   | 652   | 25 |
| - 700     | 700            | 69                                   | 2,480                                     | 115                                | 1,672               | 899   | Ø110 | 40 | 60 | 937   | 753   | 25 |
| - 800     | 800            | 79                                   | 2,800                                     | 115                                | 1,953               | 1,079 | Ø110 | 40 | 60 | 1,119 | 854   | 25 |
| - 900     | 900            | 88                                   | 3,130                                     | 115                                | 2,151               | 1,179 | Ø110 | 40 | 60 | 1,219 | 952   | 25 |
| - 1000    | 1000           | 73                                   | 3,480                                     | 92                                 | 2,351               | 1,279 | Ø110 | 40 | 60 | 1,319 | 1,052 | 25 |
| - 1200    | 1200           | 60                                   | 2,750                                     | 63                                 | 2,751               | 1,479 | Ø110 | 40 | 60 | 1,519 | 1,252 | 25 |
| - 1400    | 1400           | 41                                   | 1,910                                     | 37                                 | 3,171               | 1,689 | Ø110 | 40 | 60 | 1,729 | 1,462 | 25 |



## Hydraulic Buffers

# ISC Series

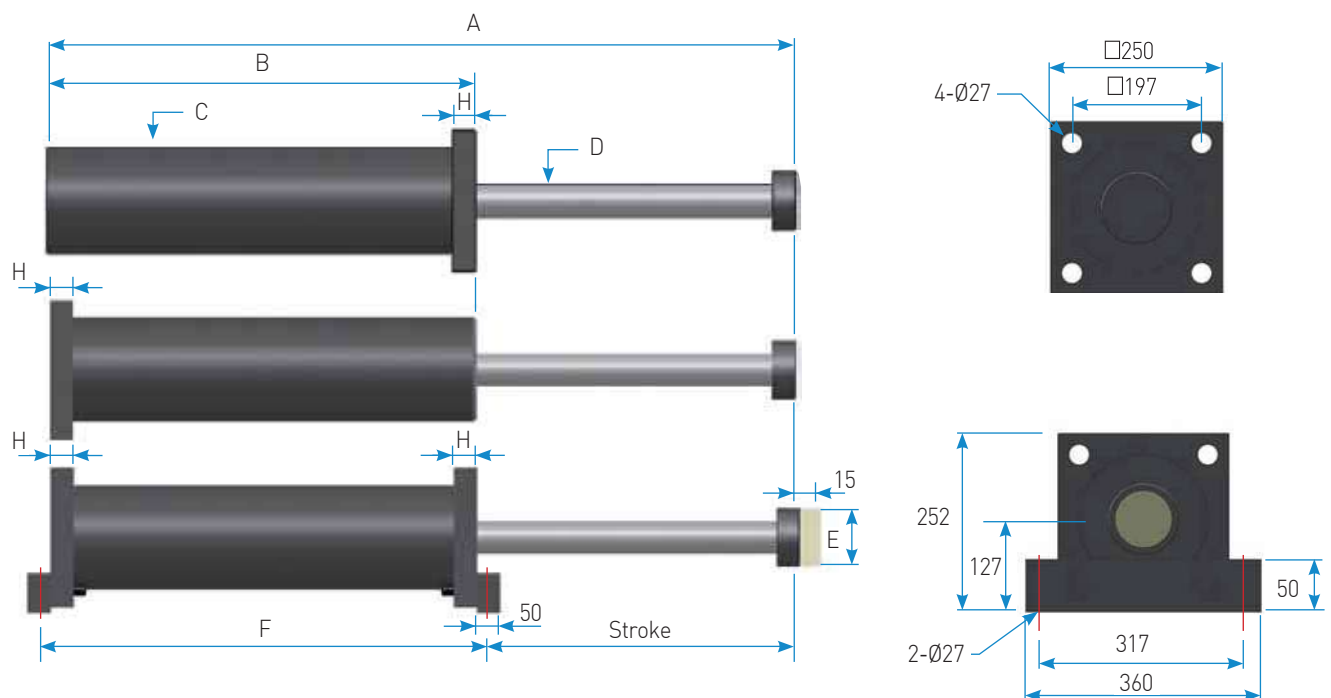




# ISC200 Series

## ■ Engineering Data & Dimensions

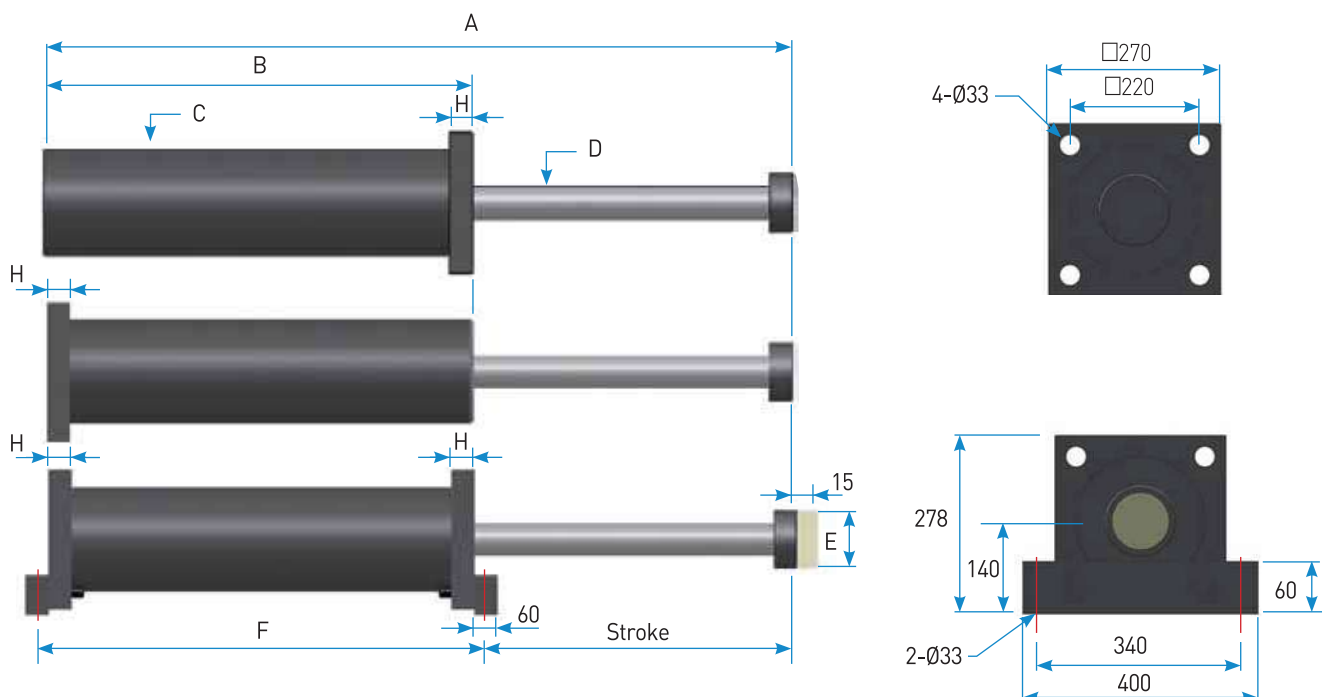
| Model       | Stroke<br>(mm) | Max. Energy /<br>Cycle (kJ)<br>$E_T$ | Max. Energy /<br>Hour (kJ/hr)<br>$E_{TC}$ | Max. Buffer<br>Force (kN)<br>$F_s$ | Dimension [unit:mm] |       |     |    |     |       |       |    |
|-------------|----------------|--------------------------------------|-------------------------------------------|------------------------------------|---------------------|-------|-----|----|-----|-------|-------|----|
|             |                |                                      |                                           |                                    | A                   | B     | C   | D  | E   | F     | G     | H  |
| ISC200 - 50 | 50             | 16                                   | 940                                       | 370                                | 430                 | 394   | 200 | 65 | 100 | 444   | 115   | 40 |
| - 100       | 100            | 31                                   | 1,530                                     | 370                                | 532                 | 445   | 200 | 65 | 100 | 495   | 164   | 40 |
| - 150       | 150            | 47                                   | 1,750                                     | 370                                | 632                 | 495   | 200 | 65 | 100 | 545   | 215   | 40 |
| - 200       | 200            | 63                                   | 1,980                                     | 370                                | 735                 | 547   | 200 | 65 | 100 | 597   | 263   | 40 |
| - 250       | 250            | 79                                   | 2,210                                     | 370                                | 836                 | 597   | 200 | 65 | 100 | 647   | 314   | 40 |
| - 300       | 300            | 93                                   | 2,850                                     | 370                                | 1,032               | 642   | 200 | 65 | 100 | 692   | 365   | 40 |
| - 400       | 400            | 126                                  | 3,300                                     | 370                                | 1,234               | 743   | 200 | 65 | 100 | 793   | 466   | 40 |
| - 500       | 500            | 157                                  | 3,750                                     | 370                                | 1,438               | 845   | 200 | 65 | 100 | 895   | 568   | 40 |
| - 600       | 600            | 188                                  | 4,210                                     | 370                                | 1,642               | 947   | 200 | 65 | 100 | 997   | 670   | 40 |
| - 700       | 700            | 220                                  | 4,660                                     | 370                                | 1,844               | 1,048 | 200 | 65 | 100 | 1,098 | 771   | 40 |
| - 800       | 800            | 251                                  | 5,110                                     | 370                                | 2,048               | 1,150 | 200 | 65 | 100 | 1,200 | 873   | 40 |
| - 900       | 900            | 283                                  | 5,560                                     | 370                                | 2,252               | 1,252 | 200 | 65 | 100 | 1,302 | 975   | 40 |
| - 1000      | 1,000          | 240                                  | 6,110                                     | 300                                | 2,454               | 1,353 | 200 | 65 | 100 | 1,403 | 1,076 | 40 |
| - 1200      | 1,200          | 210                                  | 4,920                                     | 200                                | 2,854               | 1,553 | 200 | 65 | 100 | 1,603 | 1,276 | 40 |





## ■ Engineering Data & Dimensions

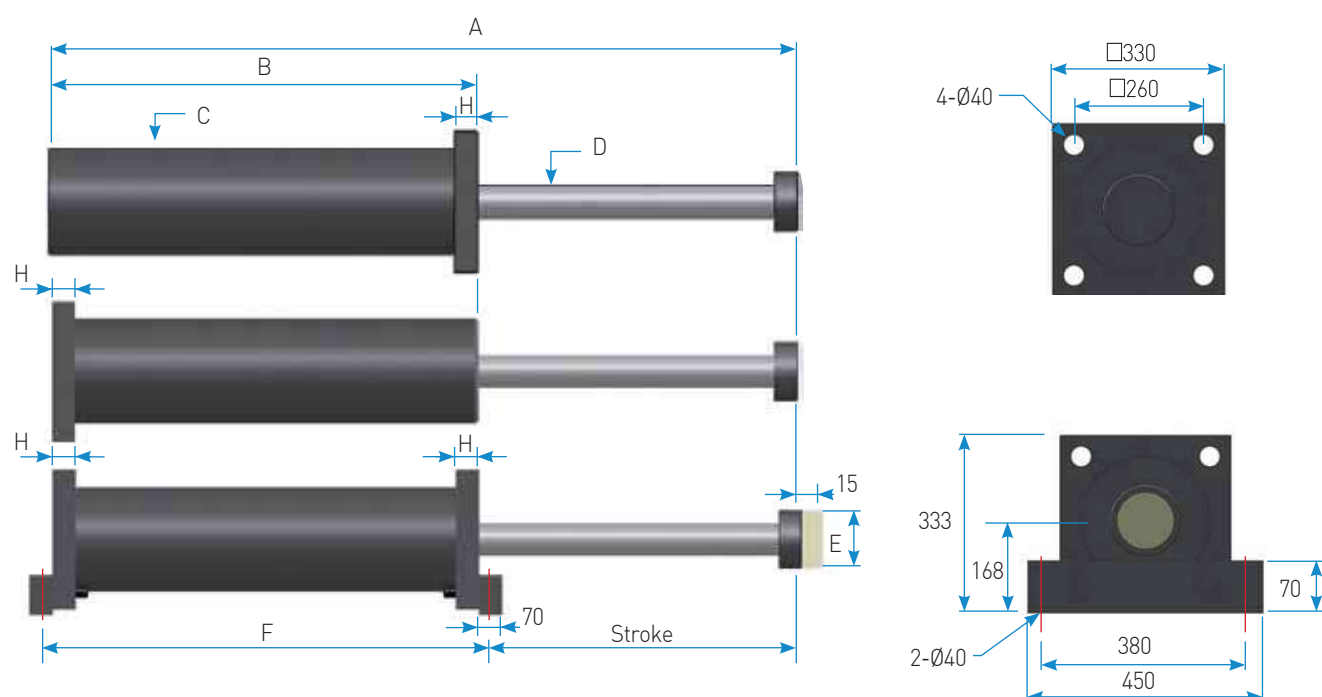
| Model        | Stroke<br>(mm) | Max. Energy /<br>Cycle (kJ)<br>$E_T$ | Max. Energy /<br>Hour (kJ/hr)<br>$E_{TC}$ | Max. Buffer<br>Force (kN)<br>$F_s$ | Dimension [unit:mm] |       |     |    |     |       |       |    |
|--------------|----------------|--------------------------------------|-------------------------------------------|------------------------------------|---------------------|-------|-----|----|-----|-------|-------|----|
|              |                |                                      |                                           |                                    | A                   | B     | C   | D  | E   | F     | G     | H  |
| ISC215 - 100 | 100            | 48                                   | 1,800                                     | 560                                | 591                 | 375   | 215 | 80 | 125 | 435   | 186   | 40 |
| - 150        | 150            | 72                                   | 2,050                                     | 560                                | 693                 | 426   | 215 | 80 | 125 | 486   | 237   | 40 |
| - 200        | 200            | 96                                   | 2,290                                     | 560                                | 795                 | 477   | 215 | 80 | 125 | 537   | 288   | 40 |
| - 250        | 250            | 120                                  | 2,530                                     | 560                                | 895                 | 527   | 215 | 80 | 125 | 587   | 338   | 40 |
| - 300        | 300            | 143                                  | 2,750                                     | 560                                | 997                 | 578   | 215 | 80 | 125 | 638   | 389   | 40 |
| - 400        | 400            | 191                                  | 3,260                                     | 560                                | 1,201               | 680   | 215 | 80 | 125 | 740   | 491   | 40 |
| - 500        | 500            | 239                                  | 4,230                                     | 560                                | 1,504               | 882   | 215 | 80 | 125 | 942   | 592   | 40 |
| - 600        | 600            | 287                                  | 4,740                                     | 560                                | 1,708               | 984   | 215 | 80 | 125 | 1,044 | 694   | 40 |
| - 700        | 700            | 334                                  | 5,200                                     | 560                                | 1,910               | 1,085 | 215 | 80 | 125 | 1,145 | 795   | 40 |
| - 800        | 800            | 382                                  | 5,690                                     | 560                                | 2,114               | 1,187 | 215 | 80 | 125 | 1,247 | 897   | 40 |
| - 1000       | 1,000          | 478                                  | 6,680                                     | 560                                | 2,520               | 1,390 | 215 | 80 | 125 | 1,450 | 1,100 | 40 |
| - 1200       | 1,200          | 417                                  | 6,250                                     | 435                                | 2,920               | 1,590 | 215 | 80 | 125 | 1,650 | 1,300 | 40 |



# ISC275 Series

## ■ Engineering Data & Dimensions

| Model        | Stroke<br>(mm) | Max. Energy /<br>Cycle (kJ)<br>$E_T$ | Max. Energy /<br>Hour (kJ/hr)<br>$E_{TC}$ | Max. Buffer<br>Force (kN)<br>$F_S$ | Dimension [unit:mm] |       |     |     |     |       |       |    |
|--------------|----------------|--------------------------------------|-------------------------------------------|------------------------------------|---------------------|-------|-----|-----|-----|-------|-------|----|
|              |                |                                      |                                           |                                    | A                   | B     | C   | D   | E   | F     | G     | H  |
| ISC275 - 100 | 100            | 78                                   | 2,440                                     | 915                                | 637                 | 391   | 275 | 100 | 160 | 461   | 211   | 50 |
| - 150        | 150            | 117                                  | 2,760                                     | 915                                | 737                 | 441   | 275 | 100 | 160 | 511   | 261   | 50 |
| - 200        | 200            | 156                                  | 3,050                                     | 915                                | 839                 | 492   | 275 | 100 | 160 | 562   | 312   | 50 |
| - 250        | 250            | 194                                  | 3,370                                     | 915                                | 941                 | 543   | 275 | 100 | 160 | 613   | 363   | 50 |
| - 300        | 300            | 233                                  | 3,760                                     | 915                                | 1,043               | 594   | 275 | 100 | 160 | 664   | 414   | 50 |
| - 400        | 400            | 311                                  | 4,300                                     | 915                                | 1,246               | 696   | 275 | 100 | 160 | 766   | 515   | 50 |
| - 500        | 500            | 389                                  | 4,930                                     | 915                                | 1,450               | 798   | 275 | 100 | 160 | 868   | 617   | 50 |
| - 600        | 600            | 467                                  | 6,180                                     | 915                                | 1,769               | 1,015 | 275 | 100 | 160 | 1,085 | 719   | 50 |
| - 750        | 750            | 583                                  | 7,110                                     | 915                                | 2,073               | 1,167 | 275 | 100 | 160 | 1,237 | 871   | 50 |
| - 900        | 900            | 700                                  | 8,040                                     | 915                                | 2,379               | 1,320 | 275 | 100 | 160 | 1,390 | 1,024 | 50 |
| - 1050       | 1,050          | 816                                  | 8,970                                     | 915                                | 2,683               | 1,472 | 275 | 100 | 160 | 1,542 | 1,176 | 50 |
| - 1200       | 1,200          | 790                                  | 8,060                                     | 827                                | 2,989               | 1,625 | 275 | 100 | 160 | 1,695 | 1,329 | 50 |



# Damper & Rate Control

## IHC Series Speed Control

NEW

### IZMAC Hydro Check

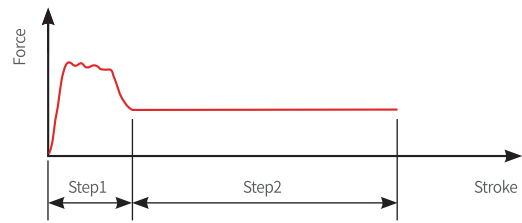
#### DESCRIPTION

IHC is an absorption function added product that absorbs the initial collision of the existing product, in addition to the capability of precise speed control according to the control range, it is possible to control the constant speed of various weights according to the stroke, so it protects the equipment and extends the life of the machine tool.



#### - Process

- ① Step1 : shock absorption
- ② Step2 : Control same speed



#### FEATURES

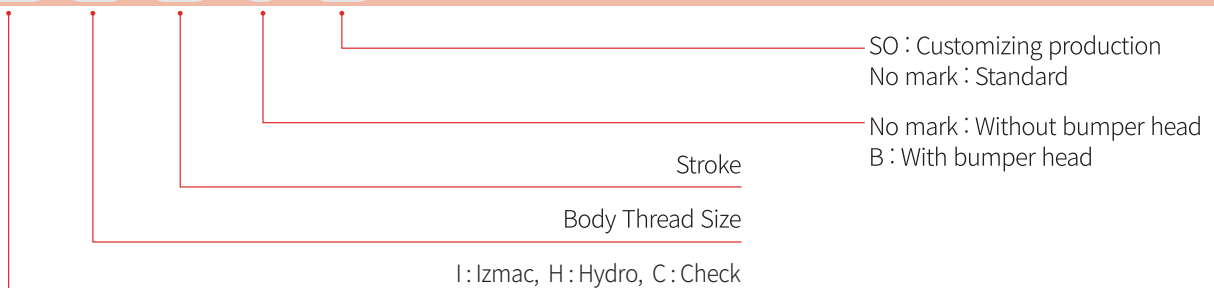
- 1 The device enables easy control precise speed according to 30 steps of adjustment Knob.
- 2 It can be installed simply by using snap ring mounting block, and it can be changed the location also. In addition it can be fixed easily.
- 3 Oil is internally filtered by itself, and speed is controlled always at the same speed.
- 4 Application : Auto feed drilling, Cutting machine, Grinding machine, Boring machine, Sawing machine etc. that require same speed control.
- 5 Application temperature : -10~80°C • Special -30~100°C
- 6 Accurate two step speed control

#### APPLICATION

Widely used where speed control is required, such as drilling, cutting, safety devices, and drilling machines.

#### IHC SERIES ORDERING INFORMATION

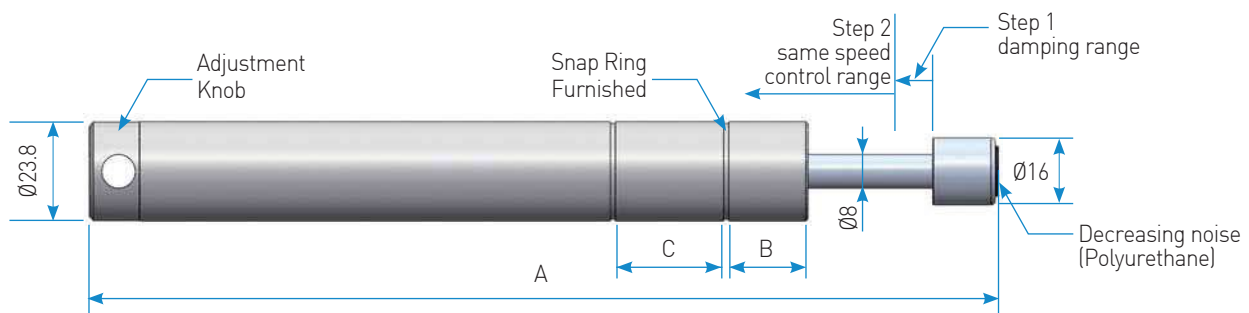
IHC - 24 - 13 - B - SO



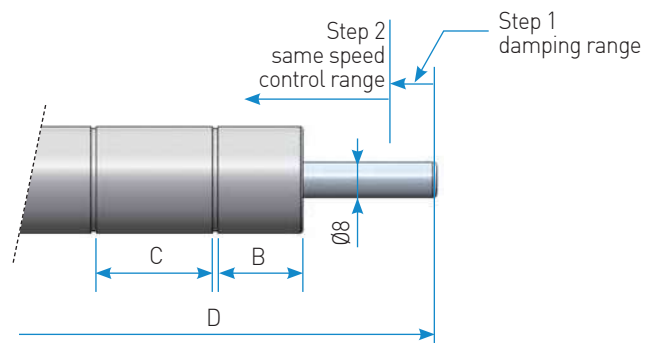
# IHC24 Series

## ■ Engineering Data

| Model       | Damping Range [mm]<br>Step1 | Stroke [mm]<br>Step2 | Dimensions [mm] |    |    |       | Max. Load Angle [°] | Speed Adjustment Range |                         | Weight (g) |
|-------------|-----------------------------|----------------------|-----------------|----|----|-------|---------------------|------------------------|-------------------------|------------|
|             |                             |                      | A               | B  | C  | D     |                     | Load Limits [N]        | Feed Rate Limits [mm/s] |            |
| IHC24 - 13B | 5                           | 13                   | 174             | 21 |    | 165.4 | 0.5                 |                        |                         | 415        |
| - 25B       | 5                           | 25                   | 213             | 19 | 26 | 204.4 | 1                   | 100~3,000              | 0.2~30                  | 465        |
| - 51B       | 5                           | 51                   | 314             | 16 |    | 305.4 | 0.5                 |                        |                         | 645        |

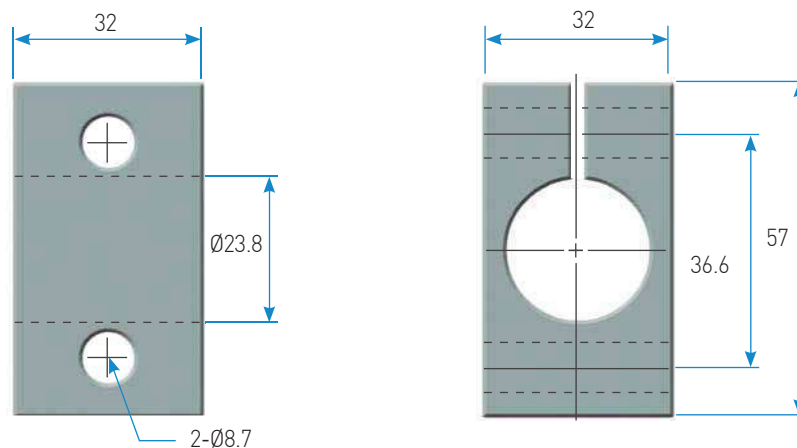


Steel head+Hard PU button  
[B TYPE : Black]



## ■ Mount Block

MB 24





# ICD Series Crash Damper

NEW

## IZMAC Crash Damper

### DESCRIPTION

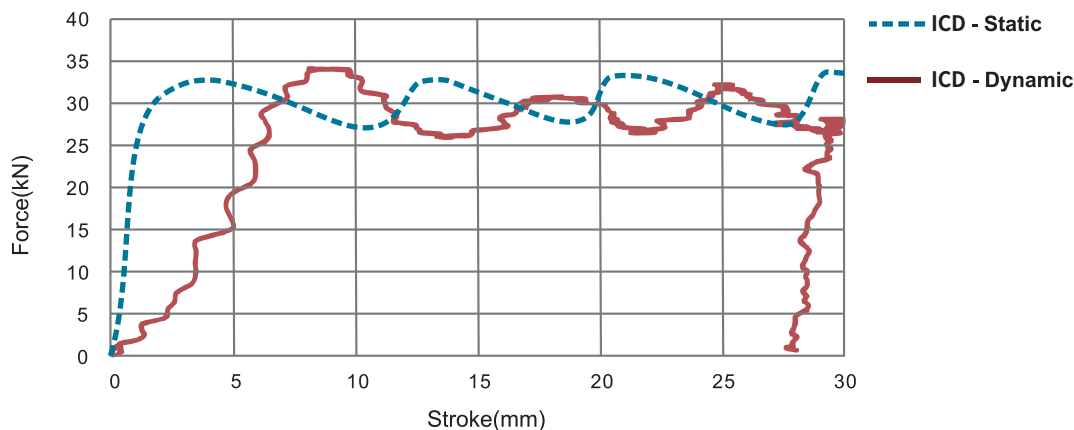
ICD has multi-steps bumpy shape and it absorbs shock energy on collision moment with compressing step by step.  
Engineered to maintain rather low peak & recoil force.

### FEATURES

- 1 Shock absorption response function is steady and absorption efficiency is high.
- 2 It has multi-steps bumpy shape and it absorb shock energy on collision moment with compressing step by step.
- 3 Compact (small and sturdy) and easy to handling.
- 4 Body surface strongly treated to prevent corrosion by nickel plating or alloy plating (black).
- 5 Various sizes, forces, strokes with customizing models are available.
- 6 Temperature range : -40 ~ 90°C



### GRAPH



### ICD SERIES ORDERING INFORMATION

ICD - 087 - 53

Nominal stroke

Tube outer diameter

I : Izmec  
C : Crash  
D : Damper

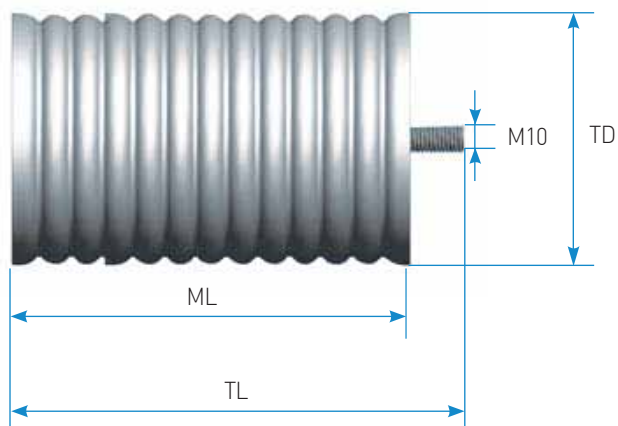
Damper & Rate Control

ICD Series

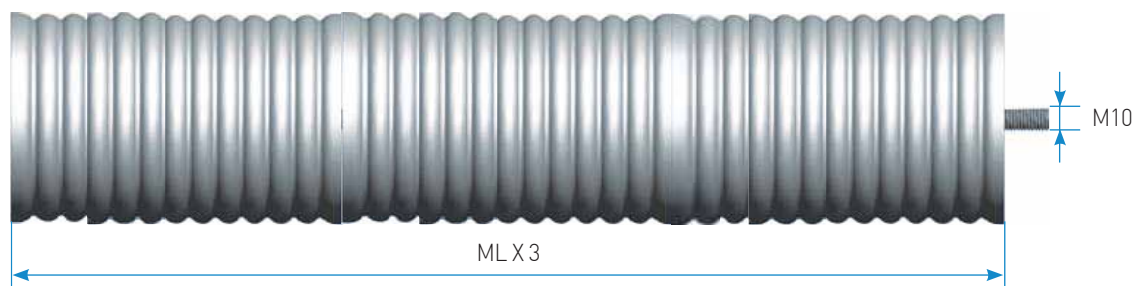
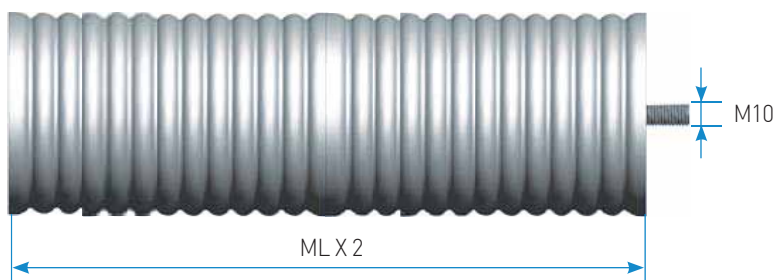
# ICD87 Series

## ■ Dimensions & Engineering Data

| Model         | TD   | ML  | TL    | Thread | stroke[mm] | Force[kN] | Energy[J] |
|---------------|------|-----|-------|--------|------------|-----------|-----------|
| ICD - 87 - 53 | 87.5 | 110 | 134.3 | M10    | 53         | 30        | 1450      |
| - 106         | 87.5 | 220 | 244   | M10    | 105        | 30        | 2900      |



## ■ How to use



# IPC Series Crash Damper

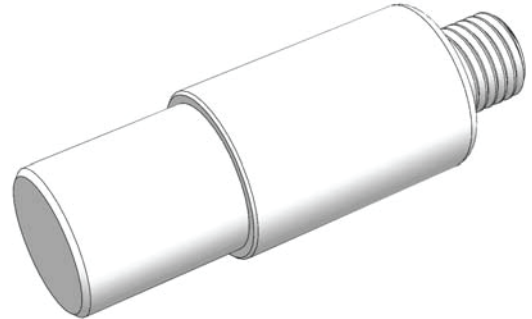
NEW

## IZMAC Plastic Type Crash Damper

### DESCRIPTION

IPC is a disposable damping product that absorbs high energy by being made of heterogeneous engineering plastics, and it has a structure that converts kinetic energy from impact into deformation energy.

We provide damping solutions to meet the best performance in various industries. This product is suitable for emergency purpose.

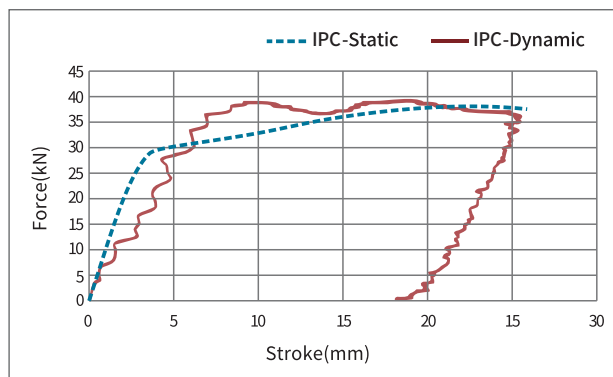


### FEATURES

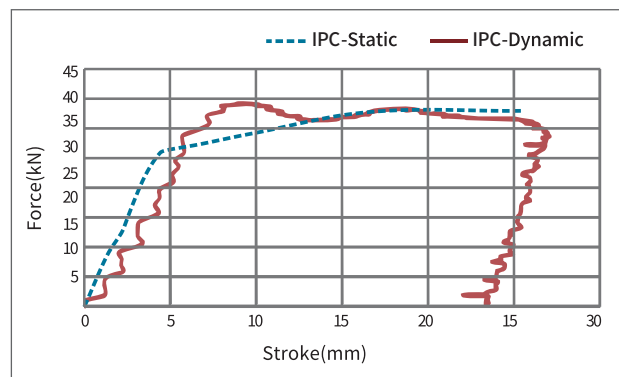
- 1 High deceleration performance.
- 2 Shock absorption response function is steady and absorption efficiency is high.
- 3 Low price, narrow space installation is available.
- 4 No maintenance required, no corrosion.
- 5 Customization is possible for various sizes, forces, strokes and capacities.
- 6 Operating temperature: -40~90°C

### GRAPH

IPC 2-038-030



IPC 2-034-030



### IPC SERIES ORDERING INFORMATION

IPC - 1 - 028 - 035

Nominal stroke

Tube outer diameter

Thread type(1 : threaded type, 2 : wrench bolt type)

I : Izmac

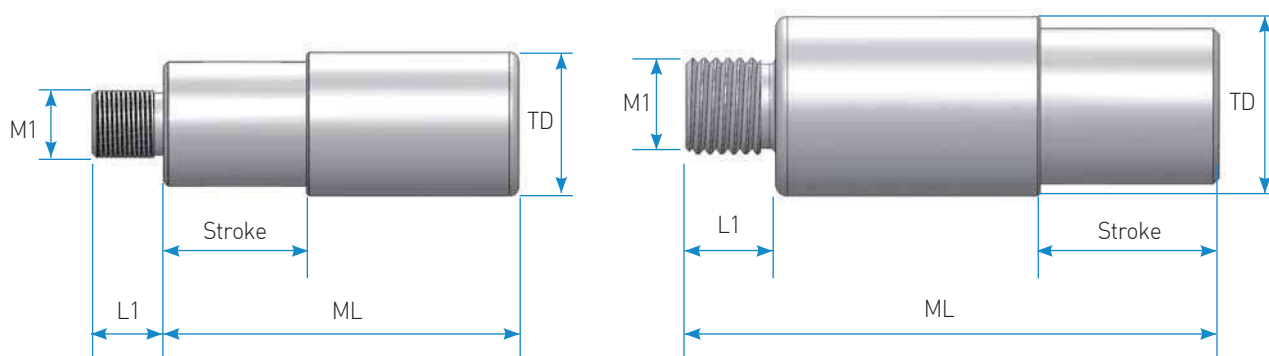
P: Plastic Type

C: Crash Damper

# IPC20 ~ 24 Series

## ■ Dimensions & Engineering Data

| Model       | Stroke[mm] | Fn[kN] | En[J]   | TD[mm] | ML[mm] | M1[mm]      | L1[mm] |
|-------------|------------|--------|---------|--------|--------|-------------|--------|
| IPC-1-20-15 | 15         | 9.1    | 122.8   | 20     | 43.0   | M12         | 10     |
| -32-25      | 25         | 28.0   | 631     | 32     | 70.0   | M16         | 15     |
| -40-30      | 30         | 46.5   | 1255.4  | 40     | 85.0   | M20         | 18     |
| -52-40      | 40         | 82.9   | 2985.6  | 52     | 111.0  | M30         | 25     |
| -64-50      | 50         | 129.9  | 5,846.2 | 64     | 136.5  | M36         | 30     |
| IPC-2-24-30 | 30         | 15.6   | 420.7   | 24     | 74.5   | M8 x 1.25P  | 14.5   |
| -34-30      | 30         | 34.8   | 940     | 34     | 79.0   | M12 x 1.75P | 18     |
| -38-30      | 30         | 44.6   | 1204.3  | 38     | 81.0   | M12 x 1.75P | 21     |
| -52-40      | 40         | 88.3   | 3180.6  | 52     | 108.0  | M16 x 2.00P | 19     |
| -64-50      | 50         | 137.5  | 6,187.1 | 64     | 132.5  | M20 x 2.50P | 23.5   |



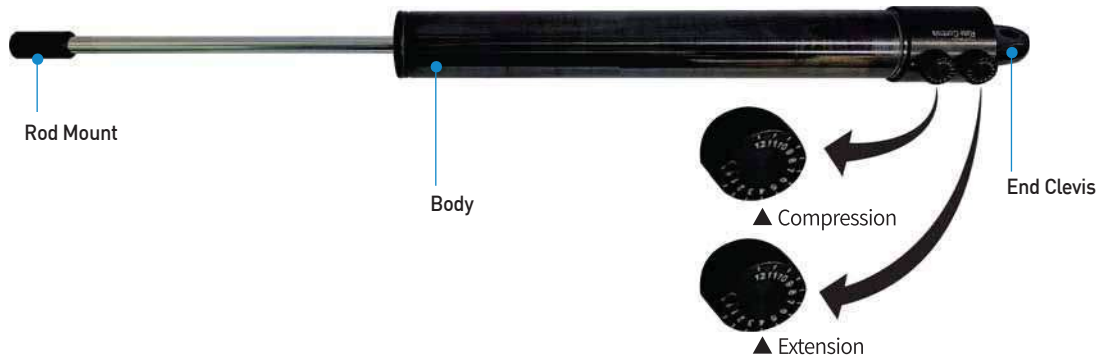
# IRD Series Rate Controls

NEW

## IZMAC Rate control Double dial

### DESCRIPTION

IRD provides stable and accurate single directional/bi-directional speed control for the object when it moves from one position to another. This product is designed to set precise adjustment dial in 12 steps according to the user's application conditions. With speed control, you can significantly reduce shock and vibration to the machine which may effect on uncontrolled & unexpected operation troubles.



### FEATURES

- 1 Guarantees product life through perfect quality control.
- 2 Reinforced materials are used to enhance durability.
- 3 Perfect speed control : For safe machine operation and longer machine life
- 4 For safety device : Equipment cover (open / close)
- 5 Operating temperature : -10 ~ 80°C • Special -30 ~ 100°C

### APPLICATION

Automotive , mechanical engineering, electronics industries.

### IRD SERIES ORDERING INFORMATION

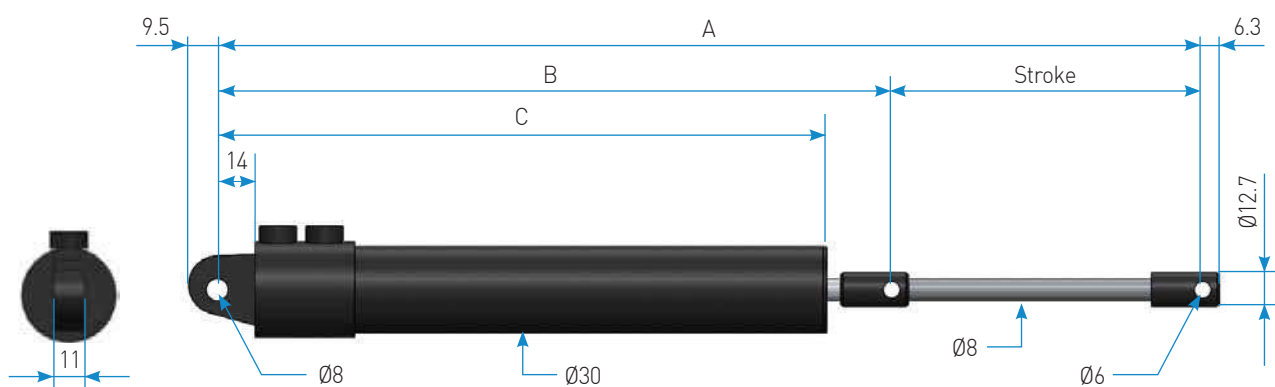
| IRD | 30 | 50 | EC |                                                  |
|-----|----|----|----|--------------------------------------------------|
|     |    |    |    | E(Extension)<br>C(Compression)<br>EC(Both Sided) |
|     |    |    |    | Stroke                                           |
|     |    |    |    | Body Thread Size                                 |
|     |    |    |    | I : Izmec<br>R : Rate control<br>D : Double dial |



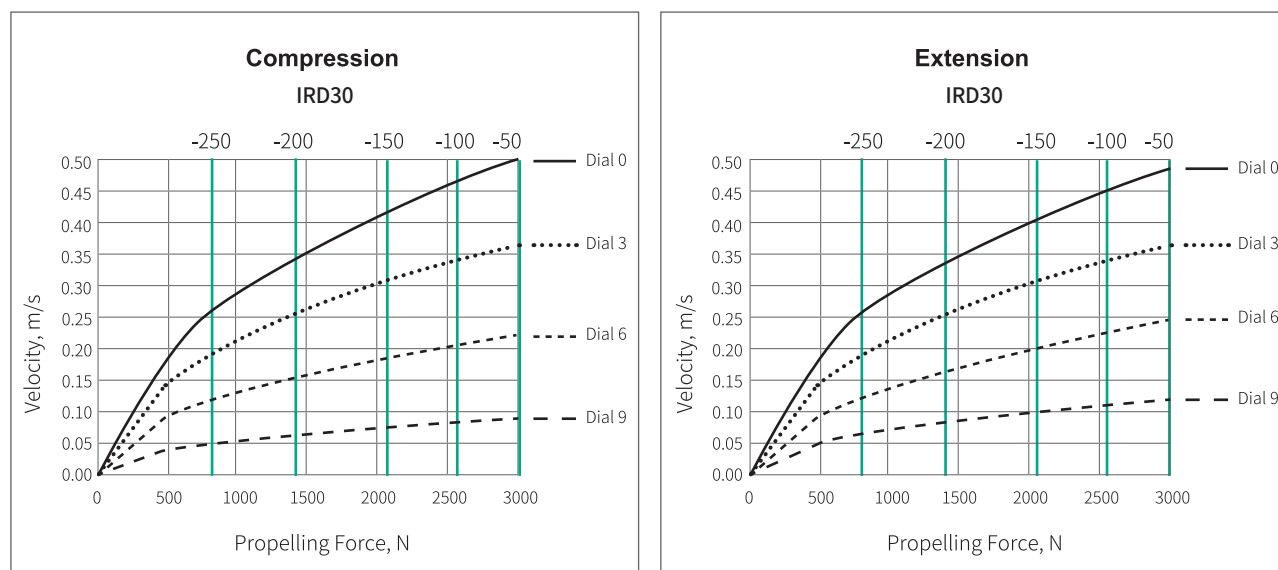
# IRD30 Series

## ■ Dimensions & Engineering Data

| Model      | Stroke (mm) s | Max. Propelling Force (N) |       | Weight (g) | Dimensions (mm) |     |     | Direction |
|------------|---------------|---------------------------|-------|------------|-----------------|-----|-----|-----------|
|            |               | Ext                       | Comp  |            | A               | B   | C   |           |
| IRD30 - 50 | 50            | 3,000                     | 3,000 | 310        | 250             | 200 | 176 | E,C,E&C   |
| - 100      | 100           | 3,000                     | 2,600 | 380        | 350             | 250 | 226 | E,C,E&C   |
| - 150      | 150           | 3,000                     | 2,100 | 450        | 450             | 300 | 276 | E,C,E&C   |
| - 200      | 200           | 3,000                     | 1,400 | 530        | 550             | 350 | 326 | E,C,E&C   |
| - 250      | 250           | 3,000                     | 800   | 600        | 650             | 400 | 376 | E,C,E&C   |



## ■ Graph



# ISR Series Shock Absorber : Short Type

**NEW**

## IZMAC Short Range Absorber

### DESCRIPTION

ISR can be installed in a limited space and is suitable for industrial applications requiring high energy absorption. And it is a suitable compact industrial hydraulic product. Unlike general absorber series, it is designed with a unique structure and by way of fast braking time, shock, and noise reduction function, it provides optimum results of best productivity improvement.

### FEATURES

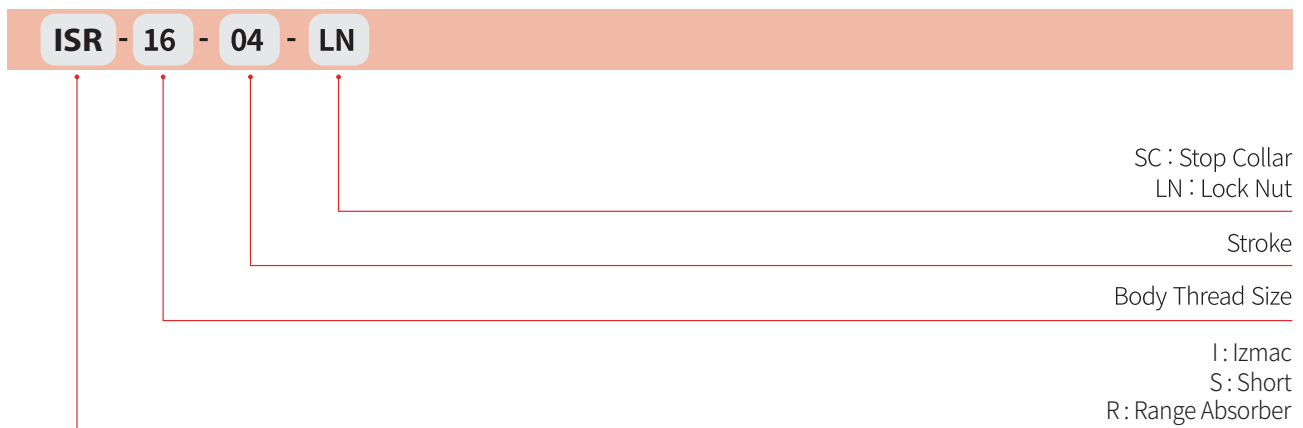
- 1 There are various models of different sizes from M16 to M32.
- 2 Specially, it has more than three times maximum energy performance compared to other existing products.
- 3 Improve noise reduction function and increase product durability
- 4 Thanks to the unique orifice structure, it enables smooth absorption characteristics to stop collision material softly.
- 5 Maximum impact speed : 0.05 ~ 3m/s
- 6 Piston rod : Hard Chrome : 25µm
- 7 Available to use without an external stopper.



### APPLICATION

Automotive , Mechanical engineering, Electronics industries

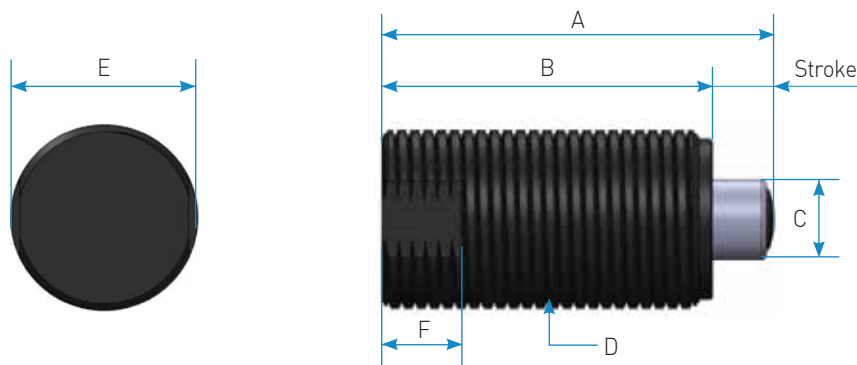
### ISR SERIES ORDERING INFORMATION



# ISR16 ~ 25 Series

## ■ Engineering Data

| Model      | Stroke (mm) | Max.Energy / Cycle(Nm) E <sub>T</sub> | Max.Energy / Hour (Nm / h) E <sub>T</sub> C | Recoil Forec(N) |       | Weight (g) |
|------------|-------------|---------------------------------------|---------------------------------------------|-----------------|-------|------------|
|            |             |                                       |                                             | Ext.            | Comp. |            |
| ISR16 - 04 | 4           | 12                                    | 18,000                                      | 8               | 16    | 1.25       |
| - 09       | 9           | 28                                    | 42,300                                      | 8               | 17    | 1.4        |
| ISR20 - 07 | 7           | 40                                    | 64,600                                      | 11              | 18    | 65         |
| - 12       | 12          | 70                                    | 110,700                                     | 12              | 18    | 75         |
| ISR25 - 08 | 8           | 70                                    | 108,900                                     | 15              | 20    | 105        |
| - 15       | 15          | 130                                   | 204,000                                     | 16              | 21    | 115        |



## ■ Dimensions

(unit : mm)

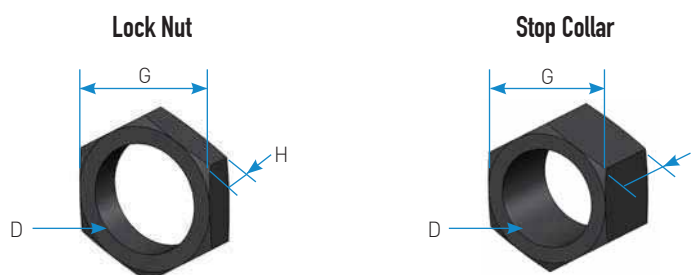
| Model      | Stroke | A    | B    | C  | D          | E  | F  |
|------------|--------|------|------|----|------------|----|----|
| ISR16 - 04 | 4      | 31   | 27   | 6  | M16 X 1.5P | 14 | 7  |
| - 09       | 9      | 49   | 41   | 6  | M16 X 1.5P | 14 | 7  |
| ISR20 - 07 | 7      | 44.5 | 37.5 | 9  | M20 X 1.5P | 18 | 9  |
| - 12       | 12     | 63.5 | 51.5 | 9  | M20 X 1.5P | 18 | 9  |
| ISR25 - 08 | 8      | 52   | 44   | 12 | M25 X 1.5P | 23 | 10 |
| - 15       | 15     | 78   | 63   | 12 | M25 X 1.5P | 23 | 10 |

## ■ Accessories

- Two Lock Nuts included for each absorber
- Other accessories to be ordered separately

(unit : mm)

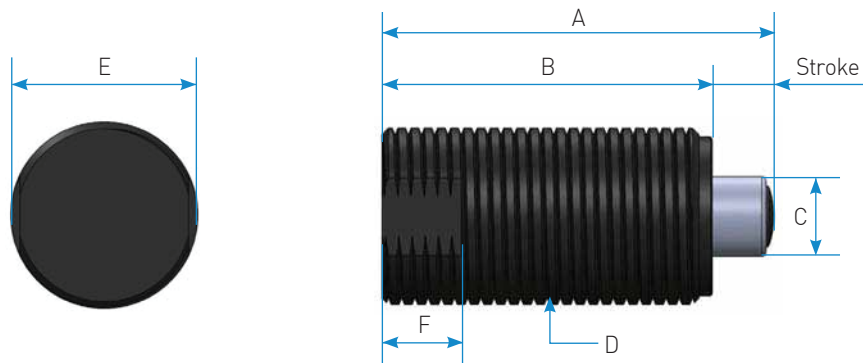
| Model      | G  | H | I  |
|------------|----|---|----|
| ISR16 - 04 | 19 | 6 | 12 |
| ISR20 - 07 | 24 | 6 | 16 |
| ISR25 - 08 | 32 | 8 | 18 |





## Engineering Data

| Model      | Stroke<br>(mm) | Max.Energy /<br>Cycle(Nm) E <sub>T</sub> | Max.Energy /<br>Hour (Nm / h) E <sub>T</sub> C | Recoil Forec(N) |       | Weight (g) |
|------------|----------------|------------------------------------------|------------------------------------------------|-----------------|-------|------------|
|            |                |                                          |                                                | Ext.            | Comp. |            |
| ISR30 - 09 | 9              | 120                                      | 186,000                                        | 17              | 30    | 200        |
| - 15       | 15             | 200                                      | 311,000                                        | 17              | 31    | 210        |
| ISR32 - 13 | 13             | 210                                      | 323,000                                        | 17              | 30    | 270        |
| - 20       | 20             | 330                                      | 498,000                                        | 17              | 31    | 280        |



## Dimensions

(unit : mm)

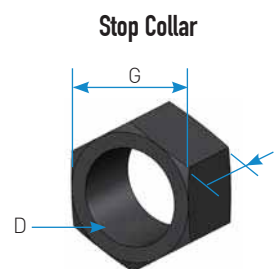
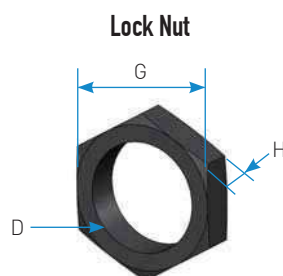
| Model      | Stroke | A    | B    | C  | D          | E  | F  |
|------------|--------|------|------|----|------------|----|----|
| ISR30 - 09 | 9      | 61.5 | 52.5 | 16 | M30 X 1.5P | 28 | 12 |
| - 15       | 15     | 88.5 | 73.5 | 16 | M30 X 1.5P | 28 | 12 |
| ISR32 - 13 | 13     | 76   | 63   | 18 | M32 X 1.5P | 30 | 13 |
| - 20       | 20     | 105  | 85   | 18 | M32 X 1.5P | 30 | 13 |

## Accessories

- Two Lock Nuts included for each absorber
- Other accessories to be ordered separately

(unit : mm)

| Model      | G  | H | I  |
|------------|----|---|----|
| ISR30 - 09 | 36 | 8 | 20 |
| ISR32 - 13 | 44 | 8 | 25 |



# Best Engineered For Energy Absorption Technology

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Shock Absorber  
IAA Series  
(Steel Head)



Shock Absorber  
IMS Series



Shock Absorber  
IHA Series



Hydraulic Buffer  
IHG Series



Hydraulic Buffer  
ISC Series



Damper & Rate control  
IHC Series



Damper & Rate control  
IRD Series



Damper & Rate control  
ICD Series



Damper & Rate control  
ISR Series



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